General Physics II

PH 212

Contents

1 Waves **1**

1.1 Introduction to Waves . . . 1

1.2 Mechanical and Non–Mechanical Waves . . . 2

1.2.1 Mechanical Waves . . . 2

1.2.2 Non–Mechanical Waves . . . 2

1.3 Transverse and Longitudinal Waves . . . 3

1.3.1 Transverse Waves . . . 3

1.3.2 Longitudinal Waves . . . 4

1.3.3 Comparison of Transverse and Longitudinal Waves . . . 6

1.4 Wavelength and Frequency . . . 6

1.4.1 Wave Function . . . 7

1.4.2 Physical Meaning of the Wave Parameters . . . 7

1.4.3 Period, Frequency, and Angular Frequency . . . 9

1.5 The Speed of a Travelling Wave . . . 15

1.6 Wave Speed on a Stretched String . . . 18

1.7 The Wave Equation . . . 24

1.8 Reflection of Waves . . . 31

1.8.1 Reflection at a Fixed End . . . 31

1.8.2 Physical Interpretation . . . 32

1.8.3 Reflection at a Free End . . . 33
1.8.4 Boundary Conditions and Phase Change . . . 34
1.8.5 Energy Considerations . . . 34
1.9 Interference of Waves . . . 34
1.9.1 Superposition of Pulses . . . 35
1.9.2 Sinusoidal Waves and Interference . . . 35
1.9.3 Resultant Wave . . . 37
1.9.4 Constructive and Destructive Interference . . . 38

2 Sound **44**
2.1 The Speed of Sound . . . 44
2.2 Travelling Sound Waves . . . 47
2.2.1 Displacement Function . . . 49
2.2.2 Pressure Variation . . . 50
2.2.3 Phase Relationship Between Displacement and Pressure . . . 51
2.3 Interference of Sound Waves . . . 52
2.3.1 Constructive and Destructive Interference . . . 53
2.3.2 Phase Difference Due to Path Difference . . . 54
2.3.3 Conditions for Interference in Terms of Path Difference . . . 56
2.4 Intensity and Sound Level . . . 57
2.4.1 Variation of Intensity with Distance . . . 58
2.4.2 The Decibel Scale . . . 60
2.4.3 Examples of Sound Levels . . . 62
2.4.4 Safe Sound Levels and Hearing Protection . . . 74
2.4.5 Physiological Effects of Noise . . . 76
2.4.6 Industrial Noise and Sound as a Diagnostic Tool . . . 78

2.4.7 Monitoring Noise . . . 81

3 Interference of Light **85**

3.1 Introduction . . . 85

3.2 Light as a Wave . . . 85

3.2.1 Huygens' Principle . . . 86

3.2.2 Physical Meaning . . . 86

3.3 Diffraction as a Prelude to Interference . . . 87

3.4 Young's Interference Experiment . . . 88

3.4.1 Experimental Arrangement . . . 89

3.4.2 Nature of the Pattern . . . 90

3.4.3 Why the Slits Must Be Coherent . . . 90

3.5 Path Difference and Phase Difference . . . 90

3.5.1 Relation Between Path Difference and Phase Difference . . . 91

3.6 Conditions for Bright and Dark Fringes . . . 92

3.6.1 Bright Fringes (Constructive Interference) . . . 92

3.6.2 Dark Fringes (Destructive Interference) . . . 92

3.7 Fringe Positions on a Distant Screen . . . 93

3.7.1 Position of Bright Fringes . . . 93

3.7.2 Position of Dark Fringes . . . 93

3.7.3 Fringe Spacing . . . 93

3.8 Worked Examples . . . 94

3.9 Further Discussion . . . 98

3.9.1 Why the Central Fringe Is Bright . . . 98

3.9.2 Effect of Increasing the Wavelength . . . 98

3.9.3 Effect of Increasing the Slit Separation . . . 98

3.9.4 Effect of Increasing the Screen Distance . . . 98
3.10 Exercises . . . 99
3.11 Solutions . . . 101

4 Diffraction **117**
4.1 Introduction . . . 117
4.2 Diffraction by a Single Slit . . . 118
4.2.1 Experimental Arrangement . . . 118
4.2.2 Origin of the Central Bright Fringe . . . 118
4.2.3 Locating the First Dark Fringe . . . 119
4.2.4 All Dark Fringes . . . 121
4.2.5 Why Integer Wavelength Path Differences Give Minima . . . 122
4.2.6 Positions of Minima on a Distant Screen . . . 122
4.2.7 Effect of Slit Width . . . 123
4.2.8 Intensity Distribution for a Single Slit . . . 124
4.3 Diffraction by a Double Slit . . . 127
4.3.1 Interference and Diffraction Together . . . 127
4.3.2 Intensity Distribution . . . 128
4.3.3 Limiting Cases . . . 130
4.3.4 Missing Orders . . . 131
4.4 Diffraction Gratings . . . 133
4.4.1 Definition and Structure . . . 133
4.4.2 Diffraction Pattern of a Grating . . . 134
4.4.3 Condition for Principal Maxima . . . 135
4.4.4 Maximum Possible Order . . . 136
4.4.5 Determination of Wavelength Using a Diffraction Grating . . . 137

4.4.6 Why a Grating Is Better Than a Double Slit for Measuring Wavelength . 137
4.4.7 Dispersion by a Diffraction Grating . . . 138
4.5 Exercises . . . 141
4.6 Solutions . . . 143

5 Electrostatics **166**
5.1 Introduction . . . 166
5.2 Electric Charge . . . 167
5.2.1 Electric Charge in Matter . . . 167
5.2.2 Conductors and Insulators . . . 168
5.2.3 Quantization of Charge . . . 169
5.2.4 Conservation of Charge . . . 170
5.3 Coulomb's Law . . . 171
5.3.1 Statement of Coulomb's Law . . . 171
5.3.2 Vector Form of Coulomb's Law . . . 172
5.3.3 Permittivity of Free Space . . . 173
5.3.4 Principle of Superposition . . . 173
5.4 The Electric Field . . . 174
5.4.1 Definition of Electric Field . . . 175
5.4.2 Direction of the Electric Field . . . 175
5.4.3 Electric Field Lines . . . 177
5.4.4 Electric Field Due to a Point Charge . . . 177
5.4.5 Electric Field Due to Several Charges . . . 179
5.5 Electric Potential and Electric Potential Energy . . . 180
5.5.1 Electric Potential Energy . . . 180
5.5.2 Electric Potential . . . 180

5.5.3 Potential Difference . . . 182
5.5.4 Electron-Volt . . . 182
5.5.5 Equipotential Surfaces . . . 183
5.6 Calculating Potential from the Electric Field . . . 184
5.6.1 Uniform Electric Field . . . 186
5.7 Potential Due to a Charged Particle . . . 187
5.7.1 Potential Due to a Group of Point Charges . . . 189
5.8 Relationship Between Electric Field and Electric Potential . . . 192
5.9 Exercise Problems . . . 192

Chapter 1

Waves

1.1 Introduction to Waves

Waves are among the most fundamental mechanisms by which energy and information are transmitted in nature. From the propagation of sound in air, ripples on the surface of water, and vibrations of a stretched string, to the transmission of light, radio signals, and X-rays through space, wave phenomena pervade nearly every branch of physics and engineering.

In its most general sense, a *wave* is a disturbance or oscillation that propagates through space and time, transferring energy from one point to another without the net transport of matter. While the particles of the medium (if a medium is present) may undergo oscillatory motion about their equilibrium positions, they do not travel with the wave itself. Instead, it is the *disturbance* that moves.

The study of waves provides a unifying framework for understanding diverse physical systems. Many concepts introduced in this chapter—such as wavelength, frequency, wave speed, superposition, interference, and standing waves—will recur throughout this course in the study of sound, optics, and electromagnetic phenomena.

1.2 Mechanical and Non–Mechanical Waves

Waves can be broadly classified according to whether or not they require a material medium for their propagation.

1.2.1 Mechanical Waves

Mechanical waves require a material medium in order to propagate. The wave motion arises from the elastic properties of the medium and the restoring forces that act when the medium is disturbed. Common examples include waves on a stretched string, sound waves in air, and water waves on the surface of a liquid.

The speed of a mechanical wave depends on the physical properties of the medium, such as its density, elasticity, and tension. For example, sound travels at different speeds in air, water, and solids due to differences in compressibility and mass density.

1.2.2 Non–Mechanical Waves

Non–mechanical waves do not require a material medium and can propagate through vacuum. The most important examples are *electromagnetic waves*, which include radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays.

Electromagnetic waves consist of oscillating electric and magnetic fields that sustain one another as they propagate through space. All electromagnetic waves travel in vacuum at the same speed, known as the speed of light,

$$c \approx 3.00 \times 10^{8} \text{ m s}^{-1}.$$

1.3 Transverse and Longitudinal Waves

One of the most important ways of classifying waves is according to the direction in which the particles of the medium oscillate relative to the direction of propagation of the wave. On this basis, waves are classified as *transverse* or *longitudinal*.

1.3.1 Transverse Waves

A **transverse wave** is a wave in which the particles of the medium oscillate in a direction *perpendicular* to the direction in which the wave propagates.

If the wave travels along the x–direction, then the displacement of the particles occurs in a direction perpendicular to x, commonly along the y– or z–axis. Each particle oscillates about its equilibrium position while the disturbance moves through the medium.

Transverse waves are characterized by alternating points of maximum and minimum displacement known as:

- **Crests** – points of maximum positive displacement,
- **Troughs** – points of maximum negative displacement.

Examples of Transverse Waves

- **Waves on a stretched string or rope**: When one end of a string is shaken up and down, a transverse wave travels along the string.
- **Water surface waves**: The water particles move mainly up and down while the wave propagates horizontally across the surface.
- **Electromagnetic waves**: Light, radio waves, X-rays, and other electromagnetic waves are transverse waves. The oscillating electric and magnetic fields are perpendicular to the direction of propagation and to each other.

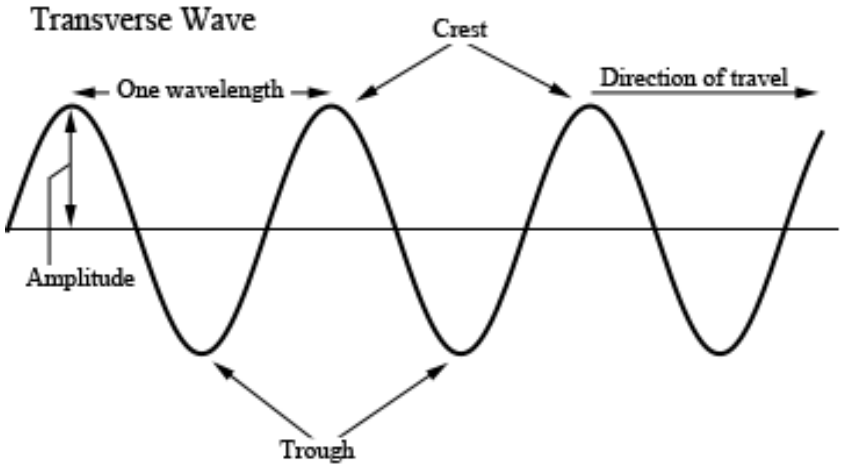


Figure 1.1: A transverse wave propagating along a medium. The particles of the medium oscillate perpendicular to the direction of wave propagation. Crests and troughs correspond to points of maximum positive and negative displacement, respectively.

An important consequence of the transverse nature of electromagnetic waves is that they can be *polarized*, a property that will be discussed later.

1.3.2 Longitudinal Waves

A **longitudinal wave** is a wave in which the particles of the medium oscillate in a direction *parallel* to the direction of wave propagation.

As the wave travels, particles of the medium move back and forth along the same line as the wave motion. This produces alternating regions of:

- **Compression**: regions where the particles are closer together and the pressure is higher,
- **Rarefaction**: regions where the particles are farther apart and the pressure is lower.

Unlike transverse waves, longitudinal waves do not exhibit crests and troughs.

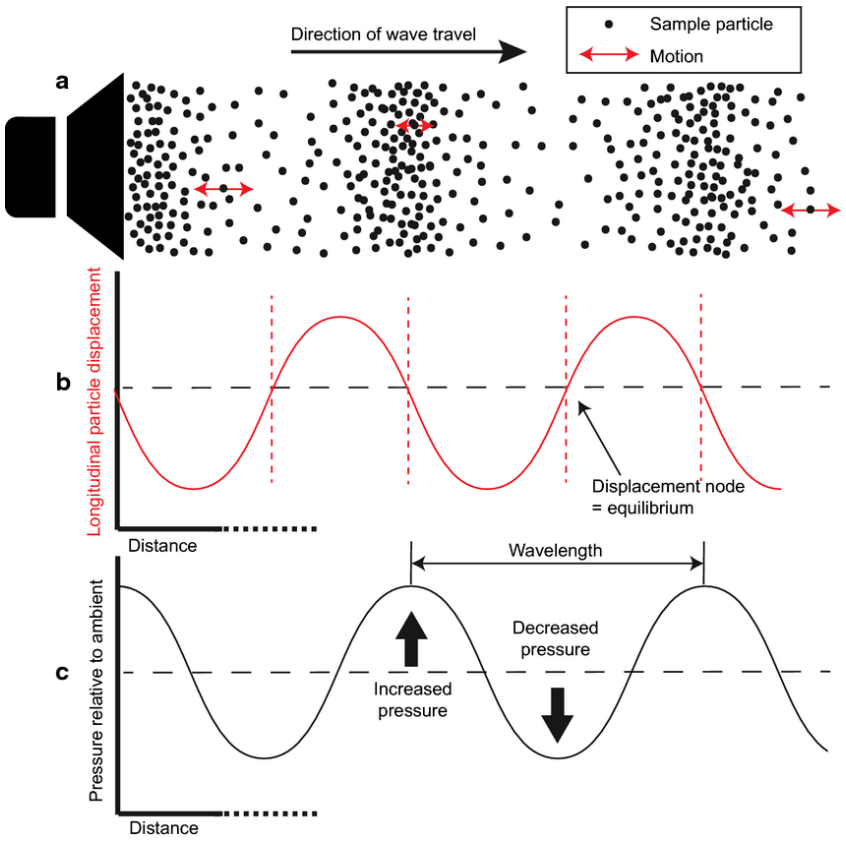


Figure 1.2: A longitudinal wave propagating through a medium. The particles oscillate parallel to the direction of propagation, producing alternating regions of compression (high pressure) and rarefaction (low pressure). Sound waves in air are a common example.

Examples of Longitudinal Waves

- **Sound waves in air**: Sound propagates through successive compressions and rarefactions of air molecules.
- **Sound waves in liquids and solids**: Although the speed differs, the longitudinal nature of sound remains the same.
- **Compression waves in a slinky**: When a slinky is pushed and pulled along its length, compressions travel along the spring.

1.3.3 Comparison of Transverse and Longitudinal Waves

The essential differences between transverse and longitudinal waves are summarized below:

Feature	**Transverse Waves**	**Longitudinal Waves**
Particle motion	Perpendicular to propagation	Parallel to propagation
Wave features	Crests and troughs	Compressions and rarefactions
Polarization	Possible	Not possible
Typical examples	String waves, light	Sound waves

Understanding the distinction between transverse and longitudinal waves is essential for the study of sound, optics, and electromagnetic phenomena. Many wave properties such as reflection, refraction, diffraction, and interference apply to both types, while polarization is exclusive to transverse waves.

1.4 Wavelength and Frequency

To describe a wave completely, it is not sufficient to state that a disturbance is propagating along a medium. We must specify both the *shape of the wave* and how this shape evolves with time. For a wave travelling along a stretched string, this information is provided by a

mathematical function that gives the transverse displacement of each element of the string at any position and time.

1.4.1 Wave Function

Let the wave propagate along the x–axis. The transverse displacement y of a small element of the string located at position x at time t is described by a function of the form

$$y = h(x, t),$$

where y measures the displacement of the string element from its equilibrium position.

In many physical situations, waves exhibit a smooth, repeating shape that can be accurately described by a sinusoidal function. A sinusoidal wave travelling in the positive x–direction may be written as

$$y(x, t) = y_m \sin(kx - \omega t),$$

where y_m, k, and ω are constants that characterize the wave.

Because this equation explicitly depends on the position x, it allows us to determine the displacement of *every* point on the string at a given instant of time. Consequently, it completely specifies the shape of the wave at any moment.

1.4.2 Physical Meaning of the Wave Parameters

The various quantities appearing in the wave equation have clear physical interpretations, as illustrated in Fig. 1.3.

Amplitude

The constant y_m is the **amplitude** of the wave. It represents the maximum transverse displacement of any element of the string from its equilibrium position. The amplitude is a measure of the wave's intensity or strength; larger amplitudes correspond to more energetic waves.

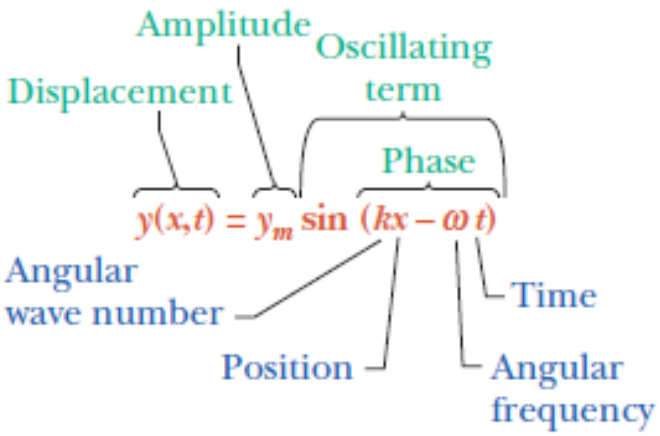


Figure 1.3: Definition of wave quantities for a sinusoidal transverse wave, showing the amplitude y_m, wavelength λ, and the spatial variation of the wave.

Phase

The quantity

$$\phi = kx - \omega t$$

is called the **phase** of the wave. The phase determines the state of oscillation of a particular element of the string at a given position and time. Two points on the string that have the same phase are oscillating in exactly the same way, even if they are located at different positions.

Wavelength

The **wavelength**, denoted by λ, is the distance between two successive points on the wave that are in phase, such as two adjacent crests or two adjacent troughs. It represents the spatial periodicity of the wave.

The wavelength is related to the wave number k by

$$k = \frac{2\pi}{\lambda},$$

where k is called the **angular wave number** and has units of radians per meter.

1.4.3 Period, Frequency, and Angular Frequency

The time-dependent behavior of the wave is governed by the angular frequency ω.

Period

The **period** T of the wave is the time required for a single element of the string to complete one full oscillation. During one period, the phase changes by 2π. Setting

$$\omega T = 2\pi,$$

we obtain

$$T = \frac{2\pi}{\omega}.$$

Frequency

The **frequency** f is the number of complete oscillations per second and is the reciprocal of the period:

$$f = \frac{1}{T} = \frac{\omega}{2\pi}.$$

The SI unit of frequency is the hertz (Hz).

Angular Frequency

The quantity ω is known as the **angular frequency** of the wave. It specifies how rapidly the phase of the wave changes with time and has units of radians per second. While f counts cycles per second, ω measures the rate of phase change in angular terms.

Phase Constant

In general, the wave function may include an additional constant phase term. The most general form of a sinusoidal wave travelling in the positive x–direction is therefore written as

$$y(x,t) = y_m \sin\big(kx - \omega t + \phi_c\big),$$

where ϕ_c is called the **phase constant** (or initial phase).

The phase constant ϕ_c determines the displacement of the wave at the origin ($x = 0$) at time $t = 0$. Different values of ϕ_c correspond to different initial conditions of the wave, such as whether a string element begins its motion at a crest, a trough, or somewhere in between.

Importantly, the phase constant does *not* affect the wavelength, frequency, or speed of the wave. Instead, it specifies the horizontal shift of the wave along the x–axis or, equivalently, the time at which a given phase of the wave passes a particular point.

Exercise Problems

1. A wave travelling along a stretched string is described by

$$y(x,t) = (6.0 \text{ mm}) \sin\big(kx + (600 \text{ rad s}^{-1})t + \phi\big).$$

   Determine the time interval required for a fixed point on the string to move from a transverse displacement of $y = +2.0$ mm to $y = -2.0$ mm.

2. A sinusoidal wave propagates along a string. It is observed that a particular point on the string takes 0.170 s to move from its maximum displacement to the equilibrium position.

   (a) Determine the period of the wave.

   (b) Find the frequency of the wave.

3. A transverse sinusoidal wave travels along a string in the positive x–direction with a speed of 80 m s^{-1}. At time $t = 0$, the particle of the string located at $x = 0$ has a transverse

displacement of 4.0 cm from its equilibrium position and is momentarily at rest. The maximum transverse speed of this particle is 16 m s^{-1}.

(a) Determine the frequency of the wave.

(b) Calculate the wavelength of the wave.

(c) If the wave is described by the equation

$$y(x,t) = y_m \sin\big(kx \pm \omega t + \phi\big),$$

find:

i. the amplitude y_m,

ii. the wave number k,

iii. the angular frequency ω,

iv. the phase constant ϕ,

v. and state the correct choice of sign in front of ω.

Solutions

Problem 1

Problem. A wave travelling along a stretched string is described by

$$y(x,t) = (6.0 \text{ mm}) \sin\big(kx + (600 \text{ rad s}^{-1})t + \phi\big).$$

Determine the time interval required for a fixed point on the string to move from a transverse displacement of $y = +2.0$ mm to $y = -2.0$ mm.

Solution. Let $y_1 = 2.0$ mm (corresponding to time t_1) and $y_2 = -2.0$ mm (corresponding to time t_2).

We find

$$kx + 600t_1 + \phi = \sin^{-1}(2.0/6.0)$$

and

$$kx + 600t_2 + \phi = \sin^{-1}(-2.0/6.0)$$

Subtraction gives

$$600(t_1 - t_2) = \sin^{-1}(2.0/6.0) - \sin^{-1}(-2.0/6.0).$$

And thus $t_1 - t_2 = 0.011$ s.

Problem 2

Problem. A sinusoidal wave propagates along a string. A particular point on the string takes 0.170 s to move from its maximum displacement to the equilibrium position.

1. Determine the period of the wave.
2. Find the frequency of the wave.

Solution. For simple harmonic motion, moving from maximum displacement ($y = +y_m$) to equilibrium ($y = 0$) corresponds to a quarter of a cycle:

$$\Delta t = \frac{T}{4}.$$

Given $\Delta t = 0.170$ s,

$$T = 4(0.170) = 0.680 \text{ s}.$$

The frequency is

$$f = \frac{1}{T} = \frac{1}{0.680} = 1.4706 \text{ Hz} \approx 1.47 \text{ Hz}.$$

$\boxed{T = 0.680 \text{ s}}$ $\boxed{f \approx 1.47 \text{ Hz}}$

Problem 3

A transverse sinusoidal wave travels along a string in the positive x–direction with speed $v = 80$ m s^{-1}. At $t = 0$, the particle at $x = 0$ has displacement $y = 4.0$ cm and is not moving. The maximum transverse speed of that particle is $v_{y,\max} = 16$ m s^{-1}.

1. Find the frequency f.
2. Find the wavelength λ.
3. For $y(x,t) = y_m \sin(kx \pm \omega t + \phi)$ determine y_m, k, ω, ϕ, and the correct sign.

Use maximum transverse speed to find ω

At a fixed position x, the motion is SHM:

$$y(t) = y_m \sin(\pm\omega t + \phi) \quad \text{(at fixed } x\text{)}.$$

The transverse velocity is

$$v_y = \frac{\partial y}{\partial t} = \pm\omega y_m \cos(\pm\omega t + \phi).$$

Thus the *maximum* transverse speed is

$$v_{y,\max} = \omega y_m.$$

We are told that at $t = 0$ and $x = 0$, the particle is at $y = 4.0$ cm and is momentarily at rest. A particle is at rest at maximum displacement, so

$$y_m = 4.0 \text{ cm} = 0.040 \text{ m}.$$

Then

$$\omega = \frac{v_{y,\max}}{y_m} = \frac{16}{0.040} = 400 \text{ rad s}^{-1}.$$

(a) Frequency

$$f = \frac{\omega}{2\pi} = \frac{400}{2\pi} = \frac{200}{\pi} \approx 63.66 \text{ Hz}.$$

$$\boxed{f \approx 63.7 \text{ Hz}}$$

(b) Wavelength

Wave speed relates to wavelength and frequency by $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{80}{63.66} \approx 1.257 \text{ m}.$$

$$\boxed{\lambda \approx 1.26 \text{ m}}$$

(c) Identify y_m, k, ω, ϕ, and the sign

Amplitude.

$$\boxed{y_m = 0.040 \text{ m}}$$

Wave number k.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{1.257} \approx 5.00 \text{ rad m}^{-1}.$$

(Indeed, since $v = \omega/k$, we can also check $k = \omega/v = 400/80 = 5.00$ rad m^{-1}.)

$$\boxed{k = 5.00 \text{ rad m}^{-1}}$$

Angular frequency.

$$\boxed{\omega = 400 \text{ rad s}^{-1}}$$

Phase constant ϕ. Because the wave travels in the *positive x direction*, the standard form is

$$y(x,t) = y_m \sin(kx - \omega t + \phi).$$

Now apply the initial condition at $x = 0, t = 0$:

$$y(0,0) = y_m \sin(\phi) = 0.040 \text{ m}.$$

But $y_m = 0.040$ m, so

$$\sin\phi = 1 \quad \Rightarrow \quad \phi = \frac{\pi}{2} \pmod{2\pi}.$$

Also, the particle is not moving at $t = 0$:

$$v_y(0,0) = \left.\frac{\partial y}{\partial t}\right|_{0,0} = -\omega y_m \cos(\phi) = 0 \quad \Rightarrow \quad \cos\phi = 0,$$

which is consistent with $\phi = \pi/2$ (or $3\pi/2$). Since $y(0,0)$ is positive maximum, choose $\phi = \pi/2$.

$$\boxed{\phi = \frac{\pi}{2}}$$

Correct sign. A wave moving in the $+x$ direction has phase $kx - \omega t + \phi$ (constant phase implies x increases as t increases). Thus the correct sign is the **minus** sign:

$$\boxed{y(x,t) = y_m \sin(kx - \omega t + \phi)}.$$

Final wave function:

$$\boxed{y(x,t) = 0.040 \, \sin\Big(5.00\, x - 400\, t + \frac{\pi}{2}\Big)}$$

(with x in meters, t in seconds, and y in meters).

1.5 The Speed of a Travelling Wave

A *travelling (or progressive) wave* is a disturbance that moves through space, transferring energy from one point to another without transporting matter. To understand how fast such a wave

propagates, we examine how the *waveform itself* moves.

Figure 1.4 illustrates a transverse travelling wave moving along the positive x-direction. As the wave propagates, identifiable points on the *waveform* - for example, a crest labelled point A - move along the string while maintaining the same vertical displacement y. It is important to distinguish between:

- **Points on the medium (string):** individual particles of the string oscillate about their equilibrium positions and do *not* move along the string.
- **Points on the waveform:** features such as crests and troughs move along the string and therefore represent the propagation of the wave.

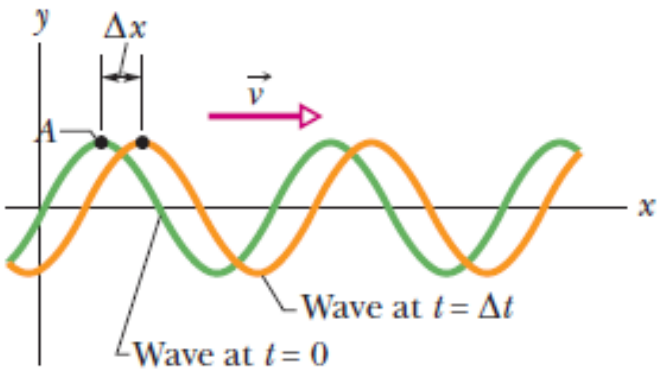


Figure 1.4: A travelling wave moving in the positive x-direction. A point such as A on the waveform (e.g. a crest) moves along the string while retaining the same displacement y.

Phase Condition for a Travelling Wave

A sinusoidal travelling wave moving in the positive x-direction may be written as

$$y(x,t) = y_m \sin(kx - \omega t + \phi),$$

where k is the wave number and ω is the angular frequency.

If a particular point on the waveform (such as point A) retains its displacement as it moves,

then the *phase* associated with that point must remain constant. Thus,

$$kx - \omega t = \text{constant}.$$

Although this quantity is constant, both x and t are changing. As time increases, the position x of that point must also increase in order to keep the phase unchanged. This confirms that the wave pattern is moving in the positive x-direction.

Derivation of the Wave Speed

To find the speed at which the waveform propagates, we differentiate the phase condition with respect to time:

$$\frac{d}{dt}(kx - \omega t) = 0.$$

This gives

$$k\frac{dx}{dt} - \omega = 0,$$

or

$$\frac{dx}{dt} = v = \frac{\omega}{k},$$

where v is the *speed of the travelling wave*.

Using the definitions

$$k = \frac{2\pi}{\lambda}, \qquad \omega = \frac{2\pi}{T} = 2\pi f,$$

we obtain the commonly used expressions for wave speed:

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f.$$

This fundamental relationship applies to all types of periodic waves and shows that, for a given medium, the wave speed is determined by how frequently the wave oscillates and how far the disturbance advances in one cycle.

1.6 Wave Speed on a Stretched String

One of the most important results in the study of mechanical waves is that the speed of a transverse wave on a stretched string is determined entirely by the physical properties of the string and the tension applied to it. In this section, we derive the wave speed using *Newton's Second Law*, closely following the physical argument illustrated in Fig. 1.5.

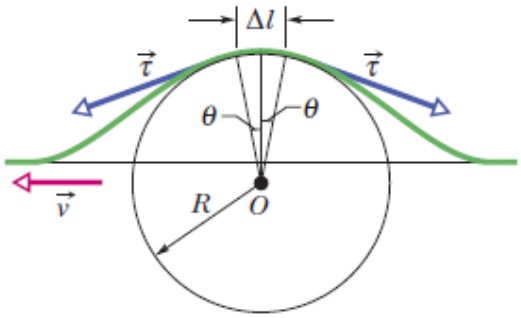


Figure 1.5: A small element of a stretched string forming an arc of a circle as a wave pulse passes. The string tension provides the restoring force responsible for wave propagation.

Physical Setup and Reference Frame

Consider a single, symmetrical transverse pulse traveling from left to right along a stretched string with speed v. To simplify the analysis, we choose a reference frame that moves along with the pulse. In this *co-moving frame*, the pulse appears stationary, while the string itself flows past the pulse from right to left with speed v.

Now focus on a very small element of the string of length Δl located within the curved region of the pulse. At a given instant, this element forms part of an arc of a circle of radius R, subtending an angle 2θ at the center of curvature, as shown in Fig. 1.5.

Forces on the String Element

The string is under uniform tension τ. At each end of the small element Δl, the tension force acts tangentially to the string. Although these two forces are equal in magnitude, they act in slightly different directions.

- The horizontal components of the tension forces cancel by symmetry.
- The vertical components add together, producing a net force directed toward the center of curvature of the arc.

The magnitude of this net restoring (radial) force is

$$F = 2\tau \sin\theta. \tag{1.1}$$

For small angles (which is valid for gentle curvature of the string), we may use the approximation

$$\sin\theta \approx \theta.$$

Thus,

$$F \approx 2\tau\theta. \tag{1.2}$$

From the geometry of the arc,

$$2\theta = \frac{\Delta l}{R}, \tag{1.3}$$

so the restoring force becomes

$$F = \tau \frac{\Delta l}{R}. \tag{1.4}$$

Mass and Acceleration of the Element

If μ denotes the linear mass density of the string (mass per unit length), then the mass of the small element is

$$\Delta m = \mu \Delta l. \tag{1.5}$$

In the co-moving frame, the string element is momentarily moving along a circular arc of radius R with speed v. Therefore, it experiences a centripetal acceleration directed toward the center of curvature:

$$a = \frac{v^2}{R}. \tag{1.6}$$

Application of Newton's Second Law

Newton's Second Law states that

$$\text{force} = \text{mass} \times \text{acceleration}.$$

Applying this law to the string element gives

$$\tau \frac{\Delta l}{R} = (\mu \Delta l) \frac{v^2}{R}. \tag{1.7}$$

The factors Δl and R cancel, leaving

$$\tau = \mu v^2. \tag{1.8}$$

Solving for the wave speed v, we obtain the fundamental result:

$$\boxed{v = \sqrt{\frac{\tau}{\mu}}.} \tag{1.9}$$

This expression shows that:

- The wave speed increases with increasing tension in the string.
- The wave speed decreases as the string becomes heavier (larger linear density).

Importantly, the speed of the wave does *not* depend on the amplitude or shape of the pulse, provided the string remains within the regime of small oscillations. This result applies to any transverse wave traveling along a stretched string and forms a cornerstone of wave mechanics.

Exercise

1. A transverse wave traveling on a stretched string is described by

$$y(x,t) = 0.002 \sin(20x - 600t).$$

The string is under a tension of 15 N.

(a) Determine the wave speed.

(b) Calculate the string's linear mass density, expressing your answer in g/m.

2. A string under tension has a linear mass density of 5.00 g/cm and a tension of 10.0 N. A sinusoidal transverse wave on the string has amplitude 0.12 mm and frequency 100 Hz, and it propagates in the negative x-direction. If the wave can be written in the form

$$y(x,t) = y_m \sin(kx \pm \omega t),$$

find:

(a) y_m,

(b) k,

(c) ω, and

(d) the correct sign in front of ω.

3. A transverse wave travels along a string with speed 170 m/s when the tension is 120 N. What tension is required to increase the wave speed to 180 m/s?

4. A string has linear mass density 1.6×10^{-4} kg/m. A transverse wave on the string is given by

$$y(x,t) = 0.021 \sin(2.0x + 30t).$$

Determine:

(a) the wave speed, and (b) the tension in the string.

Solutions

Problem 1

Given

$$y(x,t) = 0.002\sin(20x - 600t),$$

we identify the wave number and angular frequency as

$$k = 20 \text{ rad/m}, \qquad \omega = 600 \text{ rad/s}.$$

(a) The wave speed is

$$v = \frac{\omega}{k} = \frac{600}{20} = 30 \text{ m/s}.$$

(b) For a wave on a stretched string,

$$v = \sqrt{\frac{\tau}{\mu}} \quad \Longrightarrow \quad \mu = \frac{\tau}{v^2}.$$

With $\tau = 15$ N and $v = 30$ m/s,

$$\mu = \frac{15}{(30)^2} = \frac{15}{900} = 1.6667 \times 10^{-2} \text{ kg/m} = 16.7 \text{ g/m}.$$

Problem 2

Given:

$$\mu = 5.00 \text{ g/cm}, \quad \tau = 10.0 \text{ N}, \quad y_m = 0.12 \text{ mm}, \quad f = 100 \text{ Hz},$$

and the wave travels in the *negative* x-direction.

Convert the linear density:

$$5.00 \text{ g/cm} = 5.00 \times 100 \text{ g/m} = 500 \text{ g/m} = 0.500 \text{ kg/m}.$$

Wave speed on a string:

$$v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{10.0}{0.500}} = \sqrt{20} = 4.47 \text{ m/s}.$$

Wavelength:

$$\lambda = \frac{v}{f} = \frac{4.47}{100} = 4.47 \times 10^{-2} \text{ m}.$$

Now answer the requested quantities:

(a) Amplitude:

$$y_m = 0.12 \text{ mm} = 1.20 \times 10^{-4} \text{ m}.$$

(b) Wave number:

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{4.47 \times 10^{-2}} \approx 1.41 \times 10^{2} \text{ rad/m}.$$

(c) Angular frequency:

$$\omega = 2\pi f = 2\pi(100) = 200\pi \approx 6.28 \times 10^{2} \text{ rad/s}.$$

(d) Direction and sign: for

$$y(x,t) = y_m \sin(kx \pm \omega t),$$

the form $\sin(kx - \omega t)$ represents motion in the $+x$ direction, while $\sin(kx + \omega t)$ represents motion in the $-x$ direction. Since the wave travels in the negative x-direction, choose the *plus* sign:

$$y(x,t) = y_m \sin(kx + \omega t).$$

Problem 3

For waves on a string,

$$v = \sqrt{\frac{\tau}{\mu}} \quad \Rightarrow \quad \tau \propto v^2 \quad (\mu = \text{constant}).$$

Thus,

$$\tau_2 = \tau_1 \left(\frac{v_2}{v_1}\right)^2 = 120 \left(\frac{180}{170}\right)^2.$$

Compute:

$$\left(\frac{180}{170}\right)^2 = \left(\frac{18}{17}\right)^2 = \frac{324}{289} \approx 1.1211,$$

so

$$\tau_2 \approx 120(1.1211) = 134.5 \text{ N} \approx 1.35 \times 10^2 \text{ N}.$$

Problem 4

Given

$$\mu = 1.6 \times 10^{-4} \text{ kg/m}, \qquad y(x,t) = 0.021 \sin(2.0x + 30t),$$

we identify

$$k = 2.0 \text{ rad/m}, \qquad \omega = 30 \text{ rad/s}.$$

(a) Wave speed:

$$v = \frac{\omega}{k} = \frac{30}{2.0} = 15 \text{ m/s}.$$

(b) Tension in the string:

$$v = \sqrt{\frac{\tau}{\mu}} \quad \Longrightarrow \quad \tau = \mu v^2 = (1.6 \times 10^{-4})(15^2).$$

Since $15^2 = 225$,

$$\tau = (1.6 \times 10^{-4})(225) = 3.6 \times 10^{-2} \text{ N} = 0.036 \text{ N}.$$

1.7 The Wave Equation

The motion and propagation of waves in a wide variety of physical systems are governed by a single fundamental differential equation known as the *wave equation*. This equation applies to

many different types of waves, including transverse waves on strings, longitudinal sound waves in air, surface waves on water, and electromagnetic waves in free space.

Without going into its derivation, we state the one-dimensional wave equation in its standard form as

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}. \tag{1.10}$$

In this equation:

- $y(x,t)$ is the *wave function*, representing the displacement of the medium from its equilibrium position.
  - For a stretched string, y is the transverse displacement of a point on the string.
  - For sound waves, y may represent pressure variation or particle displacement.
  - For electromagnetic waves, y represents a component of the electric or magnetic field.
- x is the spatial coordinate along the direction of wave propagation.
- t is time.
- v is the *wave speed*, which depends on the physical properties of the medium (for example, tension and linear density for a string).

The left-hand side of the wave equation,

$$\frac{\partial^2 y}{\partial x^2},$$

measures the *curvature* of the wave profile in space. Large curvature corresponds to regions where the wave is strongly bent.

The right-hand side,

$$\frac{1}{v^2}\frac{\partial^2 y}{\partial t^2},$$

represents how rapidly the displacement changes with time, scaled by the square of the wave speed.

The wave equation therefore states that *spatial curvature of the wave produces temporal acceleration*. This balance between spatial variation and time variation allows the disturbance to propagate through space with a constant speed v.

Any function $y(x,t)$ that satisfies this differential equation represents a physically possible wave motion in the given medium. A particularly important class of solutions is the *traveling waves*, which have the general form

$$y(x,t) = f(x - vt) \quad \text{or} \quad y(x,t) = f(x + vt),$$

representing waves traveling in the positive or negative x-direction, respectively.

Thus, the wave equation provides a unified mathematical description of wave propagation across many branches of physics, making it one of the most fundamental equations in the study of waves.

Exercise

1. Use the one-dimensional wave equation to determine the speed of the wave described by each of the following expressions:

   (a)
   $$y(x,t) = 0.003 \sin(4.00x - 7.00t).$$

   (b)
   $$y(x,t) = 0.002 \sqrt{20x - 4.0t}\,.$$

2. By explicit differentiation, determine whether each of the following functions satisfies the wave equation:

(a)

$$f(x,t) = A\,\exp\big[-b(x-vt)^2\big].$$

(b)

$$f(x,t) = A\,\sin(bx)\,\cos\big[(bvt)^3\big].$$

Solutions

We take the (one-dimensional) wave equation in the form

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2},$$

where v is the wave speed.

Problem 1

(a) $y(x,t) = 0.003\sin(4.00x - 7.00t)$.

This is of the standard traveling-wave form

$$y(x,t) = y_m \sin(kx - \omega t),$$

so we identify

$$k = 4.00 \text{ rad/m}, \qquad \omega = 7.00 \text{ rad/s}.$$

Hence

$$v = \frac{\omega}{k} = \frac{7.00}{4.00} = 1.75 \text{ m/s}.$$

(b) $y(x,t) = 0.002\sqrt{20x - 4.0t}$.

Let

$$u = 20x - 4t, \qquad y(x,t) = 0.002\,u^{1/2}.$$

Compute first derivatives:

$$\frac{\partial y}{\partial x} = 0.002 \cdot \frac{1}{2} u^{-1/2} \cdot \frac{\partial u}{\partial x} = 0.002 \cdot \frac{1}{2} u^{-1/2} \cdot 20 = 0.02\, u^{-1/2},$$

$$\frac{\partial y}{\partial t} = 0.002 \cdot \frac{1}{2} u^{-1/2} \cdot \frac{\partial u}{\partial t} = 0.002 \cdot \frac{1}{2} u^{-1/2} \cdot (-4) = -0.004\, u^{-1/2}.$$

Now the second derivatives:

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x}\left(0.02 u^{-1/2}\right) = 0.02 \left(-\frac{1}{2}\right) u^{-3/2} \frac{\partial u}{\partial x} = 0.02 \left(-\frac{1}{2}\right) u^{-3/2}(20) = -0.2\, u^{-3/2},$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial t}\left(-0.004 u^{-1/2}\right) = -0.004 \left(-\frac{1}{2}\right) u^{-3/2} \frac{\partial u}{\partial t} = -0.004 \left(-\frac{1}{2}\right) u^{-3/2}(-4) = -0.008\, u^{-3/2}.$$

Substitute into the wave equation:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \quad \Longrightarrow \quad -0.2\, u^{-3/2} = \frac{1}{v^2} \left(-0.008\, u^{-3/2}\right).$$

Cancel the common factor $(-u^{-3/2})$ (for $u \neq 0$):

$$0.2 = \frac{0.008}{v^2} \quad \Longrightarrow \quad v^2 = \frac{0.008}{0.2} = 0.04 \quad \Longrightarrow \quad v = 0.20 \text{ m/s}.$$

Thus the function satisfies the wave equation with wave speed $v = 0.20$ m/s (away from the point $u = 0$ where derivatives become singular).

Problem 2

(a) $f(x,t) = A \exp[-b(x - vt)^2]$.

Let

$$s = x - vt, \qquad f = Ae^{-bs^2}.$$

Then

$$\frac{\partial f}{\partial x} = f \cdot (-2bs) \frac{\partial s}{\partial x} = f(-2bs),$$

so

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\Big(f(-2bs)\Big) = (-2b)\Big(s\,\frac{\partial f}{\partial x} + f\,\frac{\partial s}{\partial x}\Big) = (-2b)\Big(s\cdot f(-2bs) + f\cdot 1\Big).$$

Hence

$$\frac{\partial^2 f}{\partial x^2} = (-2b)f\Big(-2bs^2+1\Big) = f\Big(4b^2s^2 - 2b\Big).$$

Next,

$$\frac{\partial f}{\partial t} = f\cdot(-2bs)\frac{\partial s}{\partial t} = f(-2bs)(-v) = f(2bvs),$$

and thus

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\Big(f(2bvs)\Big) = 2bv\Big(s\,\frac{\partial f}{\partial t} + f\,\frac{\partial s}{\partial t}\Big) = 2bv\Big(s\cdot f(2bvs) + f\cdot(-v)\Big).$$

Therefore

$$\frac{\partial^2 f}{\partial t^2} = 2bvf\Big(2bvs^2 - v\Big) = f\Big(4b^2v^2s^2 - 2bv^2\Big) = v^2 f\Big(4b^2s^2 - 2b\Big).$$

Comparing,

$$\frac{\partial^2 f}{\partial t^2} = v^2\frac{\partial^2 f}{\partial x^2} \quad\Longrightarrow\quad \frac{\partial^2 f}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}.$$

So f **satisfies the wave equation** with wave speed v.

(b) $f(x,t) = A\sin(bx)\cos\big((bvt)^3\big)$.

Write

$$f(x,t) = A\sin(bx)\,g(t), \qquad g(t) = \cos\big((bvt)^3\big).$$

Differentiate with respect to x:

$$\frac{\partial f}{\partial x} = Ab\cos(bx)\,g(t), \qquad \frac{\partial^2 f}{\partial x^2} = -Ab^2\sin(bx)\,g(t).$$

Thus

$$\frac{\partial^2 f}{\partial x^2} = -b^2 f \quad (\text{since } f = A\sin(bx)g(t)).$$

Now differentiate with respect to t. Let

$$h(t) = (bvt)^3 = b^3v^3t^3.$$

Then

$$g'(t) = -\sin(h)\, h'(t), \qquad h'(t) = 3b^3v^3t^2,$$

so

$$g'(t) = -3b^3v^3t^2\, \sin(h).$$

Hence

$$\frac{\partial f}{\partial t} = A\sin(bx)\, g'(t) = -3Ab^3v^3t^2\, \sin(bx)\sin(h).$$

Differentiate again:

$$g''(t) = \frac{d}{dt}\Big(-3b^3v^3t^2 \sin(h)\Big) = -3b^3v^3\Big(2t\sin(h) + t^2\cos(h)\, h'(t)\Big).$$

Substitute $h'(t) = 3b^3v^3t^2$:

$$g''(t) = -3b^3v^3\Big(2t\sin(h) + 3b^3v^3t^4\cos(h)\Big).$$

Therefore

$$\frac{\partial^2 f}{\partial t^2} = A\sin(bx)\, g''(t) = -3Ab^3v^3\sin(bx)\Big(2t\sin(h) + 3b^3v^3t^4\cos(h)\Big).$$

If f were to satisfy the wave equation with speed v, we would need

$$\frac{\partial^2 f}{\partial t^2} = v^2\frac{\partial^2 f}{\partial x^2} = v^2\Big(-Ab^2\sin(bx)\cos(h)\Big) = -Ab^2v^2\sin(bx)\cos(h).$$

But our computed $\partial^2 f/\partial t^2$ contains both $\sin(h)$ and $\cos(h)$ terms, and explicit powers of t (namely t and t^4), so it cannot be proportional to $\sin(bx)\cos(h)$ for all t.

Therefore $f(x,t) = A\sin(bx)\cos((bvt)^3)$ **does not satisfy the wave equation**.

1.8 Reflection of Waves

Whenever a traveling wave reaches a boundary separating two different media - or the end of the medium itself - part or all of the wave is reflected. Reflection is a universal wave phenomenon and occurs for mechanical waves (such as waves on a string or sound waves) as well as for electromagnetic waves.

For a transverse wave on a stretched string, the nature of the reflected pulse depends critically on the type of boundary condition at the end of the string. In particular, whether the end is *fixed* or *free* determines whether the reflected pulse is inverted.

1.8.1 Reflection at a Fixed End

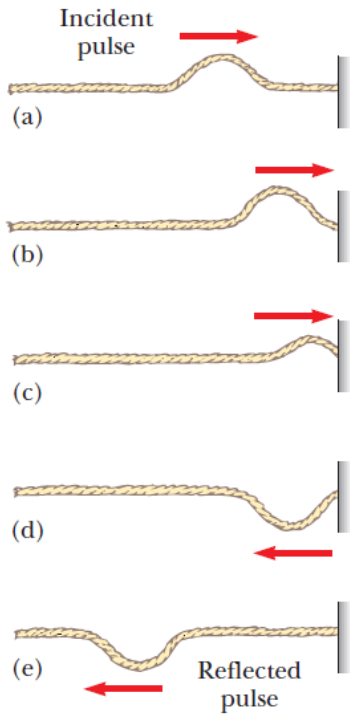


Figure 1.6: Reflection of a transverse pulse at a fixed end. The reflected pulse is inverted (phase change of π).

Consider a transverse pulse traveling along a stretched string toward a rigid wall, as shown in Fig. 1.6. The end of the string is fixed, meaning that its displacement must always remain zero. This condition imposes a strict constraint on the motion of the string at the boundary.

When the incident pulse reaches the wall, the string exerts an upward force on the wall (if the pulse is upward). According to Newton's Third Law, the wall exerts an equal and opposite force on the string. Thus:

- If the incident pulse produces an upward displacement,
- The wall exerts a downward reaction force on the string.

This downward force causes the returning pulse to be inverted. In other words:

- An upward incident pulse reflects as a downward pulse.
- A downward incident pulse reflects as an upward pulse.

Mathematically, this corresponds to a phase change of π (or $180°$) upon reflection. The boundary condition at the fixed end requires

$$y(0, t) = 0,$$

so the incident and reflected waves must superpose in such a way that their sum always vanishes at the wall.

1.8.2 Physical Interpretation

The inversion can be understood in terms of force balance and acceleration. At the fixed boundary:

- The endpoint cannot move.
- Any attempt by the incident pulse to displace the endpoint results in a restoring force from the wall.
- This restoring force reverses the direction of displacement.

Thus, the reflected pulse has opposite sign compared to the incident pulse.

1.8.3 Reflection at a Free End

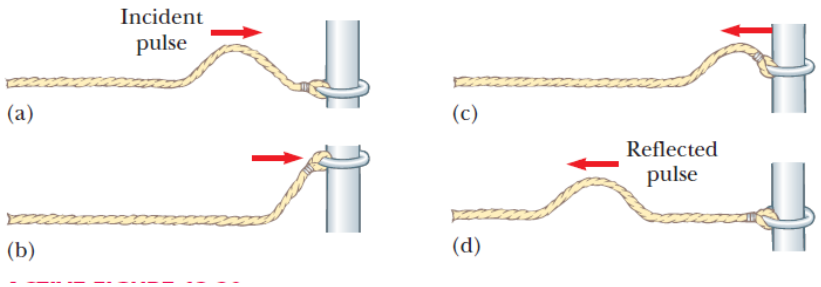


Figure 1.7: Reflection of a transverse pulse at a free end. The reflected pulse is not inverted (no phase change).

Now consider the case where the end of the string is attached to a light ring that can slide freely along a vertical post without friction, as illustrated in Fig. 1.7. In this case, the boundary is *free* rather than fixed.

When the incident pulse reaches the end:

- The pulse exerts a force on the ring.
- The ring accelerates upward (if the pulse is upward).
- The downward component of the string tension then restores the ring to its original position.

Because the ring is free to move, there is no rigid constraint forcing the displacement to vanish. Consequently:

- The reflected pulse is **not inverted**.
- An upward pulse reflects as an upward pulse.
- A downward pulse reflects as a downward pulse.

In this case, there is no phase change upon reflection.

1.8.4 Boundary Conditions and Phase Change

The essential difference between the two cases lies in the boundary condition:

- **Fixed end:** $y = 0$ at the boundary $\Rightarrow$ reflection with phase change π (inversion).
- **Free end:** $\partial y/\partial x = 0$ at the boundary $\Rightarrow$ reflection without inversion.

These boundary conditions are central to the formation of standing waves, which arise from the superposition of incident and reflected waves.

1.8.5 Energy Considerations

In both cases:

- Energy is conserved (assuming no dissipation).
- The wave speed remains unchanged.
- The reflected wave travels back along the string with the same speed as the incident wave.

The only difference is the phase relationship between the incident and reflected pulses.

Boundary Type	**Condition at End**	**Reflected Pulse**
Fixed end	$y = 0$	Inverted (phase shift π)
Free end	$\partial y/\partial x = 0$	Not inverted (no phase shift)

1.9 Interference of Waves

One of the most important and far-reaching consequences of the wave equation is the **principle of superposition**. Because the wave equation is linear, if $y_1(x,t)$ and $y_2(x,t)$ are solutions, then their sum is also a solution. Physically, this means that when two or more waves travel

simultaneously through the same medium, the resultant displacement of the medium at any point is the algebraic sum of the individual displacements.

Suppose that two waves travel simultaneously along the same stretched string. Let $y_1(x,t)$ and $y_2(x,t)$ be the displacements that the string would experience if each wave traveled alone. When the waves overlap, the actual displacement of the string is

$$y'(x,t) = y_1(x,t) + y_2(x,t). \tag{1.11}$$

This summation of displacements means that *overlapping waves algebraically add to produce a resultant (net) wave*. This is another example of the principle of superposition.

1.9.1 Superposition of Pulses

Figure 1.8 shows a sequence of snapshots of two pulses traveling in opposite directions on the same stretched string.

When the pulses overlap, the resultant shape is simply their sum at each point. After the overlap, each pulse continues its motion exactly as if the other had never been present.

Important: Overlapping waves do not in any way alter the subsequent motion of each other (provided the medium behaves linearly). Energy and momentum are temporarily redistributed during overlap, but each pulse emerges unchanged.

1.9.2 Sinusoidal Waves and Interference

Now consider two sinusoidal waves of the same wavelength and amplitude traveling in the same direction along a stretched string. According to the superposition principle, the resultant displacement is again the algebraic sum of the two waves.

Let one wave be

$$y_1(x,t) = y_m \sin(kx - \omega t), \tag{1.12}$$

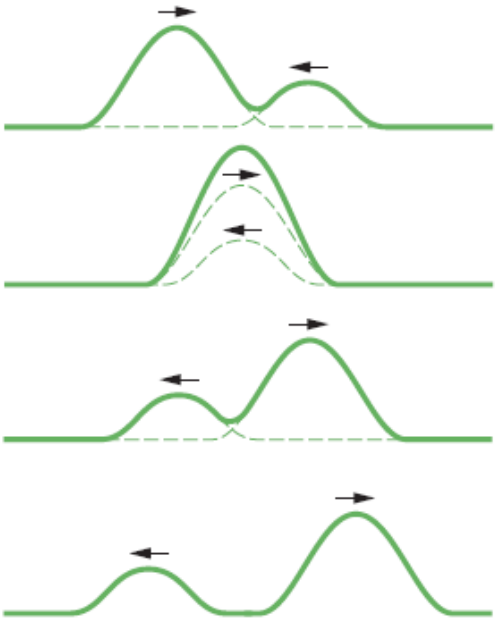


Figure 1.8: A sequence of snapshots showing two pulses traveling in opposite directions along a stretched string. As the pulses overlap, their displacements add algebraically according to the superposition principle. After overlapping, each pulse continues unchanged in its original direction.

and a second wave, shifted by a constant phase angle ϕ, be

$$y_2(x,t) = y_m \sin(kx - \omega t + \phi). \tag{1.13}$$

These waves have:

- the same angular frequency ω (hence the same frequency f),
- the same angular wave number k (hence the same wavelength λ),
- the same amplitude y_m,
- the same wave speed $v = \omega/k$,

and differ only by a constant phase ϕ.

The quantity ϕ is called the **phase difference**. The waves are said to be:

- *in phase* if $\phi = 0$,
- *out of phase* if $\phi = \pi$,
- more generally, phase-shifted by ϕ.

1.9.3 Resultant Wave

By superposition,

$$y'(x,t) = y_1(x,t) + y_2(x,t) \tag{1.14}$$

$$= y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \phi). \tag{1.15}$$

Using the trigonometric identity

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right),$$

we obtain

$$y'(x,t) = 2y_m \cos\left(\frac{\phi}{2}\right) \sin\left(kx - \omega t + \frac{\phi}{2}\right). \tag{1.16}$$

Thus, the resultant wave is also sinusoidal and travels in the positive x-direction.

It differs from the original waves in two important respects:

1. Its phase constant is $\phi/2$.
2. Its amplitude is
$$y'_m = 2y_m \left|\cos\left(\frac{\phi}{2}\right)\right|. \tag{1.17}$$

1.9.4 Constructive and Destructive Interference

The amplitude of the resultant wave depends entirely on the phase difference ϕ.

Fully Constructive Interference

If $\phi = 0$, the waves are exactly in phase. Then

$$y'(x,t) = 2y_m \sin(kx - \omega t). \tag{1.18}$$

The amplitude is doubled. This is called **fully constructive interference**. The two waves reinforce each other everywhere.

Figure 1.9 illustrates this situation.

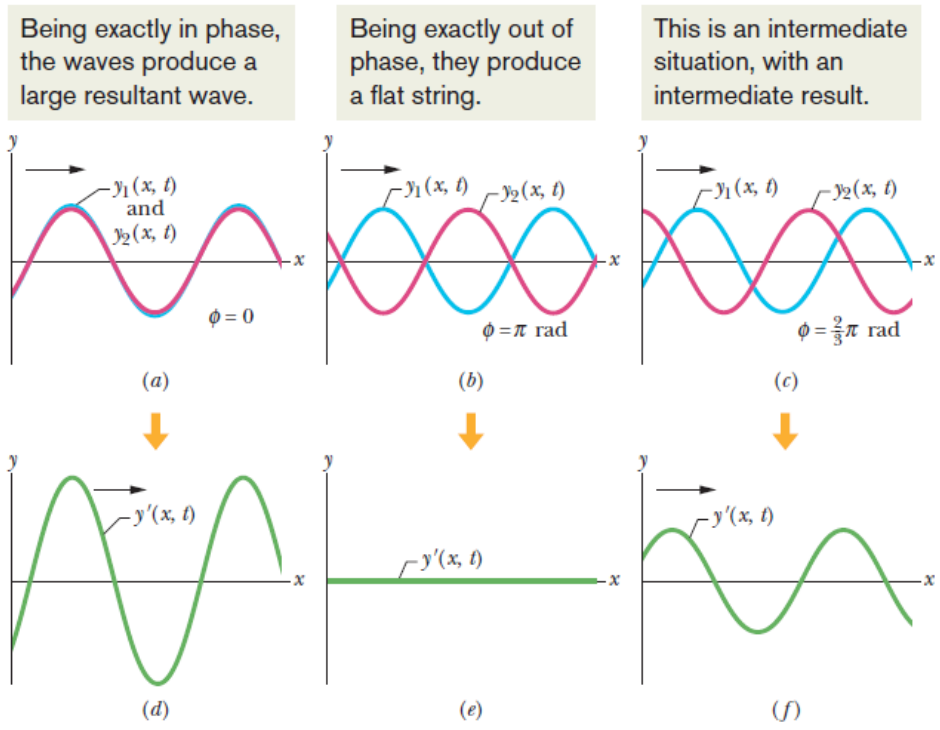


Figure 1.9: Fully constructive interference ($\phi = 0$). The two waves are in phase; crests align with crests and troughs with troughs, producing a resultant wave of double amplitude.

Fully Destructive Interference

If $\phi = \pi$ (180°), the waves are exactly out of phase. Then

$$\cos\left(\frac{\pi}{2}\right) = 0,$$

so the amplitude becomes zero:

$$y'(x,t) = 0. \tag{1.19}$$

The string remains straight at all times. This is called **fully destructive interference**.

Figure 1.10 shows how different phase differences produce different resultant amplitudes.

Phase Difference, in			Amplitude of Resultant Wave	Type of Interference
Degrees	Radians	Wavelengths		
0	0	0	$2y_m$	Fully constructive
120	$\frac{2}{3}\pi$	0.33	y_m	Intermediate
180	π	0.50	0	Fully destructive
240	$\frac{4}{3}\pi$	0.67	y_m	Intermediate
360	2π	1.00	$2y_m$	Fully constructive
865	15.1	2.40	$0.60y_m$	Intermediate

Figure 1.10: Effect of phase difference on interference. Depending on the phase shift ϕ, the resultant amplitude varies between maximum (constructive interference) and zero (destructive interference).

Interference is a universal wave phenomenon and will reappear later in this course in the study of sound, optics (Young's double-slit experiment), and electromagnetic waves.

Exercises

1. Two identical sinusoidal traveling waves move along the same stretched string in the *same direction*. Each wave has amplitude y_m, and the phase difference between them is

$\frac{\pi}{2}$ radians.

(a) Write expressions for the two waves using appropriate phase constants.

(b) Using the principle of superposition, determine the expression for the resultant wave.

(c) Express the amplitude of the resultant wave in terms of y_m.

(d) State whether the interference is constructive, destructive, or partially constructive. Justify your answer.

2. Two identical sinusoidal traveling waves of amplitude y_m propagate along a stretched string in the same direction. When they interfere, the amplitude of the combined wave is found to be $1.50\, y_m$.

(a) Starting from the general expression for the resultant amplitude of two waves with phase difference ϕ, determine the required phase difference.

(b) Express your answer in:

(i) degrees,

(ii) radians,

(iii) as a fraction of a wavelength.

(c) State whether this phase difference corresponds to constructive, destructive, or partially constructive interference.

Solutions: Superposition and Interference of Traveling Waves

Useful Result (Two-Wave Interference)

Consider two sinusoidal waves of the same frequency and wave number, traveling in the same direction:

$$y_1(x,t) = y_m \sin(kx - \omega t), \qquad y_2(x,t) = y_m \sin(kx - \omega t + \phi).$$

Using the identity

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right),$$

we obtain

$$y(x,t) = y_1 + y_2 = 2y_m\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right).$$

Hence, the **resultant amplitude** is

$$y_R = 2y_m\left|\cos\left(\frac{\phi}{2}\right)\right|.$$

The phase constant of the resultant (relative to y_1) is $\phi/2$.

Problem 1

Statement: Two identical traveling waves of amplitude y_m are out of phase by $\phi = \pi/2$ rad. Find the amplitude of the resultant wave in terms of y_m.

Solution

Let

$$y_1 = y_m\sin(kx - \omega t), \qquad y_2 = y_m\sin\big(kx - \omega t + \tfrac{\pi}{2}\big).$$

Using the general two-wave result,

$$y_R = 2y_m\left|\cos\left(\frac{\phi}{2}\right)\right| = 2y_m\left|\cos\left(\frac{\pi}{4}\right)\right|.$$

Since $\cos(\pi/4) = \frac{\sqrt{2}}{2}$,

$$\boxed{y_R = \sqrt{2}\,y_m}.$$

Because $0 < \phi < \pi$, the interference is **partially constructive**.

Problem 2

Statement: Two identical traveling waves (same direction) combine to give amplitude $y_R = 1.50\, y_m$. Find the phase difference ϕ in (a) degrees, (b) radians, and (c) wavelengths.

Solution

For two identical waves,

$$y_R = 2y_m \left|\cos\left(\frac{\phi}{2}\right)\right|.$$

Given $y_R = 1.50\, y_m$, we have

$$1.50\, y_m = 2y_m \left|\cos\left(\frac{\phi}{2}\right)\right| \quad \Rightarrow \quad \left|\cos\left(\frac{\phi}{2}\right)\right| = 0.75.$$

Taking the principal phase difference with $0 \le \phi \le \pi$ (so $\cos(\phi/2) \ge 0$),

$$\cos\left(\frac{\phi}{2}\right) = 0.75 \quad \Rightarrow \quad \frac{\phi}{2} = \cos^{-1}(0.75).$$

Thus,

$$\boxed{\phi = 2\cos^{-1}(0.75)}$$

Numerically,

$$\cos^{-1}(0.75) \approx 0.7227 \text{ rad} \quad \Rightarrow \quad \boxed{\phi \approx 1.445 \text{ rad}}$$

and in degrees,

$$1.445 \text{ rad} \times \frac{180^\circ}{\pi} \approx 82.8^\circ \quad \Rightarrow \quad \boxed{\phi \approx 82.8^\circ}$$

As a fraction of a wavelength, note that a phase difference of 2π corresponds to one wavelength λ:

$$\frac{\Delta x}{\lambda} = \frac{\phi}{2\pi} \quad \Rightarrow \quad \boxed{\frac{\Delta x}{\lambda} \approx \frac{1.445}{2\pi} \approx 0.230.}$$

So the corresponding separation (or phase lag) is about

$$\boxed{\Delta x \approx 0.230\, \lambda}$$

(with equivalent solutions $\phi = 2\pi - 1.445$ also producing the same amplitude).

The interference is **partially constructive** since $0 < \phi < \pi$.

Chapter 2

Sound

Sound is a mechanical wave produced by vibrating sources and transmitted through a material medium such as air, liquids, or solids. Unlike transverse waves on a string, sound in fluids is primarily a longitudinal wave, consisting of alternating compressions and rarefactions that travel through the medium. The study of sound is an important application of the general theory of waves developed earlier, especially the ideas of wavelength, frequency, wave speed, energy transport, and interference.

2.1 The Speed of Sound

The speed of any mechanical wave depends on two fundamental properties of the medium through which it travels:

1. an **inertial property**, which measures the resistance of the medium to acceleration and allows it to store kinetic energy, and
2. an **elastic property**, which measures how strongly the medium resists deformation and allows it to store potential energy.

This idea is already familiar from transverse waves on a stretched string, for which the wave

speed is

$$v = \sqrt{\frac{\tau}{\mu}}, \tag{2.1}$$

where τ is the tension in the string and μ is the linear mass density. In this expression, the tension τ plays the role of the elastic property, while the linear mass density μ provides the inertial property. Thus, in a very general sense, the speed of a mechanical wave may be written as

$$v = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}. \tag{2.2}$$

For sound waves in air, the situation is similar in spirit, although the wave is **longitudinal** rather than transverse. As a sound wave passes through air, small elements of the air do not move up and down; instead, they oscillate back and forth parallel to the direction of propagation, producing alternating regions of compression and rarefaction.

In this case, the appropriate inertial property is the **mass density** (or volume density) of the air, denoted by ρ. A denser medium has more inertia, and therefore tends to transmit the disturbance more slowly, all else being equal.

The relevant elastic property for a sound wave is not tension, but rather the tendency of the medium to resist compression. When a sound wave moves through air, the air is repeatedly compressed and expanded. The quantity that measures the resistance of a substance to changes in volume under pressure is the **bulk modulus** B, defined by

$$B = -\frac{\Delta p}{\Delta V/V}. \tag{2.3}$$

Here:

- Δp is the change in pressure,
- ΔV is the change in volume, and
- $\Delta V/V$ is the fractional change in volume.

The minus sign is included because an increase in pressure produces a decrease in volume,

so Δp and ΔV have opposite signs. With this convention, the bulk modulus is always positive.

A large value of B means that the medium is difficult to compress. Such a medium offers a strong restoring force when disturbed, and this tends to increase the wave speed. A small value of B means the medium is more easily compressed, so the restoring effect is weaker and the wave travels more slowly.

By analogy with the wave speed on a stretched string, we replace the string tension τ by the bulk modulus B, and the linear density μ by the volume density ρ. Hence the speed of sound in a fluid is

$$v = \sqrt{\frac{B}{\rho}}. \tag{2.4}$$

This is the general expression for the speed of longitudinal sound waves in a fluid medium such as a gas or a liquid. The equation shows clearly how the speed of sound depends on the properties of the medium:

- If the bulk modulus B increases, the medium is stiffer and resists compression more strongly, so the speed of sound increases.
- If the density ρ increases, the medium has greater inertia, so the speed of sound decreases.

Thus, sound travels fastest in media that are highly resistant to compression but have relatively small inertia. This helps explain why sound generally travels faster in solids than in liquids, and faster in liquids than in gases.

Speed of Sound in Air

For air at ordinary atmospheric conditions, the speed of sound is approximately

$$v \approx 343 \text{ m s}^{-1} \tag{2.5}$$

at a temperature of 20°C.

The speed of sound in air is not strictly constant; it depends mainly on temperature, and to a lesser extent on composition and humidity. As the temperature increases, air molecules move more rapidly, and the disturbance is transmitted more quickly. For this reason, sound travels faster in warm air than in cold air.

A useful approximate relation for dry air is

$$v \approx 331 + 0.60\,T, \tag{2.6}$$

where T is the temperature in degrees Celsius and v is in $\mathrm{m\,s^{-1}}$.

2.2 Travelling Sound Waves

We now examine the displacements and pressure variations associated with a sinusoidal sound wave travelling through air. Unlike a transverse wave on a string, a sound wave in air is *longitudinal*: the particles of the medium oscillate back and forth parallel to the direction in which the disturbance propagates.

Figure 2.1 shows a sinusoidal sound wave travelling to the right through a long air-filled tube with speed v. The wave consists of a moving, periodic pattern of *compressions* and *expansions* (or rarefactions) of the air. A compression is a region where the air particles are crowded closer together than normal, while an expansion is a region where they are farther apart.

A thin element of air of thickness Δx is shown in the expanded view of Fig. 2.1(b). As the wave passes, this air element does not travel along the tube with the wave. Instead, it oscillates left and right in simple harmonic motion about its equilibrium position. At the instant shown, the element is displaced a distance s to the right of equilibrium. Its greatest displacement on either side is the displacement amplitude s_m.

Thus, the oscillation of each air element in a travelling sound wave is analogous to the oscillation of an element of a stretched string in a transverse wave. The important difference is that, for sound, the oscillation is *longitudinal* rather than transverse. Since the motion of

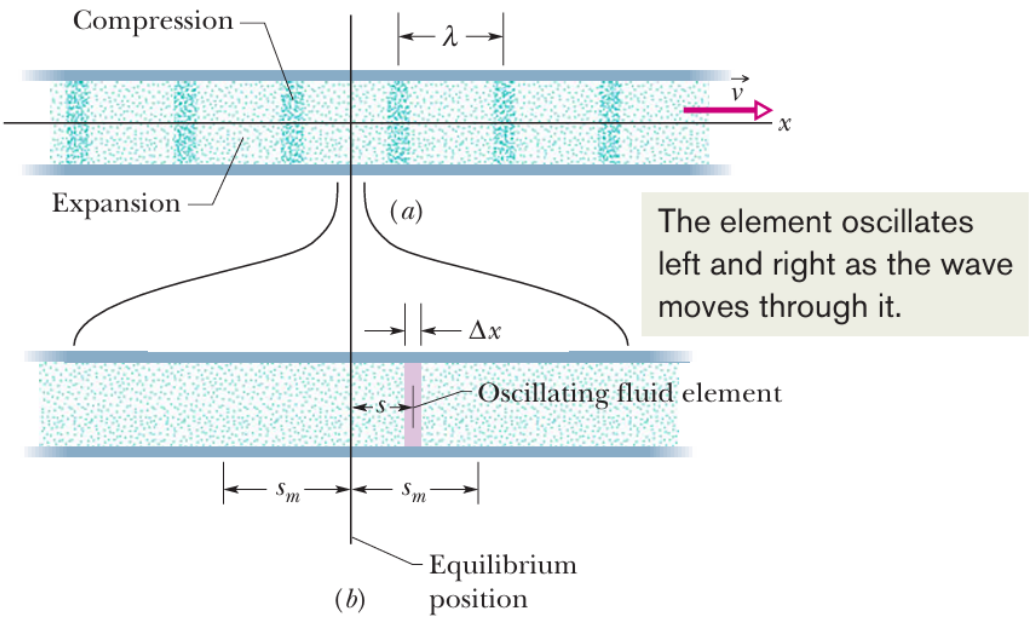


Figure 2.1: A travelling sound wave in an air-filled tube. (a) The sound wave consists of alternating compressions and expansions moving rightward with speed v. (b) A small air element of thickness Δx oscillates left and right about its equilibrium position as the wave passes.

the air elements is along the x-axis, we describe the longitudinal displacement by the function $s(x,t)$.

2.2.1 Displacement Function

For a sinusoidal sound wave travelling in the positive x-direction, the displacement of an air element can be written as

$$s(x,t) = s_m \cos(kx - \omega t), \tag{2.7}$$

where

- $s(x,t)$ is the instantaneous displacement of the air element at position x and time t,
- s_m is the displacement amplitude,
- k is the angular wave number,
- ω is the angular frequency.

Figure 2.2 identifies the parts of this equation.

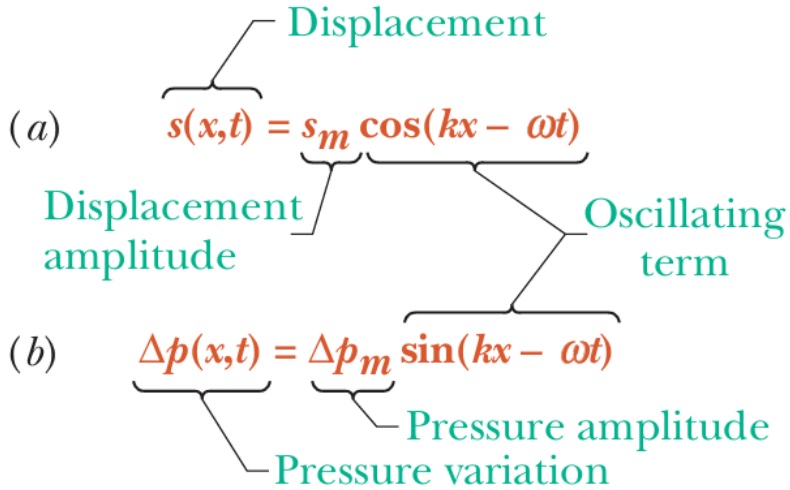


Figure 2.2: The displacement and pressure equations for a travelling sound wave.

The quantities k, ω, f, λ, T, and v are related exactly as they are for transverse waves:

$$k = \frac{2\pi}{\lambda}, \qquad \omega = 2\pi f, \qquad T = \frac{1}{f}, \qquad v = \frac{\omega}{k} = f\lambda. \tag{2.8}$$

Here, however, the wavelength λ has a slightly different physical interpretation than in the transverse case. For a sound wave, λ is the distance along the direction of propagation over which the pattern of compressions and expansions repeats itself.

2.2.2 Pressure Variation

As the sound wave moves through the tube, it produces periodic changes in the pressure of the air at each position x. The pressure variation from the equilibrium pressure is denoted by $\Delta p(x,t)$, and for a sinusoidal travelling sound wave we write

$$\Delta p(x,t) = \Delta p_m \sin(kx - \omega t), \tag{2.9}$$

where Δp_m is the pressure amplitude.

A positive value of Δp corresponds to a compression of the air, while a negative value corresponds to an expansion. Thus, Δp_m represents the maximum increase or decrease in pressure produced by the wave.

The pressure amplitude is related to the displacement amplitude by

$$\Delta p_m = v\rho\omega s_m, \tag{2.10}$$

where ρ is the density of the medium and v is the speed of sound in that medium.

This equation shows that the pressure variation produced by a sound wave becomes larger when

- the medium is denser,
- the wave frequency is higher,

- the wave speed is larger, or
- the displacement amplitude is larger.

2.2.3 Phase Relationship Between Displacement and Pressure

Figure 2.3 shows the displacement and pressure variation as functions of position at the instant $t = 0$. As time increases, both curves move rightward along the x-axis.

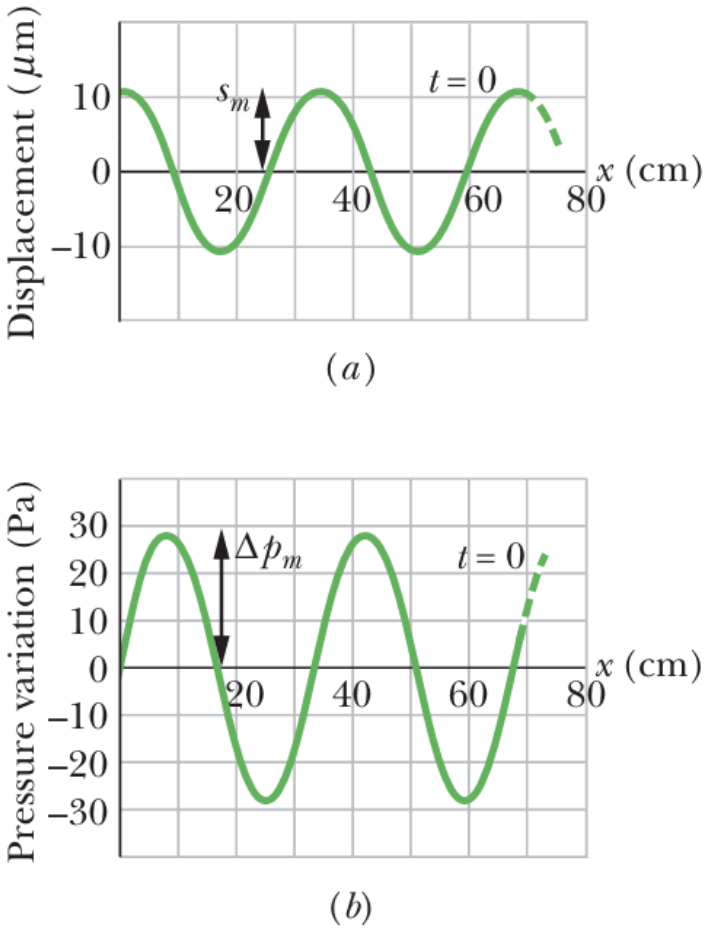


Figure 2.3: Displacement and pressure variation for a travelling sound wave at $t = 0$. The two quantities are out of phase by $\pi/2$ rad.

An important feature of travelling sound waves is that the displacement s and the pressure variation Δp are out of phase by

$$\frac{\pi}{2} \text{ rad} \quad (90^\circ). \tag{2.11}$$

This phase difference follows directly from the two equations:

$$s(x,t) = s_m \cos(kx - \omega t), \tag{2.12}$$

$$\Delta p(x,t) = \Delta p_m \sin(kx - \omega t). \tag{2.13}$$

Since sine and cosine differ in phase by $\pi/2$, the pressure variation reaches zero whenever the displacement is at a maximum or minimum. Likewise, the pressure variation is greatest when the displacement passes through zero.

Physically, this makes sense. At points where an air element is instantaneously farthest from equilibrium, it is momentarily at rest before reversing direction, and the local pressure variation is zero. On the other hand, when the element passes through equilibrium, the surrounding air is being most strongly compressed or expanded, so the pressure variation is largest in magnitude.

A travelling sound wave therefore involves two related oscillations:

1. the **longitudinal displacement** of air elements, and
2. the **pressure variation** within the medium.

These two oscillations travel together through the medium as a wave pattern moving with speed v. Although the pattern moves forward, the individual air elements merely vibrate about their equilibrium positions; they do not move along with the wave over large distances.

2.3 Interference of Sound Waves

Like transverse waves, sound waves can undergo **interference**. When two or more sound waves overlap in the same region of space, the displacement of the air particles at any point is obtained by the principle of superposition: the resultant displacement is the algebraic sum of the individual displacements.

Consider two sinusoidal sound waves of the same amplitude and wavelength, both travelling

in the positive x-direction, but differing in phase by an amount ϕ:

$$s_1(x,t) = s_m \cos(kx - \omega t), \tag{2.14}$$

and

$$s_2(x,t) = s_m \cos(kx - \omega t + \phi). \tag{2.15}$$

Here, s_m is the displacement amplitude, k is the angular wave number, and ω is the angular frequency.

When these two waves overlap, the resultant displacement is

$$s(x,t) = s_1(x,t) + s_2(x,t). \tag{2.16}$$

We thus obtain

$$s(x,t) = \left[2s_m \cos\left(\frac{\phi}{2}\right)\right] \cos\left(kx - \omega t + \frac{\phi}{2}\right). \tag{2.17}$$

Just as for transverse waves, the resultant of two sound waves of the same frequency is itself a travelling wave. Its amplitude is

$$s'_m = \left|2s_m \cos\left(\frac{\phi}{2}\right)\right|. \tag{2.18}$$

The magnitude of this amplitude depends on the phase difference ϕ. Therefore, the value of ϕ determines the type of interference that occurs.

2.3.1 Constructive and Destructive Interference

The interference is **constructive** when the two waves arrive in phase, so that their displacements reinforce one another. This happens when

$$\phi = 0,\ 2\pi,\ 4\pi,\ \ldots \tag{2.19}$$

or, more generally,

$$\phi = m(2\pi), \qquad m = 0, 1, 2, \ldots \tag{2.20}$$

In this case,

$$\cos\left(\frac{\phi}{2}\right) = \pm 1,$$

and the resultant amplitude is maximum:

$$s'_m = 2s_m. \tag{2.21}$$

The interference is **destructive** when the waves arrive exactly out of phase, so that one tends to cancel the other. This occurs when

$$\phi = \pi,\ 3\pi,\ 5\pi,\ \ldots \tag{2.22}$$

or

$$\phi = (2m + 1)\pi, \qquad m = 0, 1, 2, \ldots \tag{2.23}$$

In this case,

$$\cos\left(\frac{\phi}{2}\right) = 0,$$

and the resultant amplitude is zero:

$$s'_m = 0. \tag{2.24}$$

2.3.2 Phase Difference Due to Path Difference

One practical way of controlling the phase difference between two sound waves is to make them travel different distances before they meet. Figure 2.4 illustrates this idea.

Two point sources S_1 and S_2 emit sound waves that are **in phase** and have the same wavelength λ. Since the sources are in phase, the waves leave the two sources with identical displacements at corresponding instants.

We now consider the sound waves arriving at point P. We assume that the distance from

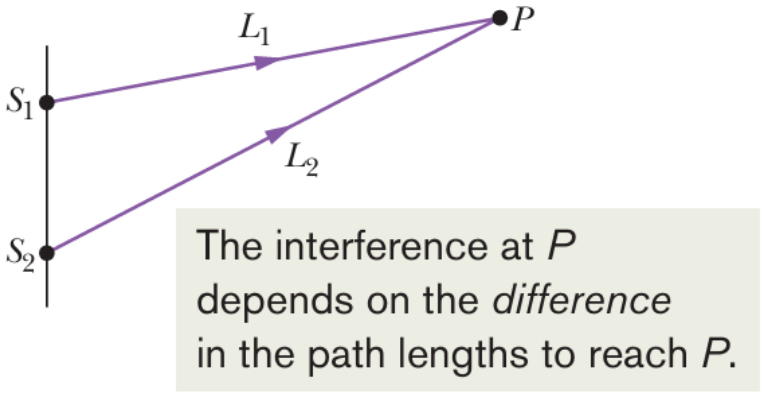


Figure 2.4: Two in-phase sound sources S_1 and S_2 send waves to point P along paths of lengths L_1 and L_2. The interference at P depends on the path length difference $\Delta L = |L_2 - L_1|$.

the sources to P is much greater than the separation of the sources, so that near P the two waves may be treated as travelling in nearly the same direction.

If both waves travelled equal distances to reach P, they would remain in phase there and would undergo fully constructive interference. However, in general, the path lengths are different. If the wave from S_1 travels a distance L_1 and the wave from S_2 travels a distance L_2, then the path difference is

$$\Delta L = |L_2 - L_1|. \tag{2.25}$$

Because one wave travels farther than the other, the two waves may not arrive at P in phase. Their phase difference ϕ at P depends on the path difference ΔL.

To relate ϕ to ΔL, recall that a phase difference of 2π radians corresponds to one full wavelength λ. Therefore,

$$\frac{\phi}{2\pi} = \frac{\Delta L}{\lambda}. \tag{2.26}$$

Hence,

$$\phi = \frac{2\pi \Delta L}{\lambda}. \tag{2.27}$$

2.3.3 Conditions for Interference in Terms of Path Difference

Using

$$\phi = \frac{2\pi\Delta L}{\lambda},$$

we can express the conditions for constructive and destructive interference directly in terms of the path difference.

Fully Constructive Interference

Fully constructive interference occurs when

$$\phi = m(2\pi), \qquad m = 0, 1, 2, \ldots \tag{2.28}$$

Substituting into the phase-path relation gives

$$\frac{2\pi\Delta L}{\lambda} = m(2\pi), \tag{2.29}$$

so that

$$\Delta L = m\lambda, \qquad m = 0, 1, 2, \ldots \tag{2.30}$$

Thus, fully constructive interference occurs when the path difference is an integer multiple of the wavelength:

$$\frac{\Delta L}{\lambda} = 0, 1, 2, \ldots \tag{2.31}$$

Fully Destructive Interference

Fully destructive interference occurs when

$$\phi = (2m + 1)\pi, \qquad m = 0, 1, 2, \ldots \tag{2.32}$$

Substituting into the phase-path relation gives

$$\frac{2\pi\Delta L}{\lambda} = (2m+1)\pi, \tag{2.33}$$

which simplifies to

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, 1, 2, \ldots \tag{2.34}$$

Thus, fully destructive interference occurs when the path difference is an odd half-integer multiple of the wavelength:

$$\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots \tag{2.35}$$

These results show that interference of sound waves depends not only on the properties of the sources, but also on the distances travelled by the waves before meeting at a point. At some positions, the waves reinforce each other and the sound is louder. At other positions, they cancel partially or completely, making the sound weak or even inaudible.

This phenomenon explains why, in places where two sound sources operate simultaneously, a listener may notice alternating regions of loud and soft sound. The pattern arises because the path difference from the sources to different points in space changes continuously, and so does the phase difference.

2.4 Intensity and Sound Level

When a sound wave travels through a medium, it carries energy. As the wave passes through a surface, this energy is transferred through that surface. The **intensity** of a sound wave is defined as the average rate at which energy is transferred per unit area. Thus,

$$I = \frac{P}{A}, \tag{2.36}$$

where P is the power of the sound wave, that is, the rate of energy transfer, and A is the area through which the sound passes.

The SI unit of intensity is

$$\mathrm{W/m^2}.$$

For a sinusoidal sound wave, the intensity is related to the displacement amplitude s_m by

$$I = \frac{1}{2}\rho v \omega^2 s_m^2, \tag{2.37}$$

where

- ρ is the density of the medium,
- v is the speed of sound in the medium,
- ω is the angular frequency of the wave, and
- s_m is the displacement amplitude.

This expression shows that the intensity of a sound wave is proportional to the square of the displacement amplitude. Therefore, if the displacement amplitude doubles, the intensity increases by a factor of four.

It is important to distinguish between **intensity** and **loudness**. Intensity is a physical quantity that can be measured objectively by an instrument or detector. Loudness, on the other hand, is a subjective sensation. It depends not only on the intensity of the sound, but also on the sensitivity of the human ear, which varies with frequency. Thus, two sounds with the same intensity may not necessarily seem equally loud to a listener.

2.4.1 Variation of Intensity with Distance

In general, the way sound intensity varies with distance from a real sound source can be quite complicated. Many real sources, such as loudspeakers, do not emit sound equally in all directions. In addition, sound waves often reflect from walls, ceilings, and other objects, producing echoes that overlap with the direct sound.

However, in some idealized situations, we can neglect reflections and model the source as a **point source** that emits sound **isotropically**, that is, equally in all directions. Figure 2.5 illustrates this situation.

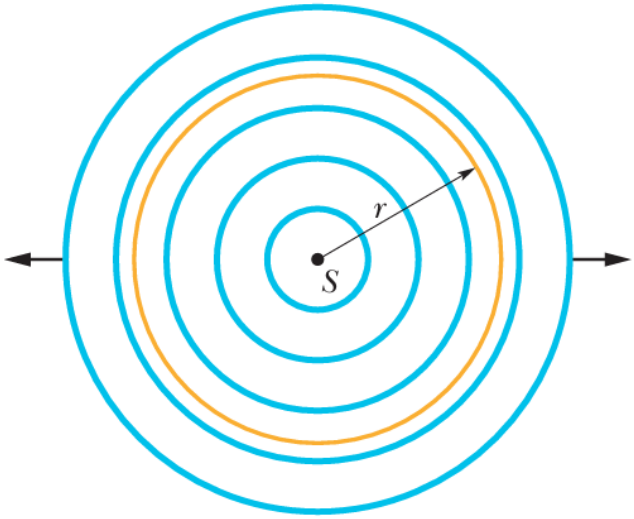


Figure 2.5: Wavefronts spreading outward from an isotropic point source S. As the distance r from the source increases, the same sound power is distributed over a larger spherical surface area.

At any instant, the sound wavefronts form concentric spheres centered on the source. Suppose we imagine a sphere of radius r centered at the source. If mechanical energy is conserved as the sound spreads outward, then all the sound energy emitted by the source must pass through the surface of that sphere.

If the source emits sound with power P_s, then the intensity at distance r is the power divided by the area of the sphere:

$$I = \frac{P_s}{4\pi r^2}. \tag{2.38}$$

Since the surface area of a sphere is $4\pi r^2$, this result shows that the intensity decreases with the square of the distance from the source. This is known as the **inverse-square law** for sound intensity:

$$I \propto \frac{1}{r^2}. \tag{2.39}$$

Thus, if the distance from the source is doubled, the intensity becomes one-fourth of its original value. If the distance is tripled, the intensity becomes one-ninth of its original value.

2.4.2 The Decibel Scale

The human ear can detect sounds over an extremely wide range of intensities. The displacement amplitude at the ear ranges approximately from

$$10^{-11}\,\mathrm{m}$$

for the faintest detectable sound to about

$$10^{-5}\,\mathrm{m}$$

for the loudest tolerable sound. This corresponds to an amplitude ratio of

$$\frac{10^{-5}}{10^{-11}} = 10^{6}.$$

Because intensity is proportional to the square of the displacement amplitude, the corresponding ratio of intensities is

$$(10^{6})^{2} = 10^{12}.$$

This means that the range of audible intensities spans a factor of one trillion. Because this range is so enormous, it is inconvenient to describe sound using intensity values directly. Instead, we use a logarithmic scale called the **decibel scale**.

The **sound level** β of a sound is defined by

$$\beta = (10\,\mathrm{dB}) \log_{10}\left(\frac{I}{I_0}\right), \tag{2.40}$$

where

- β is the sound level in decibels (dB),
- I is the intensity of the sound, and
- I_0 is a standard reference intensity.

The standard reference intensity is taken to be

$$I_0 = 10^{-12}\,\mathrm{W/m^2}, \tag{2.41}$$

which is close to the threshold of human hearing for sounds of moderate frequency.

When $I = I_0$, the sound level is

$$\beta = 10\log_{10}(1) = 0\,\mathrm{dB}.$$

Thus, the reference intensity corresponds to a sound level of 0 decibels.

Because the decibel scale is logarithmic, an increase of 10 dB corresponds to an increase in intensity by a factor of 10. Similarly:

- an increase of 20 dB corresponds to an intensity increase by a factor of $10^2 = 100$,
- an increase of 30 dB corresponds to an intensity increase by a factor of $10^3 = 1000$,
- an increase of 40 dB corresponds to an intensity increase by a factor of 10^4.

For example, if a sound has level

$$\beta = 40\,\mathrm{dB},$$

then

$$\frac{I}{I_0} = 10^4,$$

so its intensity is 10^4 times the reference intensity.

2.4.3 Examples of Sound Levels

A few typical sound levels are:

- 0 dB: threshold of hearing,
- 30 dB: whisper,
- 60 dB: normal conversation,
- 90 dB: heavy traffic,
- 120 dB: threshold of pain.

These values are approximate and may vary depending on frequency and listening conditions.

Exercises

1. When the door of a chapel is slammed shut, a person standing just inside the doorway hears the final echo 15 s later.

   (a) Assuming this last echo is caused by a single reflection from the wall directly opposite the door, determine the distance between the door and that wall.

   (b) If the opposite wall is actually 25.7 m from the door, calculate how many back-and-forth reflections occur before the final echo is heard.

2. Earthquakes generate sound waves that travel through the Earth. Since the Earth is not a gas, it can support both transverse waves (S-waves) and longitudinal waves (P-waves). Typically, S-waves travel at about 4.5 km/s, while P-waves travel at about 8.0 km/s. A seismograph records the arrival of the P-waves 3.0 min before the S-waves. Assuming both waves travel along straight-line paths, determine the distance from the seismograph to the earthquake.

3. A man strikes one end of a slender rod with a hammer. The speed of sound in the rod is 15 times greater than the speed of sound in air. At the opposite end, a woman with her ear near the rod hears two sounds separated by 0.12 s: one that travels through the rod and another that travels through the surrounding air. If the speed of sound in air is 343 m/s, find the length of the rod.

4. A stone is dropped from rest into a well, and the sound of the splash is heard 3.00 s later. Determine the depth of the well.

5. A sound wave travelling through air is described by

$$s(x,t) = (6.0\,\text{nm})\cos\big(kx + (3000\,\text{rad/s})t + \phi\big).$$

   For a fixed air molecule along the path of the wave, calculate the time required for it to move from a displacement of $s = 2.0\,\text{nm}$ to $s = -2.0\,\text{nm}$.

6. Ultrasound with frequency 4.50 MHz is used in medical diagnosis to examine tumors in soft tissue.

   (a) Find the wavelength of this sound wave in air.

   (b) If the speed of sound in soft tissue is 1500 m/s, determine its wavelength in the tissue.

7. The pressure variation in a travelling sound wave is given by

$$\Delta p = (1.50\,\text{Pa})\sin[\pi(0.900x - 315t)]\,.$$

   Determine:

   (a) the pressure amplitude,

   (b) the frequency,

   (c) the wavelength,

   (d) the wave speed.

8. Two sound waves produced by different sources have the same frequency, 540 Hz, and travel in the same direction with speed 330 m/s. The sources are in phase. Find the phase difference between the waves at a point that is 4.40 m from one source and 4.00 m from the other.

9. The sound level of a conversation decreases from a loud 70 dB to a softer 50 dB. Assume the sound frequency is 500 Hz. Determine:

   (a) the initial sound intensity,

   (b) the final sound intensity,

   (c) the initial amplitude of the sound wave,

   (d) the final amplitude of the sound wave.

10. A sound wave with frequency 300 Hz has an intensity of $1.00\,\mu\text{W/m}^2$. Calculate the amplitude of the oscillatory motion of the air particles caused by the wave.

11. The sound level produced by a certain source is increased by 30.0 dB. By what factor does this increase:

    (a) the intensity of the sound,

    (b) the pressure amplitude of the sound wave?

12. A sound source emits waves with power $1.00\,\mu$W. Treating it as an isotropic point source:

    (a) find the intensity at a distance of 3.00 m,

    (b) determine the corresponding sound level in decibels at that distance.

Solutions

1. **Echo**

   (a) If the final echo is due to one reflection from the opposite wall, then the sound travels to the wall and back. Thus,

   $$2d = vt,$$

   where $v = 343$ m/s and $t = 15$ s. Hence,

   $$d = \frac{vt}{2} = \frac{(343)(15)}{2} = 2572.5 \text{ m}.$$

   Therefore,

   $$\boxed{d \approx 2.57 \times 10^3 \text{ m}}.$$

   (b) If the wall is actually 25.7 m away, then one round trip has length

   $$2(25.7) = 51.4 \text{ m}.$$

   The time for one back-and-forth reflection is

   $$t_1 = \frac{51.4}{343} = 0.1499 \text{ s}.$$

   The number of such round trips in 15 s is

   $$N = \frac{15}{0.1499} \approx 100.$$

   Thus, the final echo is heard after approximately

   $$\boxed{\text{100 back-and-forth reflections}}.$$

2. **Earthquake waves**

Let the distance to the earthquake be d. Then

$$t_P = \frac{d}{8.0}, \qquad t_S = \frac{d}{4.5},$$

with speeds in km/s. Since the P-wave arrives 3.0 min = 180 s earlier,

$$t_S - t_P = 180.$$

Therefore,

$$\frac{d}{4.5} - \frac{d}{8.0} = 180.$$

Solving,

$$d\left(\frac{1}{4.5} - \frac{1}{8.0}\right) = 180,$$

$$d(0.2222 - 0.125) = 180,$$

$$d(0.09722) = 180,$$

$$d \approx 1851 \text{ km}.$$

Hence,

$$\boxed{d \approx 1.85 \times 10^3 \text{ km}}.$$

3. **Sound through a rod and air**

Let the rod length be L. The speed of sound in air is

$$v_a = 343 \text{ m/s},$$

and in the rod it is

$$v_r = 15v_a = 15(343) = 5145 \text{ m/s}.$$

The difference in arrival times is

$$\frac{L}{v_a} - \frac{L}{v_r} = 0.12.$$

Thus,

$$L\left(\frac{1}{343}-\frac{1}{5145}\right)=0.12.$$

Since

$$\frac{1}{343}-\frac{1}{5145}=\frac{14}{5145},$$

we obtain

$$L=\frac{0.12\times 5145}{14}=44.1\text{ m}.$$

Therefore,

$$\boxed{L=44.1\text{ m}}.$$

4. **Depth of a well**

Let the depth of the well be h. The total time is the fall time plus the time for sound to travel back up:

$$\sqrt{\frac{2h}{g}}+\frac{h}{343}=3.00.$$

Using $g=9.8\text{ m/s}^2$,

$$\sqrt{\frac{2h}{9.8}}+\frac{h}{343}=3.00.$$

Solving this equation gives

$$h\approx 40.7\text{ m}.$$

Thus,

$$\boxed{h\approx 40.7\text{ m}}.$$

5. **Motion of an air molecule**

The displacement is

$$s(x,t)=(6.0\text{ nm})\cos(kx+3000t+\phi).$$

For a fixed molecule, x is constant, so the motion is simple harmonic with angular frequency

$$\omega=3000\text{ rad/s}.$$

We want the time for the displacement to change from

$$s = 2.0 \text{ nm} \quad \text{to} \quad s = -2.0 \text{ nm}.$$

Since the amplitude is 6.0 nm,

$$\cos\theta_1 = \frac{2.0}{6.0} = \frac{1}{3}, \qquad \cos\theta_2 = \frac{-2.0}{6.0} = -\frac{1}{3}.$$

The smallest phase change is

$$\Delta\theta = \cos^{-1}\left(-\frac{1}{3}\right) - \cos^{-1}\left(\frac{1}{3}\right) \approx 0.6797 \text{ rad}.$$

Hence,

$$\Delta t = \frac{\Delta\theta}{\omega} = \frac{0.6797}{3000} \approx 2.27 \times 10^{-4} \text{ s}.$$

Therefore,

$$\boxed{\Delta t \approx 2.27 \times 10^{-4} \text{ s}}.$$

6. **Ultrasound wavelength**

Given

$$f = 4.50 \text{ MHz} = 4.50 \times 10^6 \text{ Hz}.$$

(a) In air, taking $v = 343$ m/s,

$$\lambda = \frac{v}{f} = \frac{343}{4.50 \times 10^6} = 7.62 \times 10^{-5} \text{ m}.$$

Thus,

$$\boxed{\lambda_{\text{air}} = 7.62 \times 10^{-5} \text{ m}}.$$

(b) In soft tissue, $v = 1500$ m/s, so

$$\lambda = \frac{1500}{4.50 \times 10^6} = 3.33 \times 10^{-4} \text{ m}.$$

Therefore,

$$\boxed{\lambda_{\text{tissue}} = 3.33 \times 10^{-4} \text{ m}}.$$

7. **Pressure variation wave**

The pressure variation is

$$\Delta p = (1.50 \text{ Pa}) \sin[\pi(0.900x - 315t)] .$$

Comparing with

$$\Delta p = \Delta p_m \sin(kx - \omega t),$$

we identify

$$\Delta p_m = 1.50 \text{ Pa}, \qquad k = 0.900\pi \text{ rad/m}, \qquad \omega = 315\pi \text{ rad/s}.$$

(a) Pressure amplitude:

$$\boxed{\Delta p_m = 1.50 \text{ Pa}}.$$

(b) Frequency:

$$f = \frac{\omega}{2\pi} = \frac{315\pi}{2\pi} = 157.5 \text{ Hz}.$$

Hence,

$$\boxed{f = 157.5 \text{ Hz}}.$$

(c) Wavelength:

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.900\pi} = 2.22 \text{ m}.$$

Thus,

$$\boxed{\lambda = 2.22 \text{ m}}.$$

(d) Wave speed:

$$v = \frac{\omega}{k} = \frac{315\pi}{0.900\pi} = 350 \text{ m/s}.$$

Therefore,

$$\boxed{v = 350 \text{ m/s}}.$$

8. **Phase difference from path difference**

The frequency is

$$f = 540 \text{ Hz},$$

and the wave speed is

$$v = 330 \text{ m/s}.$$

Hence the wavelength is

$$\lambda = \frac{v}{f} = \frac{330}{540} = 0.6111 \text{ m}.$$

The path difference is

$$\Delta L = 4.40 - 4.00 = 0.40 \text{ m}.$$

The phase difference is

$$\phi = \frac{2\pi \Delta L}{\lambda} = \frac{2\pi(0.40)}{0.6111} \approx 4.11 \text{ rad}.$$

Thus,

$$\boxed{\phi \approx 4.11 \text{ rad}}.$$

9. **Conversation sound level**

The sound levels are

$$\beta_1 = 70 \text{ dB}, \qquad \beta_2 = 50 \text{ dB}.$$

Using

$$\beta = 10 \log_{10}\left(\frac{I}{I_0}\right), \qquad I_0 = 10^{-12} \text{ W/m}^2,$$

we get

$$I = I_0 \, 10^{\beta/10}.$$

(a) Initial intensity:

$$I_1 = 10^{-12} \times 10^7 = 10^{-5} \text{ W/m}^2.$$

So,

$$\boxed{I_1 = 1.0 \times 10^{-5} \text{ W/m}^2}.$$

(b) Final intensity:

$$I_2 = 10^{-12} \times 10^5 = 10^{-7} \text{ W/m}^2.$$

Thus,

$$\boxed{I_2 = 1.0 \times 10^{-7} \text{ W/m}^2}.$$

(c) Initial amplitude:

Using

$$I = \frac{1}{2}\rho v \omega^2 s_m^2,$$

with $\rho = 1.20$ kg/m^3, $v = 343$ m/s, and

$$\omega = 2\pi f = 2\pi(500) = 1000\pi \text{ rad/s},$$

we solve for s_m:

$$s_m = \sqrt{\frac{2I}{\rho v \omega^2}}.$$

For $I_1 = 10^{-5}$ W/m^2,

$$s_{m1} \approx 7.02 \times 10^{-8} \text{ m}.$$

Therefore,

$$\boxed{s_{m1} \approx 7.0 \times 10^{-8} \text{ m}}.$$

(d) Final amplitude:

For $I_2 = 10^{-7}$ W/m^2,

$$s_{m2} \approx 7.02 \times 10^{-9} \text{ m}.$$

Hence,

$$\boxed{s_{m2} \approx 7.0 \times 10^{-9} \text{ m}}.$$

10. **Amplitude of air-particle motion**

Given

$$f = 300 \text{ Hz}, \qquad I = 1.00 \ \mu\text{W/m}^2 = 1.00 \times 10^{-6} \text{ W/m}^2.$$

Using

$$I = \frac{1}{2}\rho v \omega^2 s_m^2,$$

with $\rho = 1.20 \text{ kg/m}^3$, $v = 343 \text{ m/s}$, and

$$\omega = 2\pi f = 600\pi \text{ rad/s},$$

we obtain

$$s_m = \sqrt{\frac{2I}{\rho v \omega^2}}.$$

Substituting the values,

$$s_m = \sqrt{\frac{2(1.00 \times 10^{-6})}{(1.20)(343)(600\pi)^2}} \approx 1.14 \times 10^{-7} \text{ m}.$$

Therefore,

$$\boxed{s_m \approx 1.14 \times 10^{-7} \text{ m}}.$$

11. **Increase in sound level by** 30.0 dB

Since

$$\Delta\beta = 10 \log_{10}\left(\frac{I_2}{I_1}\right),$$

a 30.0 dB increase gives

$$30.0 = 10 \log_{10}\left(\frac{I_2}{I_1}\right),$$

so

$$\log_{10}\left(\frac{I_2}{I_1}\right) = 3.$$

Hence,

$$\frac{I_2}{I_1} = 10^3 = 1000.$$

(a) The intensity increases by a factor of

$$\boxed{1000}.$$

(b) Since intensity is proportional to the square of the pressure amplitude,

$$I \propto (\Delta p_m)^2,$$

the pressure amplitude increases by

$$\sqrt{1000} = 31.6.$$

Therefore,

$$\boxed{31.6}.$$

12. **Isotropic point source**

Given

$$P_s = 1.00\ \mu\text{W} = 1.00 \times 10^{-6}\ \text{W}, \qquad r = 3.00\ \text{m}.$$

(a) The intensity is

$$I = \frac{P_s}{4\pi r^2} = \frac{1.00 \times 10^{-6}}{4\pi(3.00)^2}.$$

Thus,

$$I = \frac{1.00 \times 10^{-6}}{36\pi} \approx 8.84 \times 10^{-9}\ \text{W/m}^2.$$

Hence,

$$\boxed{I \approx 8.84 \times 10^{-9}\ \text{W/m}^2}.$$

(b) The sound level is

$$\beta = 10 \log_{10}\left(\frac{I}{I_0}\right) = 10 \log_{10}\left(\frac{8.84 \times 10^{-9}}{10^{-12}}\right).$$

Therefore,

$$\beta = 10 \log_{10}(8.84 \times 10^3) \approx 39.5 \text{ dB}.$$

So,

$$\boxed{\beta \approx 39.5 \text{ dB}}.$$

2.4.4 Safe Sound Levels and Hearing Protection

The decibel scale provides a convenient way of describing the strength of sound, but in practical situations it is not enough merely to know the numerical value of the sound level. We must also ask whether the sound is safe for human exposure, how long the exposure lasts, and whether protective measures are necessary.

The human ear is a sensitive detector, capable of responding to sounds over an enormous range of intensities. However, this sensitivity also means that the ear can be damaged by excessive sound. In general, the risk associated with sound depends on two factors:

1. the **sound level**, usually measured in decibels, and
2. the **duration of exposure**.

A sound that is tolerable for a very short time may become harmful if a person is exposed to it for many minutes or hours. This is because prolonged exposure transfers more energy to the ear.

Classification of sound levels

Although exact limits may vary somewhat depending on regulations and conditions, the following ranges are useful as a guide:

- **0 dB:** threshold of hearing,
- **20–30 dB:** very quiet room, whisper,
- **40–60 dB:** ordinary indoor activity, conversation,
- **70–80 dB:** busy street, loud office equipment,
- **85–90 dB:** sustained exposure may become hazardous,
- **100 dB and above:** dangerous for prolonged exposure,
- **120 dB:** threshold of pain,
- **140 dB and above:** immediate risk of acoustic injury.

Thus, sound levels above about 85 dB are generally regarded as requiring caution, especially if exposure continues for long periods. In workshops, factories, mines, airports, construction sites, and music venues, hearing protection may be necessary.

Safe practice and hearing protection

The danger from sound exposure can be reduced by applying simple physical principles:

1. Reduce the sound at the source. Whenever possible, noise should be reduced where it is produced, for example by improving machine design, lubrication, balancing moving parts, or enclosing noisy equipment.

2. Increase the distance from the source. For an approximately isotropic source,

$$I = \frac{P_s}{4\pi r^2},$$

so the intensity decreases with the square of the distance. Doubling the distance reduces the intensity to one quarter of its value.

3. Limit exposure time. Even if the sound cannot be reduced, limiting the time spent in the noisy environment reduces the total acoustic energy reaching the ear.

4. Use hearing protectors. Earplugs and earmuffs reduce the intensity of sound entering the ear and are especially important in industrial environments.

5. Provide barriers and sound-absorbing materials. Walls, enclosures, and absorbent surfaces can reduce the sound reaching workers or the public.

The concept of safe sound level is therefore closely tied to the physics of intensity, the inverse-square law, and the physiological response of the ear.

2.4.5 Physiological Effects of Noise

Sound is detected when pressure variations in air cause the eardrum to vibrate. These vibrations are transmitted through the middle ear to the inner ear, where sensory receptors convert the mechanical disturbance into nerve impulses. When sound levels are moderate, this process allows normal hearing. When sound levels are excessive, however, the same process can stress or damage the auditory system.

The physiological effects of noise may be divided into two broad classes:

1. **auditory effects**, which act directly on hearing, and
2. **non-auditory effects**, which affect comfort, health, and performance.

Auditory effects

Temporary threshold shift. After spending time in a noisy place, a person may notice that sounds seem less clear or less loud than usual. Hearing may appear muffled for a short period. This temporary loss of sensitivity is called a **temporary threshold shift**. It is often a warning that the ear has been exposed to excessive sound.

Permanent hearing loss. If exposure to loud noise is repeated over long periods, the sensitive structures of the inner ear may suffer irreversible damage. The result is **permanent hearing loss**. This often develops gradually and may first become noticeable at higher frequencies.

Tinnitus. A person exposed to excessive noise may hear a ringing, buzzing, or whistling sound in the ears even when no external sound is present. This condition is known as **tinnitus** and is often associated with hearing damage or overstress of the auditory system.

Acoustic trauma. A very intense sound, such as an explosion, gunshot, or sudden industrial impact, may cause immediate injury. Such injury is called **acoustic trauma**. It may damage the eardrum, the middle ear, or the sensory structures of the inner ear.

Non-auditory effects

Noise affects the body and mind even when it is not strong enough to cause direct hearing loss. Important non-auditory effects include:

- annoyance,
- distraction,
- reduced concentration,
- fatigue,
- stress and irritability,
- interference with sleep,
- reduced efficiency in work and learning.

Interference with communication. If background noise is large, speech becomes difficult to understand. In such cases, a speaker must raise their voice, and listeners may miss important information. This is especially important in classrooms, hospitals, workshops, airports, and control rooms.

Effect on learning and performance. In educational settings, excessive noise may reduce concentration and hinder understanding. In workplaces, it may reduce efficiency and increase the probability of mistakes.

Sleep disturbance. Noise at night may interrupt normal sleep patterns even when it does not fully awaken the person. Repeated disturbance of sleep can produce tiredness, irritability, and reduced daytime performance.

Stress response. Persistent noise may trigger physiological stress responses, including increased tension, faster heart rate, and general discomfort. Thus, the effects of noise are not limited to the ear alone.

For these reasons, noise control is important not only in preventing hearing loss but also in maintaining comfort, health, and productivity.

2.4.6 Industrial Noise and Sound as a Diagnostic Tool

In industry, sound has a dual role. On the one hand, industrial noise is often an unwanted by-product of machines and processes, and excessive noise may be a serious occupational hazard. On the other hand, sound can also provide useful information about the condition of equipment. For this reason, sound is not only something to be controlled but also something that can be used diagnostically.

Industrial noise as a hazard

Industrial plants often contain many sources of strong sound, including:

- electric motors,
- turbines,
- pumps and compressors,
- cutting and drilling machines,
- stamping presses,
- conveyor systems,
- ventilation equipment,
- moving vehicles and generators.

These may produce continuous sound, intermittent sound, or sudden impulsive noise. In enclosed spaces, reflections from walls and metal surfaces may make the effective sound level even larger.

The control of industrial noise may be considered at three stages:

1. **Control at the source** – reduce vibration, improve machine balance, lubricate moving parts, redesign noisy systems.
2. **Control along the path** – use enclosures, barriers, silencers, and sound-absorbing materials.
3. **Control at the receiver** – reduce exposure time and use ear protection.

Sound as a diagnostic tool

Every working machine produces a characteristic pattern of sound and vibration. When the machine is operating normally, this pattern tends to remain fairly stable. If a defect develops, the sound may change in level, frequency content, rhythm, or waveform. Careful listening or instrumental analysis can therefore reveal faults at an early stage.

Examples include the following.

Worn bearings. A worn bearing may produce rough, rattling, or periodic clicking sounds.

Misalignment of shafts. If rotating shafts are not properly aligned, abnormal vibration and tonal noise may appear.

Mechanical imbalance. A rotating component with uneven mass distribution produces periodic vibration, often at the rotation frequency.

Leaks in pipes or vessels. Escaping gas, steam, or compressed air often produces a characteristic hissing sound. Acoustic methods can help identify the location of the leak.

Structural defects. Cracks, loose joints, or changes in stiffness may alter the resonance behavior of a structure, changing the sound it emits when struck or when in operation.

Physical basis of acoustic diagnosis

The diagnostic use of sound is based on the fact that vibrating systems emit waves whose properties depend on the state of the source. A change in the source changes the emitted sound. In particular, an engineer may look for changes in:

- amplitude,
- dominant frequencies,
- periodicity,
- resonance peaks,
- or sudden transient signals.

In more advanced work, the sound is analyzed into its frequency components using spectrum analysis. Peaks at particular frequencies may correspond to specific rotating parts or structural resonances. Thus, wave theory finds a direct application in engineering maintenance and fault diagnosis.

Audible and ultrasonic diagnostics

The diagnostic use of sound is not limited to the audible range. High-frequency sound, especially ultrasound, is widely used for inspection in engineering. Ultrasonic methods can be used to:

- detect cracks inside solids,
- test weld quality,
- measure thickness,
- locate internal flaws,
- and assess material integrity.

Thus, in industrial physics, sound may be both a hazard to workers and a powerful tool for investigation.

2.4.7 Monitoring Noise

Because sound levels vary from place to place and often fluctuate with time, it is important to measure noise systematically. **Noise monitoring** is the process of measuring and evaluating sound levels in order to assess exposure, ensure safety, and guide noise control measures.

Purpose of noise monitoring

Noise monitoring is carried out for several reasons:

1. to protect workers and the public from excessive sound exposure,
2. to identify particularly noisy areas or machines,
3. to check compliance with health and safety regulations,
4. to determine whether noise-control measures are effective,

5. and to assess the environmental impact of traffic, construction, airports, mines, or factories.

Sound level meter

The most common instrument for monitoring sound is the **sound level meter**. It consists essentially of:

- a **microphone**, which converts pressure fluctuations into electrical signals,
- an **electronic processing system**, which amplifies and analyzes the signal,
- and a **display**, which shows the sound level in decibels.

The microphone responds to the alternating pressure variations of the sound wave. Since pressure variation is related to intensity, and intensity determines sound level, the instrument can be calibrated to read directly in dB.

Noise dosimeter

In many industrial situations, a worker moves between areas with different sound levels. In such cases, a fixed sound level meter may not represent the worker's actual total exposure. A **noise dosimeter** is therefore used. This is a small instrument worn by the worker that records cumulative exposure over time.

A dosimeter is especially useful when:

- the sound level changes throughout the day,
- work is carried out near several different sound sources,
- or the concern is total daily exposure rather than the sound level at a single location.

Quantities of interest in noise monitoring

When monitoring sound, several quantities may be considered:

- the **instantaneous sound level**,
- the **maximum sound level**,
- the **average sound level** over a time interval,
- the **duration of exposure**,
- and the **total daily noise dose**.

These quantities are important because many real environments are not acoustically steady. A machine may start and stop, traffic may vary with time, and sudden impacts may briefly raise the sound level far above the background value.

Noise mapping

In large environments such as factories, campuses, city districts, or transport corridors, sound levels may be measured at many positions and presented on a **noise map**. Such a map shows how the sound level varies across the area and helps identify regions of especially high exposure.

Noise maps are useful in:

- urban planning,
- industrial safety,
- transport studies,
- and environmental impact assessments.

Interpreting measurements

The interpretation of measured sound levels requires care because the decibel scale is logarithmic. If two independent identical sound sources act together, the decibel level does not simply double. Instead, the intensities must first be added, and only then converted to decibels. Thus, the physics of intensity must always be kept in mind when interpreting acoustic measurements.

Furthermore, the significance of a measurement depends on context. A sound level that may be acceptable in a busy factory may be unsuitable in a hospital ward, library, or classroom. Noise monitoring therefore involves both physical measurement and practical judgment.

Chapter 3

Interference of Light

3.1 Introduction

Interference is one of the most important phenomena in wave physics. It arises whenever two or more waves overlap in space and combine according to the principle of superposition. The resulting disturbance depends on the relative phases of the combining waves. If the waves arrive in phase, they reinforce one another and produce a larger resultant amplitude. If they arrive out of phase, they partially or completely cancel.

For light, interference provides direct evidence that light behaves as a wave. This was of great historical importance because, for a long time, scientists debated whether light was composed of particles or waves. The observation of stable bright and dark bands in optical experiments showed that light can undergo constructive and destructive interference just as water waves and sound waves do.

3.2 Light as a Wave

The first convincing wave theory of light was proposed in 1678 by the Dutch physicist **Christiaan Huygens**. Although Maxwell's electromagnetic theory later gave a deeper description of light, Huygens' wave theory was a major step forward because it explained reflection and

refraction in terms of wave motion and gave physical meaning to the index of refraction.

3.2.1 Huygens' Principle

Huygens' theory is based on the following principle:

> *Every point on a wavefront acts as a source of secondary spherical wavelets. After a time interval Δt, the new position of the wavefront is the surface tangent to these secondary wavelets.*

This idea is illustrated in Fig. 3.1. Suppose that at time $t = 0$ the wavefront of a plane light wave is represented by the plane ab. Each point on this plane acts as a source of a secondary spherical wavelet. After a short time Δt, each wavelet has expanded to a radius

$$c\Delta t,$$

where c is the speed of light in vacuum. The common tangent plane de drawn to these wavelets is the new wavefront at time Δt. Thus, the plane wavefront advances forward by a distance $c\Delta t$.

3.2.2 Physical Meaning

Huygens' principle provides a simple geometrical construction for predicting how a wavefront moves. It is especially useful because it helps explain several important wave phenomena:

1. **Rectilinear propagation of light:** a plane wave remains a plane wave as it advances through a uniform medium.
2. **Reflection:** the reflected wavefront can be constructed from secondary wavelets, leading to the law of reflection.
3. **Refraction:** when wave speed changes in a new medium, the wavefront bends, giving Snell's law.

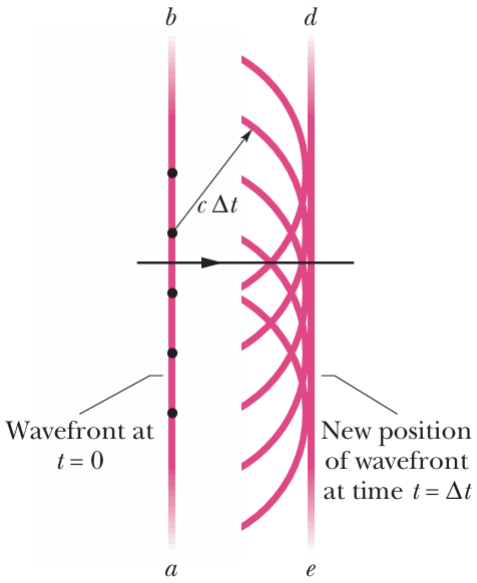


Figure 3.1: Illustration of Huygens' principle. The initial plane wavefront ab gives rise to secondary wavelets. After a time Δt, the tangent surface de represents the new wavefront.

4. **Diffraction:** when a wave passes through a narrow opening, the secondary wavelets spread into the region beyond the opening.

Thus, Huygens' principle forms the foundation for understanding interference and diffraction in optics.

3.3 Diffraction as a Prelude to Interference

Before discussing Young's experiment, we must introduce the idea of **diffraction**. Diffraction is the spreading of a wave when it passes through an aperture or around an obstacle.

If the width a of an opening is much larger than the wavelength λ, the transmitted wave spreads only slightly. However, if the opening is comparable to the wavelength, the wave spreads out strongly into the region beyond the barrier. This spreading is precisely what Huygens' principle predicts, because each point across the opening acts as a source of secondary wavelets.

Figures 3.2(a) and 3.2(b) illustrate the effect schematically. For a fixed wavelength λ, diffraction becomes more pronounced as the slit width a becomes smaller.

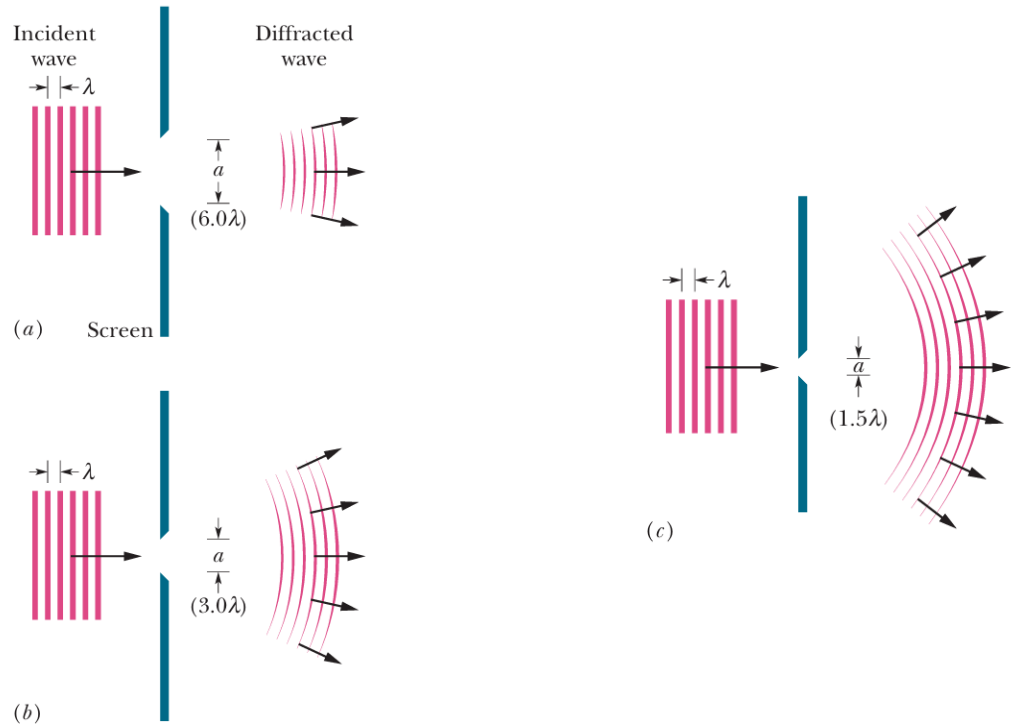


Figure 3.2: Diffraction at a slit. For the same wavelength λ, the spreading is greater when the slit width a is smaller.

Diffraction is essential in Young's experiment because each slit must spread light into the region beyond it. Without diffraction, the waves from the two slits would not overlap sufficiently to produce an observable interference pattern.

3.4 Young's Interference Experiment

In 1801, Thomas Young performed the experiment that first convincingly demonstrated that light is a wave. In addition to showing interference, he was able to estimate the average wavelength of sunlight. His measured value of about 570 nm was impressively close to the modern value near 555 nm for peak visible sensitivity.

3.4.1 Experimental Arrangement

The basic arrangement is shown in Fig. 3.3. Monochromatic light first passes through a narrow slit S_0. Because of diffraction, S_0 acts as a point source and produces semicircular wavefronts. These wavefronts then illuminate two narrow slits, S_1 and S_2, in a second screen.

The two slits S_1 and S_2 act as coherent sources because they are illuminated by the same incoming wave. The light emerging from them spreads out and overlaps on a viewing screen C, where an interference pattern of bright and dark fringes is formed.

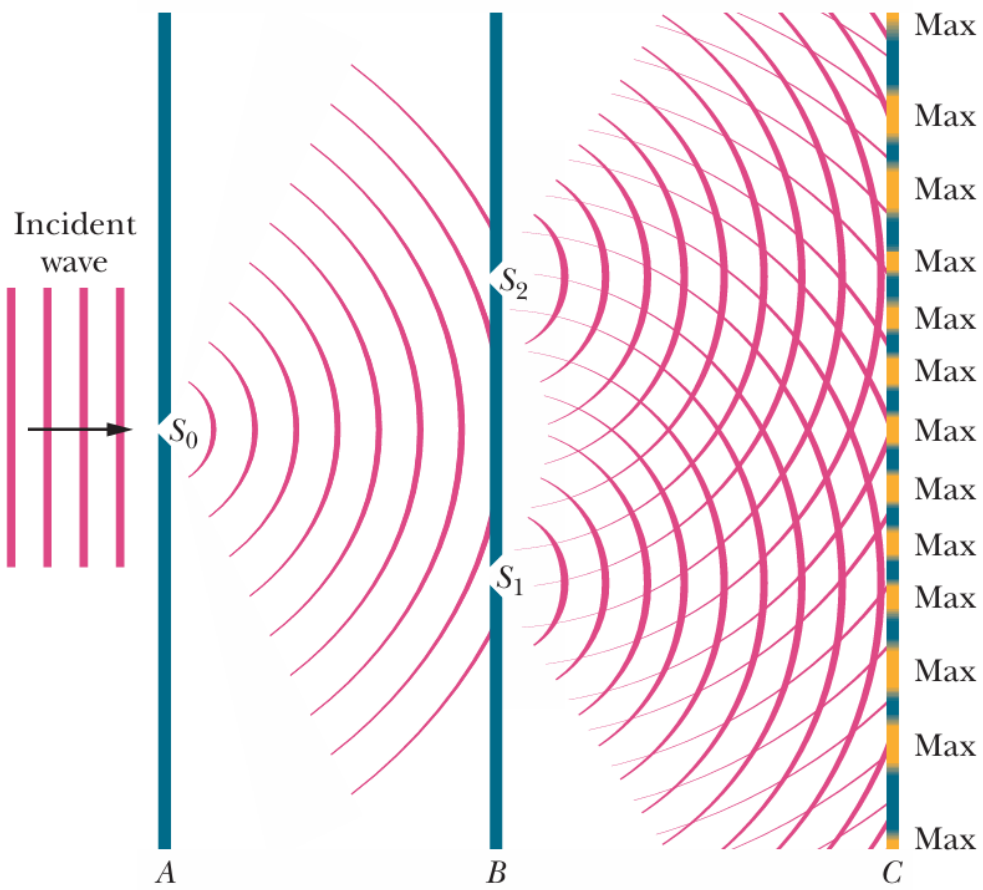


Figure 3.3: Young's double-slit experiment. Light diffracted by slit S_0 illuminates the two slits S_1 and S_2, which act as coherent sources and produce an interference pattern on screen C.

3.4.2 Nature of the Pattern

The interference pattern on the screen consists of alternating bright and dark bands:

- **Bright fringes (maxima):** regions where the light waves reinforce one another.
- **Dark fringes (minima):** regions where the light waves cancel one another.

These bright and dark regions are observable only where a screen intercepts the light. Without the screen, the waves still interfere in space, but the pattern is not directly visible.

3.4.3 Why the Slits Must Be Coherent

For a stable interference pattern to be observed, the sources must be **coherent**; that is, they must maintain a fixed phase relationship. Two independent light bulbs cannot produce a stable interference pattern because their phases fluctuate randomly. In Young's experiment, coherence is achieved by deriving both slit waves from the same initial slit S_0.

3.5 Path Difference and Phase Difference

To determine where bright and dark fringes occur, consider the geometry shown in Fig. 3.4. The two slits S_1 and S_2 are separated by a distance d. A point P on the viewing screen is located at an angle θ relative to the central axis.

The waves passing through S_1 and S_2 are initially in phase, because at the slits they are just parts of the same incident wavefront. However, after leaving the slits, they travel different distances to reach point P. If the paths have lengths r_1 and r_2, then the path difference is

$$\Delta L = r_2 - r_1.$$

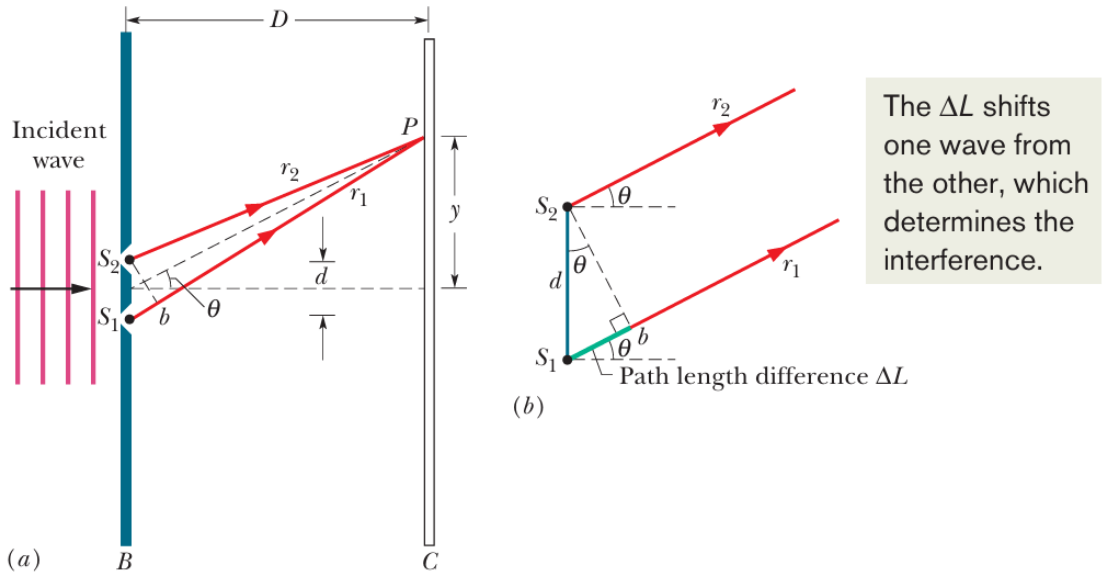


Figure 3.4: Geometry for locating the fringes in Young's double-slit experiment. The path length difference between the rays from S_1 and S_2 determines the interference at point P.

From the geometry of the figure, we obtain

$$\Delta L = d \sin\theta.$$

This relation is fundamental. It shows that the path difference, and therefore the phase difference, changes from point to point across the screen.

3.5.1 Relation Between Path Difference and Phase Difference

A path difference of one wavelength corresponds to a phase difference of 2π. Therefore,

$$\phi = \frac{2\pi}{\lambda}\Delta L.$$

Since $\Delta L = d \sin\theta$, we may also write

$$\phi = \frac{2\pi d \sin\theta}{\lambda}.$$

Thus, the interference at point P is determined entirely by the slit separation, the wavelength, and the direction θ.

3.6 Conditions for Bright and Dark Fringes

3.6.1 Bright Fringes (Constructive Interference)

Constructive interference occurs when the path difference is an integer multiple of the wavelength:

$$\Delta L = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots$$

Since $\Delta L = d \sin\theta$, the condition for bright fringes becomes

$$d \sin\theta = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{3.1}$$

The fringe with $m = 0$ is the **central bright fringe**. It lies on the central axis.

3.6.2 Dark Fringes (Destructive Interference)

Destructive interference occurs when the path difference is an odd half-integer multiple of the wavelength:

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots$$

Hence the condition for dark fringes is

$$d \sin\theta = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{3.2}$$

Each dark fringe lies between two adjacent bright fringes.

3.7 Fringe Positions on a Distant Screen

Suppose the screen is a distance D from the double slit, and let y be the perpendicular distance of a fringe from the central axis. In most practical cases, the angle θ is small. Then

$$\sin\theta \approx \tan\theta \approx \frac{y}{D}.$$

3.7.1 Position of Bright Fringes

Using Eq. (3.1),

$$d\sin\theta = m\lambda.$$

For small angles,

$$d\frac{y_m}{D} = m\lambda,$$

so

$$y_m = \frac{m\lambda D}{d}. \tag{3.3}$$

3.7.2 Position of Dark Fringes

Using Eq. (3.2),

$$d\frac{y_m^{(\text{dark})}}{D} = \left(m + \frac{1}{2}\right)\lambda,$$

thus

$$y_m^{(\text{dark})} = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}. \tag{3.4}$$

3.7.3 Fringe Spacing

The separation between successive bright fringes is

$$\Delta y = y_{m+1} - y_m = \frac{\lambda D}{d}.$$

This quantity is called the **fringe spacing**. It is very important experimentally because it allows the wavelength to be determined from directly measurable quantities.

3.8 Worked Examples

Example 1: Locating the First Bright Fringe

Problem. In a Young's double-slit experiment, the slit separation is $d = 0.300\,\text{mm}$, the wavelength is $\lambda = 600\,\text{nm}$, and the viewing screen is $D = 2.00\,\text{m}$ away. Find the distance from the central axis to the first bright fringe.

Solution.

The first bright fringe corresponds to $m = 1$. Using

$$y_m = \frac{m\lambda D}{d},$$

we have

$$y_1 = \frac{(1)(600 \times 10^{-9}\,\text{m})(2.00\,\text{m})}{0.300 \times 10^{-3}\,\text{m}}.$$

Thus,

$$y_1 = \frac{1.20 \times 10^{-6}}{3.00 \times 10^{-4}} = 4.00 \times 10^{-3}\,\text{m}.$$

Therefore,

$$\boxed{y_1 = 4.00\,\text{mm}}.$$

Example 2: Position of the Second Dark Fringe

Problem. Using the same apparatus as in Example 1, determine the position of the second dark fringe above the central maximum.

Solution.

The dark fringes are given by

$$y_m^{(\text{dark})} = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}.$$

The second dark fringe above the center corresponds to $m = 1$, so

$$y_1^{(\text{dark})} = \frac{\left(1 + \frac{1}{2}\right)(600 \times 10^{-9})(2.00)}{0.300 \times 10^{-3}}.$$

Hence,

$$y_1^{(\text{dark})} = \frac{1.5(1.20 \times 10^{-6})}{3.00 \times 10^{-4}} = 6.00 \times 10^{-3}\,\text{m}.$$

So,

$$\boxed{y_1^{(\text{dark})} = 6.00\,\text{mm}}.$$

Example 3: Determining the Wavelength of Light

Problem. In a double-slit experiment, slits are separated by $d = 0.250\,\text{mm}$. The screen is $1.50\,\text{m}$ away, and the spacing between adjacent bright fringes is measured to be $3.60\,\text{mm}$. Find the wavelength of the light.

Solution.

The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

Thus,

$$\lambda = \frac{\Delta y\, d}{D}.$$

Substituting,

$$\lambda = \frac{(3.60 \times 10^{-3}\,\text{m})(0.250 \times 10^{-3}\,\text{m})}{1.50\,\text{m}}.$$

Therefore,

$$\lambda = \frac{9.00 \times 10^{-7}}{1.50} = 6.00 \times 10^{-7}\,\text{m}.$$

Hence,

$$\boxed{\lambda = 600\,\text{nm}}.$$

Example 4: Finding the Angle of a Bright Fringe

Problem. In a Young's experiment, the slit separation is $d = 0.500\,\text{mm}$ and the wavelength is $\lambda = 500\,\text{nm}$. Find the angle of the third bright fringe.

Solution.

For a bright fringe,

$$d\sin\theta = m\lambda.$$

For the third bright fringe, take $m = 3$:

$$\sin\theta = \frac{3\lambda}{d} = \frac{3(500\times10^{-9})}{0.500\times10^{-3}}.$$

Thus,

$$\sin\theta = 3.00\times10^{-3}.$$

Since the angle is small,

$$\theta \approx 3.00\times10^{-3}\,\text{rad}.$$

Converting to degrees,

$$\theta \approx 3.00\times10^{-3}\times\frac{180^\circ}{\pi} \approx 0.172^\circ.$$

So,

$$\boxed{\theta \approx 3.00\times10^{-3}\,\text{rad} \approx 0.172^\circ.}$$

Example 5: Missing Orders in the Pattern

Problem. A double-slit arrangement uses slit separation $d = 0.18\,\text{mm}$ and light of wavelength $\lambda = 600\,\text{nm}$. How many bright fringes occur between the central maximum and the angle $\theta = 1.0°$?

Solution.

Bright fringes satisfy

$$d \sin\theta = m\lambda.$$

So the largest possible order is

$$m_{\max} = \frac{d \sin\theta}{\lambda}.$$

Substitute:

$$m_{\max} = \frac{(0.18 \times 10^{-3})(\sin 1.0°)}{600 \times 10^{-9}}.$$

Using $\sin 1.0° \approx 0.01745$,

$$m_{\max} = \frac{0.18 \times 10^{-3} \times 0.01745}{600 \times 10^{-9}} \approx 5.24.$$

Thus the largest integer order is $m = 5$. Therefore, the bright fringes between the center and $\theta = 1.0°$ are

$$m = 1, 2, 3, 4, 5.$$

Hence,

$$\boxed{\text{5 bright fringes}}$$

occur on one side of the central maximum.

3.9 Further Discussion

3.9.1 Why the Central Fringe Is Bright

At $\theta = 0$, the two paths from the slits to the center of the screen are equal, so

$$\Delta L = 0.$$

Therefore the waves arrive in phase, and the central fringe is always bright.

3.9.2 Effect of Increasing the Wavelength

From

$$\Delta y = \frac{\lambda D}{d},$$

we see that if the wavelength increases, the fringe spacing increases. Thus red light (longer wavelength) produces more widely spaced fringes than blue light (shorter wavelength).

3.9.3 Effect of Increasing the Slit Separation

If the slit separation d increases, the fringe spacing decreases. The pattern becomes more closely packed.

3.9.4 Effect of Increasing the Screen Distance

If the screen distance D increases, the fringe spacing increases. The pattern spreads out and becomes easier to measure.

3.10 Exercises

1. State Huygens' principle and explain how it accounts for the propagation of a plane wavefront.

2. What is diffraction? Why is diffraction essential in Young's double-slit experiment?

3. In your own words, explain why two independent light bulbs do not produce a stable interference pattern on a screen.

4. A double-slit arrangement has slit separation $d = 0.200\,\text{mm}$ and uses light of wavelength $500\,\text{nm}$. The screen is $1.20\,\text{m}$ away.

   (a) Find the fringe spacing.

   (b) Find the position of the second bright fringe.

   (c) Find the position of the first dark fringe.

5. In a Young's experiment, the slit separation is $0.350\,\text{mm}$ and the screen is $2.50\,\text{m}$ away. If the wavelength is $630\,\text{nm}$, calculate the distance from the central maximum to the third bright fringe.

6. A double-slit interference pattern has adjacent bright fringes separated by $2.80\,\text{mm}$. If the slits are $0.150\,\text{mm}$ apart and the screen is $1.40\,\text{m}$ away, determine the wavelength of the light.

7. In a Young's experiment, light of wavelength $550\,\text{nm}$ passes through slits separated by $0.250\,\text{mm}$.

   (a) Find the angle to the second bright fringe.

   (b) Find the angle to the first dark fringe.

8. A screen is placed $3.00\,\text{m}$ from a double slit. The slit separation is $0.300\,\text{mm}$, and the light used has wavelength $600\,\text{nm}$.

   (a) Determine the position of the central maximum.

(b) Determine the positions of the first three bright fringes above the center.

(c) Determine the positions of the first two dark fringes above the center.

9. Show from the geometry of the double-slit arrangement that the path difference to a point at angle θ is $\Delta L = d \sin\theta$.

10. Derive the condition for constructive interference in Young's experiment.

11. Derive the condition for destructive interference in Young's experiment.

12. A certain double-slit arrangement produces a fringe spacing of 4.50 mm for red light of wavelength 650 nm. What fringe spacing would you expect if the red light were replaced by blue light of wavelength 450 nm, all other parameters remaining unchanged?

13. In a Young's experiment, the wavelength is doubled while d and D are kept fixed. What happens to

    (a) the fringe spacing,

    (b) the position of the mth bright fringe,

    (c) the angle to the first dark fringe?

14. The slit separation in a double-slit experiment is halved while the wavelength and screen distance remain unchanged. Explain the effect on the interference pattern.

15. White light is used instead of monochromatic light in a double-slit experiment. Describe qualitatively what would be observed on the screen.

16. Two slits are separated by 0.400 mm and illuminated by light of wavelength 480 nm. How many bright fringes are found between the center and the angle $\theta = 0.50°$?

17. In a Young's experiment, the fourth bright fringe is observed at a distance 7.20 mm from the central maximum. If the screen is 1.80 m from the slits and the slit separation is 0.200 mm, find the wavelength of the light.

18. A student measures the distance between the sixth bright fringe on one side of the central maximum and the sixth bright fringe on the other side to be 18.0 mm. The screen is 2.00 m away, and the slit separation is 0.250 mm. Determine the wavelength of the light.

19. Why does increasing the screen distance make it easier to measure the fringe spacing?

20. Explain the difference between a bright fringe and a dark fringe in terms of phase difference and path difference.

3.11 Solutions

Solution 1

Question. State Huygens' principle and explain how it accounts for the propagation of a plane wavefront.

Solution. Huygens' principle states that every point on a wavefront acts as a source of secondary spherical wavelets, and after a time interval Δt, the new wavefront is the surface tangent to these wavelets.

For a plane wavefront, all points on the original plane emit secondary wavelets of equal radius after the same time Δt. The common tangent surface to these wavelets is another plane, parallel to the original one. Thus, the wavefront advances forward while remaining plane. This explains the propagation of a plane wave in a uniform medium.

Solution 2

Question. What is diffraction? Why is diffraction essential in Young's double-slit experiment?

Solution. Diffraction is the spreading of a wave when it passes through a narrow opening or around an obstacle. It becomes especially noticeable when the size of the opening is of the same order as the wavelength.

In Young's double-slit experiment, diffraction is essential because each slit must spread the light into the region beyond the slit. This spreading allows the waves from the two slits to overlap on the screen. Without diffraction, the two beams would not sufficiently overlap, and an interference pattern would not be formed.

Solution 3

Question. In your own words, explain why two independent light bulbs do not produce a stable interference pattern on a screen.

Solution. A stable interference pattern requires *coherent* sources, meaning that the waves must maintain a fixed phase relationship. Two independent light bulbs emit light from many atoms acting randomly. Their phases change continuously and unpredictably. As a result, any momentary interference pattern changes too rapidly to be observed, and the screen shows only uniform illumination rather than stable bright and dark fringes.

Solution 4

Question. A double-slit arrangement has slit separation $d = 0.200\,\text{mm}$ and uses light of wavelength $500\,\text{nm}$. The screen is $1.20\,\text{m}$ away.

1. Find the fringe spacing.
2. Find the position of the second bright fringe.
3. Find the position of the first dark fringe.

Solution.

Given:

$$d = 0.200 \times 10^{-3}\,\text{m}, \qquad \lambda = 500 \times 10^{-9}\,\text{m}, \qquad D = 1.20\,\text{m}.$$

(a) Fringe spacing

The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

Thus,

$$\Delta y = \frac{(500 \times 10^{-9})(1.20)}{0.200 \times 10^{-3}} = 3.00 \times 10^{-3}\,\text{m}.$$

Therefore,

$$\boxed{\Delta y = 3.00\,\text{mm}}.$$

(b) Position of the second bright fringe

For bright fringes,

$$y_m = \frac{m\lambda D}{d}.$$

For the second bright fringe, $m = 2$:

$$y_2 = 2\Delta y = 2(3.00\,\text{mm}) = 6.00\,\text{mm}.$$

Hence,

$$\boxed{y_2 = 6.00\,\text{mm}}.$$

(c) Position of the first dark fringe

For dark fringes,

$$y_m^{(\text{dark})} = \left(m + \frac{1}{2}\right)\frac{\lambda D}{d}.$$

The first dark fringe above the center corresponds to $m = 0$:

$$y_0^{(\text{dark})} = \frac{1}{2}(3.00\,\text{mm}) = 1.50\,\text{mm}.$$

So,

$$\boxed{y_0^{(\text{dark})} = 1.50\,\text{mm}}.$$

Solution 5

Question. In a Young's experiment, the slit separation is 0.350 mm and the screen is 2.50 m away. If the wavelength is 630 nm, calculate the distance from the central maximum to the third bright fringe.

Solution.

Given:

$$d = 0.350 \times 10^{-3}\,\text{m}, \qquad D = 2.50\,\text{m}, \qquad \lambda = 630 \times 10^{-9}\,\text{m}.$$

For the third bright fringe, $m = 3$:

$$y_3 = \frac{3\lambda D}{d}.$$

Substituting,

$$y_3 = \frac{3(630 \times 10^{-9})(2.50)}{0.350 \times 10^{-3}}.$$

Calculate:

$$3(630 \times 10^{-9})(2.50) = 4.725 \times 10^{-6},$$

so

$$y_3 = \frac{4.725 \times 10^{-6}}{3.50 \times 10^{-4}} = 1.35 \times 10^{-2}\,\text{m}.$$

Therefore,

$$\boxed{y_3 = 13.5\,\text{mm}}.$$

Solution 6

Question. A double-slit interference pattern has adjacent bright fringes separated by 2.80 mm. If the slits are 0.150 mm apart and the screen is 1.40 m away, determine the wavelength of the light.

Solution.

The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

Hence,

$$\lambda = \frac{\Delta y\, d}{D}.$$

Given:

$$\Delta y = 2.80 \times 10^{-3}\,\text{m}, \qquad d = 0.150 \times 10^{-3}\,\text{m}, \qquad D = 1.40\,\text{m}.$$

Thus,

$$\lambda = \frac{(2.80 \times 10^{-3})(0.150 \times 10^{-3})}{1.40} = 3.00 \times 10^{-7}\,\text{m}.$$

Therefore,

$$\boxed{\lambda = 300\,\text{nm}}.$$

Solution 7

Question. In a Young's experiment, light of wavelength 550 nm passes through slits separated by 0.250 mm.

1. Find the angle to the second bright fringe.
2. Find the angle to the first dark fringe.

Solution.

Given:

$$\lambda = 550 \times 10^{-9}\,\text{m}, \qquad d = 0.250 \times 10^{-3}\,\text{m}.$$

(a) Second bright fringe

For bright fringes,

$$d \sin\theta = m\lambda.$$

For the second bright fringe, $m = 2$:

$$\sin\theta = \frac{2\lambda}{d} = \frac{2(550 \times 10^{-9})}{0.250 \times 10^{-3}} = 4.40 \times 10^{-3}.$$

Thus,

$$\theta \approx 4.40 \times 10^{-3}\,\text{rad}.$$

In degrees,

$$\theta \approx 4.40 \times 10^{-3} \times \frac{180^\circ}{\pi} \approx 0.252^\circ.$$

Hence,

$$\boxed{\theta \approx 4.40 \times 10^{-3}\,\text{rad} \approx 0.252^\circ.}$$

(b) First dark fringe

For dark fringes,

$$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda.$$

For the first dark fringe, $m = 0$:

$$\sin\theta = \frac{\lambda/2}{d} = \frac{0.5(550 \times 10^{-9})}{0.250 \times 10^{-3}} = 1.10 \times 10^{-3}.$$

Thus,

$$\theta \approx 1.10 \times 10^{-3}\,\text{rad}.$$

In degrees,

$$\theta \approx 1.10 \times 10^{-3} \times \frac{180^\circ}{\pi} \approx 0.0630^\circ.$$

Therefore,

$$\boxed{\theta \approx 1.10 \times 10^{-3}\,\text{rad} \approx 0.0630^\circ.}$$

Solution 8

Question. A screen is placed 3.00 m from a double slit. The slit separation is 0.300 mm, and the light used has wavelength 600 nm.

1. Determine the position of the central maximum.
2. Determine the positions of the first three bright fringes above the center.
3. Determine the positions of the first two dark fringes above the center.

Solution.

Given:

$$D = 3.00\,\text{m}, \qquad d = 0.300 \times 10^{-3}\,\text{m}, \qquad \lambda = 600 \times 10^{-9}\,\text{m}.$$

First compute the fringe spacing:

$$\Delta y = \frac{\lambda D}{d} = \frac{(600 \times 10^{-9})(3.00)}{0.300 \times 10^{-3}} = 6.00 \times 10^{-3}\,\text{m} = 6.00\,\text{mm}.$$

(a) Central maximum

The central maximum corresponds to $m = 0$, so

$$\boxed{y = 0.}$$

(b) First three bright fringes above the center

Bright fringes occur at

$$y_m = m\Delta y.$$

Thus,

$$y_1 = 6.00\,\text{mm}, \qquad y_2 = 12.0\,\text{mm}, \qquad y_3 = 18.0\,\text{mm}.$$

So the positions are

$$\boxed{6.00\,\text{mm},\ 12.0\,\text{mm},\ 18.0\,\text{mm}.}$$

(c) First two dark fringes above the center

Dark fringes occur at

$$y_m^{(\text{dark})} = \left(m + \frac{1}{2}\right)\Delta y.$$

Therefore,

$$y_0^{(\text{dark})} = 3.00\,\text{mm}, \qquad y_1^{(\text{dark})} = 9.00\,\text{mm}.$$

Hence,

$$\boxed{3.00\,\text{mm},\ 9.00\,\text{mm}.}$$

Solution 9

Question. Show from the geometry of the double-slit arrangement that the path difference to a point at angle θ is $\Delta L = d\sin\theta$.

Solution. Let the two slits S_1 and S_2 be separated by a perpendicular distance d. Consider a point P on the screen such that the rays from the slits to P make an angle θ with the central axis. If the screen is far from the slits, the rays to P are nearly parallel.

The extra distance traveled by the ray from the upper slit relative to the lower slit is the projection of the slit separation d onto the direction of propagation toward P. From basic trigonometry, this projection is

$$\Delta L = d\sin\theta.$$

Thus,

$$\boxed{\Delta L = d\sin\theta.}$$

Solution 10

Question. Derive the condition for constructive interference in Young's experiment.

Solution. Constructive interference occurs when the two waves arrive at a point in phase. A phase difference of 2π corresponds to a path difference of one wavelength λ. Therefore, the

condition for constructive interference is that the path difference be an integer multiple of the wavelength:

$$\Delta L = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots$$

Since in Young's experiment

$$\Delta L = d \sin\theta,$$

the condition becomes

$$\boxed{d \sin\theta = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots}$$

Solution 11

Question. Derive the condition for destructive interference in Young's experiment.

Solution. Destructive interference occurs when the two waves arrive exactly out of phase, that is, with a phase difference of π, 3π, 5π, and so on. This corresponds to a path difference equal to an odd half-integer multiple of the wavelength:

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots$$

Since

$$\Delta L = d \sin\theta,$$

the condition for destructive interference is

$$\boxed{d \sin\theta = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots}$$

Solution 12

Question. A certain double-slit arrangement produces a fringe spacing of 4.50 mm for red light of wavelength 650 nm. What fringe spacing would you expect if the red light were replaced by blue light of wavelength 450 nm, all other parameters remaining unchanged?

Solution. The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

Thus, with D and d unchanged, the fringe spacing is directly proportional to the wavelength:

$$\frac{\Delta y_{\text{blue}}}{\Delta y_{\text{red}}} = \frac{\lambda_{\text{blue}}}{\lambda_{\text{red}}}.$$

Hence,

$$\Delta y_{\text{blue}} = 4.50\,\text{mm} \times \frac{450}{650} = 3.12\,\text{mm}.$$

Therefore,

$$\boxed{\Delta y_{\text{blue}} \approx 3.12\,\text{mm}.}$$

Solution 13

Question. In a Young's experiment, the wavelength is doubled while d and D are kept fixed. What happens to

1. the fringe spacing,
2. the position of the mth bright fringe,
3. the angle to the first dark fringe?

Solution.

(a) Fringe spacing

Since

$$\Delta y = \frac{\lambda D}{d},$$

doubling λ doubles the fringe spacing.

$$\boxed{\text{The fringe spacing doubles.}}$$

(b) Position of the mth bright fringe

Since

$$y_m = \frac{m\lambda D}{d},$$

doubling λ doubles y_m.

The position of the mth bright fringe doubles.

(c) Angle to the first dark fringe

For the first dark fringe,

$$d \sin\theta = \frac{1}{2}\lambda.$$

If λ doubles, the right-hand side doubles, so $\sin\theta$ doubles. For small angles, θ approximately doubles.

The angle to the first dark fringe increases, approximately doubling for small angles.

Solution 14

Question. The slit separation in a double-slit experiment is halved while the wavelength and screen distance remain unchanged. Explain the effect on the interference pattern.

Solution. The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

If d is halved, then

$$\Delta y' = \frac{\lambda D}{d/2} = 2\frac{\lambda D}{d} = 2\Delta y.$$

Therefore, the bright and dark fringes become twice as far apart. The pattern spreads out on the screen.

Halving the slit separation doubles the fringe spacing.

Solution 15

Question. White light is used instead of monochromatic light in a double-slit experiment. Describe qualitatively what would be observed on the screen.

Solution. White light contains many wavelengths. At the center of the screen, the path difference is zero for all wavelengths, so all colors interfere constructively there. Thus, the central fringe is white.

Away from the center, different wavelengths satisfy the bright-fringe condition at different positions. As a result, colored fringes appear on either side of the central white fringe. Because shorter wavelengths and longer wavelengths spread differently, the colored fringes overlap more and more farther from the center, eventually washing out the pattern.

Thus the screen shows:

- a **central white fringe**,
- **colored fringes** on either side,
- and gradually diminishing distinctness farther from the center.

Solution 16

Question. Two slits are separated by 0.400 mm and illuminated by light of wavelength 480 nm. How many bright fringes are found between the center and the angle $\theta = 0.50°$?

Solution.

Bright fringes satisfy

$$d \sin \theta = m\lambda.$$

Thus,

$$m_{\max} = \frac{d \sin \theta}{\lambda}.$$

Given:

$$d = 0.400 \times 10^{-3}\,\text{m}, \qquad \lambda = 480 \times 10^{-9}\,\text{m}, \qquad \theta = 0.50^\circ.$$

Using $\sin 0.50^\circ \approx 0.008727$,

$$m_{\max} = \frac{(0.400 \times 10^{-3})(0.008727)}{480 \times 10^{-9}} \approx 7.27.$$

Thus the largest whole-number order is $m = 7$. Therefore, the bright fringes between the center and that angle are the orders

$$m = 1, 2, 3, 4, 5, 6, 7.$$

Hence,

$$\boxed{\text{7 bright fringes.}}$$

Solution 17

Question. In a Young's experiment, the fourth bright fringe is observed at a distance 7.20 mm from the central maximum. If the screen is 1.80 m from the slits and the slit separation is 0.200 mm, find the wavelength of the light.

Solution.

For bright fringes,

$$y_m = \frac{m\lambda D}{d}.$$

Thus,

$$\lambda = \frac{y_m d}{mD}.$$

Given:

$$y_4 = 7.20 \times 10^{-3}\,\text{m}, \qquad d = 0.200 \times 10^{-3}\,\text{m}, \qquad m = 4, \qquad D = 1.80\,\text{m}.$$

So,

$$\lambda = \frac{(7.20 \times 10^{-3})(0.200 \times 10^{-3})}{4(1.80)} = \frac{1.44 \times 10^{-6}}{7.20} = 2.00 \times 10^{-7}\,\text{m}.$$

Therefore,

$$\boxed{\lambda = 200\,\text{nm}.}$$

Solution 18

Question. A student measures the distance between the sixth bright fringe on one side of the central maximum and the sixth bright fringe on the other side to be 18.0 mm. The screen is 2.00 m away, and the slit separation is 0.250 mm. Determine the wavelength of the light.

Solution.

The sixth bright fringe on each side is at positions $\pm y_6$, so the measured separation is

$$2y_6 = 18.0\,\text{mm}.$$

Hence,

$$y_6 = 9.00\,\text{mm} = 9.00 \times 10^{-3}\,\text{m}.$$

For bright fringes,

$$y_m = \frac{m\lambda D}{d}.$$

Thus,

$$\lambda = \frac{y_m d}{mD}.$$

Substitute:

$$\lambda = \frac{(9.00 \times 10^{-3})(0.250 \times 10^{-3})}{6(2.00)} = \frac{2.25 \times 10^{-6}}{12} = 1.875 \times 10^{-7}\,\text{m}.$$

Therefore,

$$\boxed{\lambda = 1.88 \times 10^{-7}\,\text{m} \approx 188\,\text{nm}.}$$

Solution 19

Question. Why does increasing the screen distance make it easier to measure the fringe spacing?

Solution. The fringe spacing is

$$\Delta y = \frac{\lambda D}{d}.$$

Thus, if the screen distance D increases, the fringe spacing increases proportionally. The bright and dark fringes become farther apart, making their positions easier to distinguish and measure accurately on the screen.

Solution 20

Question. Explain the difference between a bright fringe and a dark fringe in terms of phase difference and path difference.

Solution.

A **bright fringe** occurs when the two waves arrive in phase. This means:

$$\text{phase difference} = 2m\pi,$$

which corresponds to the path-difference condition

$$\Delta L = m\lambda, \qquad m = 0, 1, 2, \ldots$$

A **dark fringe** occurs when the waves arrive exactly out of phase:

$$\text{phase difference} = (2m + 1)\pi,$$

which corresponds to the path-difference condition

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda, \qquad m = 0, 1, 2, \ldots$$

Thus:

- bright fringe $\rightarrow$ constructive interference,
- dark fringe $\rightarrow$ destructive interference.

Chapter 4

Diffraction

4.1 Introduction

In the preceding chapter, we studied the interference of light, especially Young's double-slit experiment. There we saw that when two coherent light waves overlap, they may reinforce each other to produce bright fringes or cancel each other to produce dark fringes. In this chapter we extend the same wave ideas to a closely related phenomenon called **diffraction**.

Diffraction is the bending or spreading of waves when they pass through an aperture or around an obstacle. It is observed with all types of waves, including water waves, sound waves, and light waves. For light, diffraction becomes especially noticeable when the size of the opening or obstacle is comparable with the wavelength of the light.

The phenomenon of diffraction follows naturally from Huygens' principle. Every point on a wavefront acts as a source of secondary wavelets. When a wavefront passes through a narrow opening, the wavelets emerging from different points of the opening spread out and overlap. Since these wavelets have different path lengths to a point on a screen, they may arrive with different phases. The resulting pattern of brightness and darkness is therefore produced by interference among the many wavelets originating from different parts of the same wavefront.

Although interference and diffraction are often discussed separately, they are not different physical processes. Both arise from the principle of superposition. The distinction is mainly

one of convenience:

- Interference usually refers to the superposition of waves from a small number of coherent sources, such as two narrow slits.
- Diffraction usually refers to the superposition of many wavelets from different parts of the same aperture or wavefront.

4.2 Diffraction by a Single Slit

4.2.1 Experimental Arrangement

Consider plane waves of monochromatic light of wavelength λ incident normally on a single long, narrow slit of width a in an otherwise opaque screen B, as shown in Fig. 4.1. The light that passes through the slit is diffracted and reaches a viewing screen C placed a distance D from the slit.

If light travelled strictly in straight lines, the screen would show only a sharp bright region directly behind the slit. Instead, the light spreads into the region beyond the slit. On the screen we observe a diffraction pattern consisting of a broad central bright fringe and a set of weaker bright fringes separated by dark fringes.

The central bright fringe lies on the central axis, the line drawn from the center of the slit perpendicular to the screen. It is much wider and more intense than the other bright fringes.

4.2.2 Origin of the Central Bright Fringe

The central bright fringe can be understood using Huygens' principle. Every point across the slit acts as a source of secondary wavelets. At the center of the screen, the wavelets from all points of the slit travel approximately equal distances. They therefore arrive in phase and reinforce one another. This produces the central maximum.

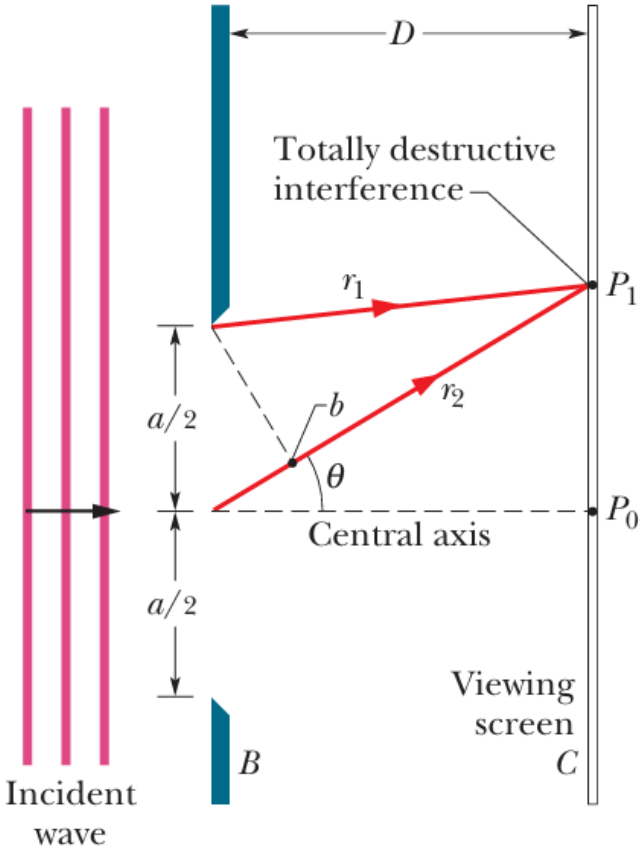


Figure 4.1: Diffraction of plane light waves by a single slit of width a. The diffracted light reaches a viewing screen C, where a pattern of bright and dark fringes is formed.

Away from the central axis, the wavelets from different parts of the slit travel different distances. At some angles the path differences cause complete cancellation, producing dark fringes. At other angles the cancellation is incomplete, producing weaker bright fringes.

4.2.3 Locating the First Dark Fringe

We now find the condition for the first dark fringe. Let P_1 be a point on the viewing screen at an angle θ from the central axis. To simplify the analysis, suppose the screen is far from the slit, so that

$$D \gg a.$$

This is called the **far-field** or **Fraunhofer diffraction** approximation. Under this condition, the rays from different points of the slit to the point P_1 may be treated as approximately parallel, as shown in Fig. 4.2.

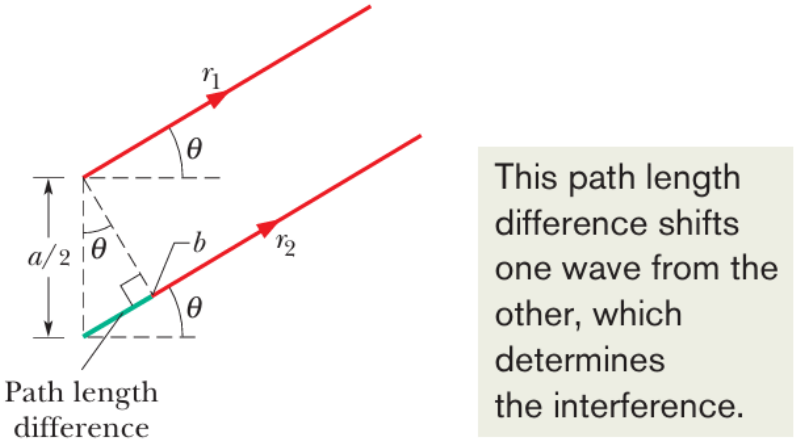


Figure 4.2: Geometry used to locate the first dark fringe in single-slit diffraction. The slit is divided into two equal zones. Rays from corresponding points in the two zones cancel when their path difference is $\lambda/2$.

To locate the first dark fringe, mentally divide the slit into two equal zones, each of width $a/2$. Consider two rays:

- ray r_1 from the top of the upper half of the slit,
- ray r_2 from the top of the lower half of the slit.

These two rays originate from points separated by a distance $a/2$. Since the incident wave is a plane wave, the wavelets at these two points are initially in phase. However, when the rays travel to the point P_1, one ray travels farther than the other.

For the far-field geometry, the path difference between these two rays is

$$\Delta L = \frac{a}{2} \sin\theta.$$

For these two rays to cancel at P_1, they must arrive out of phase by π, which corresponds to a path difference of $\lambda/2$. Therefore,

$$\frac{a}{2}\sin\theta = \frac{\lambda}{2}.$$

Multiplying both sides by 2 gives

$$a\sin\theta = \lambda. \tag{4.1}$$

This is the condition for the first dark fringe.

The reason the entire slit produces darkness at this angle is that every ray from the upper half of the slit can be paired with a corresponding ray from the lower half. Each pair has a path difference of $\lambda/2$, so the two waves in that pair arrive exactly out of phase and cancel.

Thus, the first minimum is not caused only by the two rays shown in the figure. Those rays are representative of all corresponding pairs of rays from the two halves of the slit.

4.2.4 All Dark Fringes

The same reasoning can be extended to locate the other dark fringes. For example, to find the second dark fringe, we divide the slit into four equal zones. The wavelets from the first and second zones cancel, and the wavelets from the third and fourth zones also cancel. For the third dark fringe, we divide the slit into six equal zones, and so on.

In general, the dark fringes of a single-slit diffraction pattern occur at angles satisfying

$$a\sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots \tag{4.2}$$

where m is the order number of the minimum.

It is important to note that $m = 0$ is not included in this equation. If $m = 0$, then $\theta = 0$, which corresponds to the central bright fringe, not a dark fringe.

4.2.5 Why Integer Wavelength Path Differences Give Minima

At first, Eq. (4.2) may seem surprising. In double-slit interference, a path difference equal to an integer multiple of the wavelength gives constructive interference. Why, then, does the condition

$$a \sin\theta = m\lambda$$

give darkness for a single slit?

The answer is that in single-slit diffraction the relevant cancellation is not between only the top and bottom rays. Instead, each ray from one part of the slit is cancelled by another ray from a different part of the slit. Even if the top and bottom rays are in phase with each other, each of them may still be out of phase with some other ray from the slit. The total sum of all the wavelets can therefore be zero.

4.2.6 Positions of Minima on a Distant Screen

If the screen is far from the slit and the diffraction angles are small, then

$$\sin\theta \approx \tan\theta \approx \frac{y}{D},$$

where y is the distance of a fringe from the center of the pattern on the screen. Substituting this into Eq. (4.2) gives

$$a\frac{y_m}{D} = m\lambda.$$

Thus, the positions of the dark fringes are approximately

$$y_m = \frac{m\lambda D}{a}, \qquad m = 1, 2, 3, \ldots \tag{4.3}$$

The first dark fringes occur at

$$y_1 = \pm \frac{\lambda D}{a}.$$

Therefore, the width of the central bright fringe is the distance between the first dark fringe above the center and the first dark fringe below the center:

$$W_{\text{central}} = 2y_1 = \frac{2\lambda D}{a}. \tag{4.4}$$

This equation shows that the central bright fringe becomes wider when:

- the wavelength λ is increased,
- the screen distance D is increased,
- the slit width a is decreased.

4.2.7 Effect of Slit Width

From

$$a \sin\theta = \lambda$$

for the first minimum, we have

$$\sin\theta = \frac{\lambda}{a}.$$

Thus, if the slit width a is made smaller while λ is kept fixed, the angle θ to the first minimum increases. The diffraction pattern spreads out.

If $a = \lambda$, then

$$\sin\theta = 1,$$

so

$$\theta = 90^\circ.$$

In this limiting case, the first dark fringes occur at $\pm 90^\circ$, meaning that the central bright fringe spreads over the entire forward region.

This result illustrates an important general rule:

The narrower the aperture, the greater the diffraction.

4.2.8 Intensity Distribution for a Single Slit

A complete mathematical treatment gives the intensity distribution for single-slit diffraction as

$$I(\theta) = I_m \left(\frac{\sin\alpha}{\alpha} \right)^2, \tag{4.5}$$

where

$$\alpha = \frac{\pi a}{\lambda} \sin\theta. \tag{4.6}$$

Here I_m is the intensity at the center of the pattern. The minima occur when

$$\sin\alpha = 0,$$

except at $\alpha = 0$. Thus,

$$\alpha = m\pi, \qquad m = 1, 2, 3, \ldots$$

which gives

$$\frac{\pi a}{\lambda}\sin\theta = m\pi.$$

Therefore,

$$a\sin\theta = m\lambda,$$

as obtained earlier by the pairing argument.

Worked Example 1: Width of the Central Maximum

Problem. Light of wavelength 600 nm passes through a single slit of width 0.200 mm. A screen is placed 2.00 m from the slit. Find the width of the central bright fringe.

Solution. The width of the central maximum is

$$W_{\text{central}} = \frac{2\lambda D}{a}.$$

Given

$$\lambda = 600\,\text{nm} = 600\times10^{-9}\,\text{m},$$

$$a = 0.200\,\text{mm} = 0.200\times10^{-3}\,\text{m},$$

and

$$D = 2.00\,\text{m}.$$

Thus,

$$W_{\text{central}} = \frac{2(600 \times 10^{-9})(2.00)}{0.200 \times 10^{-3}}.$$

Hence,

$$W_{\text{central}} = 1.20 \times 10^{-2}\,\text{m}.$$

Therefore,

$$\boxed{W_{\text{central}} = 12.0\,\text{mm}.}$$

Worked Example 2: Angle of the First Minimum

Problem. A single slit has width $a = 0.0500\,\text{mm}$. It is illuminated with light of wavelength $\lambda = 500\,\text{nm}$. Find the angle of the first diffraction minimum.

Solution. For the first minimum,

$$a \sin\theta = \lambda.$$

Therefore,

$$\sin\theta = \frac{\lambda}{a}.$$

Substitute

$$\lambda = 500 \times 10^{-9}\,\text{m}, \qquad a = 0.0500 \times 10^{-3}\,\text{m}.$$

Then

$$\sin\theta = \frac{500 \times 10^{-9}}{0.0500 \times 10^{-3}} = 0.0100.$$

Thus,

$$\theta = \sin^{-1}(0.0100).$$

For small angles,

$$\theta \approx 0.0100\,\text{rad}.$$

Converting to degrees,

$$\theta \approx 0.0100\left(\frac{180^\circ}{\pi}\right) = 0.573^\circ.$$

Therefore,

$$\boxed{\theta \approx 0.573^\circ.}$$

4.3 Diffraction by a Double Slit

4.3.1 Interference and Diffraction Together

In the previous chapter, we treated the double slits as if they were very narrow. Under that approximation, each slit acted like a point source, and the intensity pattern was described by the ordinary double-slit interference factor. In reality, each slit has a finite width. Therefore, each slit also produces its own single-slit diffraction pattern.

The actual double-slit pattern is therefore a combination of:

- interference between waves from the two slits, and

- diffraction due to the finite width of each slit.

Figure 4.3 illustrates this idea. Part (a) shows a double-slit pattern in which both interference and diffraction are present. Part (b) shows the broader single-slit diffraction pattern obtained when one slit is covered.

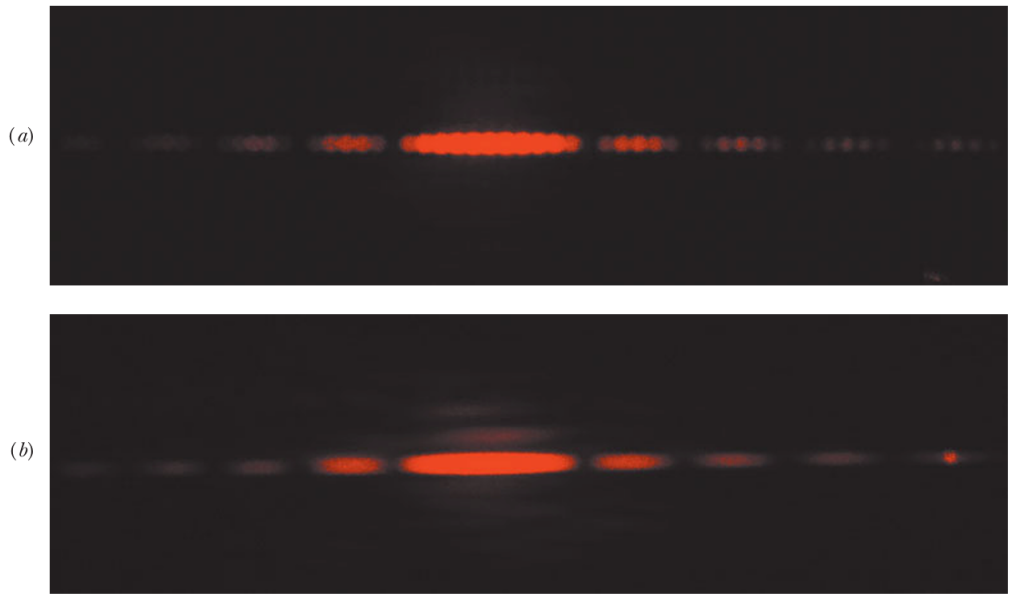


Figure 4.3: Comparison of double-slit interference with diffraction effects included and the corresponding single-slit diffraction pattern.

4.3.2 Intensity Distribution

For a double slit in which each slit has width a and the center-to-center separation of the slits is d, the intensity at an angle θ is

$$I(\theta) = I_m \cos^2 \beta \left(\frac{\sin \alpha}{\alpha} \right)^2 , \tag{4.7}$$

where

$$\beta = \frac{\pi d}{\lambda} \sin \theta \tag{4.8}$$

and

$$\alpha = \frac{\pi a}{\lambda} \sin\theta. \tag{4.9}$$

The two factors in Eq. (4.7) have different physical meanings.

The Interference Factor

The factor

$$\cos^2 \beta$$

comes from interference between the waves emerging from the two slits. It determines the positions of the narrow bright and dark interference fringes. The interference maxima occur when

$$d \sin\theta = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{4.10}$$

The Diffraction Factor

The factor

$$\left(\frac{\sin\alpha}{\alpha}\right)^2$$

comes from diffraction by each individual slit. It acts as an envelope that controls the overall brightness of the interference fringes. The diffraction minima occur when

$$a \sin\theta = p\lambda, \qquad p = 1, 2, 3, \ldots \tag{4.11}$$

Thus, in the actual double-slit pattern, the interference fringes are not all equally bright. They are brightest near the center and become weaker as the diffraction envelope decreases.

4.3.3 Limiting Cases

The double-slit intensity formula gives the correct results in two important limiting cases.

Case 1: Very Narrow Slits

If the slit width becomes very small, so that

$$a \to 0,$$

then

$$\alpha \to 0.$$

Using the limiting result

$$\lim_{\alpha \to 0} \frac{\sin \alpha}{\alpha} = 1,$$

the diffraction factor becomes 1. Equation (4.7) reduces to

$$I(\theta) = I_m \cos^2 \beta.$$

This is the ordinary double-slit interference pattern for two very narrow slits.

Case 2: Slits Merging into One Slit

If the separation between the two slits is reduced so that effectively $d = 0$, then

$$\beta = 0$$

and

$$\cos^2 \beta = 1.$$

The intensity formula becomes

$$I(\theta) = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2.$$

This is the single-slit diffraction formula.

4.3.4 Missing Orders

An interesting consequence of combining double-slit interference with single-slit diffraction is the possibility of missing interference maxima.

Interference maxima occur when

$$d \sin \theta = m\lambda.$$

Diffraction minima occur when

$$a \sin \theta = p\lambda.$$

If an angle satisfies both conditions at the same time, then the interference maximum is suppressed by the diffraction minimum. In that case, the corresponding bright fringe is missing.

Dividing the two equations gives

$$\frac{d}{a} = \frac{m}{p}.$$

Thus, missing orders occur when the order number m of an interference maximum satisfies

$$m = p\frac{d}{a}, \qquad p = 1, 2, 3, \ldots \tag{4.12}$$

provided that m is an integer.

Worked Example 3: Missing Orders

Problem. A double-slit arrangement has slit separation $d = 0.300\,\text{mm}$ and slit width $a = 0.100\,\text{mm}$. Which interference orders are missing?

Solution. The ratio is

$$\frac{d}{a} = \frac{0.300\,\text{mm}}{0.100\,\text{mm}} = 3.$$

Missing orders occur when

$$m = p\frac{d}{a}.$$

Therefore,

$$m = 3p.$$

For

$$p = 1, 2, 3, \ldots,$$

the missing orders are

$$m = 3, 6, 9, \ldots$$

Thus,

The third, sixth, ninth, …interference maxima are missing.

4.4 Diffraction Gratings

4.4.1 Definition and Structure

A diffraction grating is an optical device consisting of a large number of equally spaced parallel slits or grooves. It is one of the most useful instruments in the study of light because it can separate light into its component wavelengths with high precision.

A grating may consist of:

- many transparent slits ruled on an opaque screen, or
- many parallel grooves on a reflecting surface.

In a transmission grating, light passes through the slits. In a reflection grating, light is reflected or scattered from the grooves.

Figure 4.4 shows an idealized grating with only a few slits. Real gratings may have hundreds or thousands of rulings per millimeter.

The distance between adjacent slits or rulings is called the **grating spacing** and is denoted by d.

If a grating has N rulings distributed over a total width w, then the grating spacing is

$$d = \frac{w}{N}. \tag{4.13}$$

If the grating has n rulings per unit length, then

$$d = \frac{1}{n}. \tag{4.14}$$

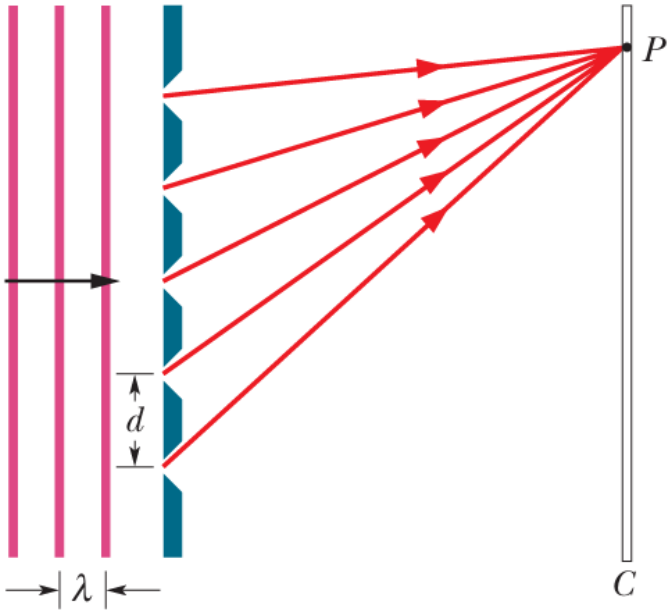


Figure 4.4: An idealized diffraction grating consisting of several equally spaced slits. The spacing between adjacent slits is d.

For example, if a grating has 600 lines per millimeter, then

$$n = 600\,\text{mm}^{-1}.$$

Therefore,

$$d = \frac{1}{600}\,\text{mm} = 1.67 \times 10^{-3}\,\text{mm} = 1.67 \times 10^{-6}\,\text{m}.$$

4.4.2 Diffraction Pattern of a Grating

When monochromatic light is incident on a diffraction grating, the emerging waves from the many slits interfere. At certain angles, the waves from all the slits arrive in phase and produce very sharp bright lines. Between these bright lines are broad dark regions.

As the number of slits increases, the bright fringes become:

- narrower,
- sharper,
- more intense,
- and more widely separated by dark regions.

Figure 4.5 shows the intensity distribution and the corresponding pattern on a screen.

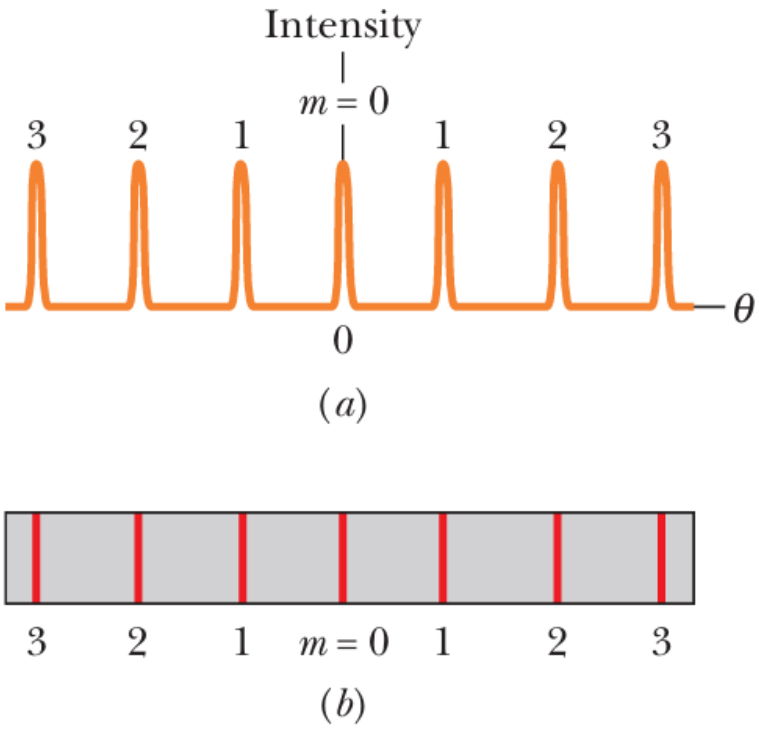


Figure 4.5: Diffraction grating pattern for monochromatic light. The principal maxima are narrow and bright, and are separated by broad dark regions.

4.4.3 Condition for Principal Maxima

Let the grating spacing be d. Consider light diffracted at an angle θ from the central axis. The path difference between waves from adjacent slits is

$$\Delta L = d \sin\theta.$$

For a bright line to occur, waves from adjacent slits must arrive in phase. Therefore, the path difference must be an integer multiple of the wavelength:

$$d \sin\theta = m\lambda.$$

Thus, the grating equation is

$$d \sin\theta = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{4.15}$$

The integer m is called the **order number**.

- $m = 0$ gives the central or zeroth-order maximum.
- $m = 1$ gives the first-order maximum.
- $m = 2$ gives the second-order maximum.
- and so on.

For each positive order on one side of the central maximum, there is a corresponding negative order on the other side.

4.4.4 Maximum Possible Order

Since

$$\sin\theta \leq 1,$$

the grating equation implies

$$m\lambda \leq d.$$

Therefore, the largest possible order is

$$m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor, \tag{4.16}$$

where the brackets mean that only the integer part is taken.

4.4.5 Determination of Wavelength Using a Diffraction Grating

The grating equation may be written as

$$\lambda = \frac{d \sin \theta}{m}. \tag{4.17}$$

Thus, if the grating spacing d is known and the angle θ to the mth-order bright line is measured, the wavelength of the light can be determined.

This is one of the most important uses of a diffraction grating. A grating can separate different wavelengths because the angle of a given order depends on λ:

$$\theta = \sin^{-1}\left(\frac{m\lambda}{d}\right).$$

Longer wavelengths are diffracted through larger angles than shorter wavelengths. Therefore, red light is deviated more than blue light in a given order.

4.4.6 Why a Grating Is Better Than a Double Slit for Measuring Wavelength

The double-slit equation

$$d \sin \theta = m\lambda$$

has the same mathematical form as the grating equation. However, a diffraction grating has many slits instead of only two. This makes the bright lines much narrower and sharper. As a result, the angles can be measured more accurately.

Also, when light contains several wavelengths, a grating separates the different wavelengths more clearly than a double slit. This is why gratings are used in spectrometers and other optical instruments.

4.4.7 Dispersion by a Diffraction Grating

Suppose light of two different wavelengths, λ_1 and λ_2, is incident on a grating. For the same order m, the angular positions are given by

$$d \sin \theta_1 = m\lambda_1$$

and

$$d \sin \theta_2 = m\lambda_2.$$

If $\lambda_2 > \lambda_1$, then

$$\sin \theta_2 > \sin \theta_1,$$

so

$$\theta_2 > \theta_1.$$

Thus, longer wavelengths appear at larger angles. This spreading of different wavelengths

is called **angular dispersion**. It is the basis of the use of diffraction gratings in optical spectroscopy.

Worked Example 4: Wavelength from a Grating

Problem. A diffraction grating has 500 lines per millimeter. The first-order maximum for a certain monochromatic light is observed at an angle of 15.0°. Find the wavelength of the light.

Solution. First find the grating spacing:

$$d = \frac{1}{500}\,\text{mm}.$$

Thus,

$$d = 2.00 \times 10^{-3}\,\text{mm} = 2.00 \times 10^{-6}\,\text{m}.$$

For the first order,

$$m = 1.$$

Using the grating equation,

$$d\sin\theta = m\lambda,$$

we get

$$\lambda = \frac{d\sin\theta}{m}.$$

Therefore,

$$\lambda = (2.00 \times 10^{-6})\sin 15.0^\circ.$$

Since

$$\sin 15.0^\circ = 0.2588,$$

we find

$$\lambda = 5.18 \times 10^{-7}\,\text{m}.$$

Thus,

$$\boxed{\lambda = 518\,\text{nm}.}$$

Worked Example 5: Maximum Order

Problem. A diffraction grating has spacing $d = 2.00 \times 10^{-6}\,\text{m}$. It is illuminated with light of wavelength 600 nm. What is the highest observable order?

Solution. The maximum order is determined by

$$m\lambda \leq d.$$

Thus,

$$m \leq \frac{d}{\lambda}.$$

Substitute

$$d = 2.00 \times 10^{-6}\,\text{m}, \qquad \lambda = 600 \times 10^{-9}\,\text{m}.$$

Then

$$m \leq \frac{2.00 \times 10^{-6}}{600 \times 10^{-9}} = 3.33.$$

The largest integer order is therefore

$$\boxed{m_{\max} = 3.}$$

Thus, the zeroth, first, second, and third orders can be observed, but the fourth order cannot.

4.5 Exercises

1. Define diffraction. Explain why diffraction is most noticeable when the size of an aperture is comparable with the wavelength.

2. Explain, using Huygens' principle, why light spreads after passing through a narrow slit.

3. In a single-slit diffraction experiment, explain why the central bright fringe is wider than the other bright fringes.

4. Derive the condition

   $$a \sin\theta = \lambda$$

   for the first dark fringe in single-slit diffraction.

5. Derive the general condition

   $$a \sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots$$

   for the dark fringes in single-slit diffraction.

6. Light of wavelength 550 nm passes through a slit of width 0.100 mm. Find the angle of the first diffraction minimum.

7. A single slit of width 0.250 mm is illuminated by light of wavelength 600 nm. A screen is placed 1.50 m away. Find:

   (a) the distance from the center to the first dark fringe,

   (b) the width of the central bright fringe.

8. A single-slit diffraction pattern is formed on a screen 2.00 m away. The slit width is 0.150 mm and the wavelength is 500 nm. Find the positions of the first three dark fringes on one side of the central maximum.

9. Explain why narrowing a slit increases the spreading of the diffraction pattern.

10. In a double-slit experiment, the slit width is a and the slit separation is d. Explain the physical meaning of the factors

$$\cos^2 \beta$$

and

$$\left(\frac{\sin \alpha}{\alpha}\right)^2$$

in the intensity expression.

11. A double-slit arrangement has $d = 0.400\,\text{mm}$ and $a = 0.100\,\text{mm}$. Determine the missing interference orders.

12. Define a diffraction grating. How does it differ from a double slit?

13. A diffraction grating has 600 lines per millimeter. Calculate the grating spacing in meters.

14. Light of wavelength 500 nm is incident normally on a grating with 400 lines per millimeter. Find the angle of the first-order maximum.

15. A grating has spacing $d = 1.67 \times 10^{-6}\,\text{m}$. The second-order maximum is observed at 40.0°. Determine the wavelength of the light.

16. A diffraction grating has 500 lines per millimeter and is illuminated by light of wavelength 600 nm. Determine the maximum observable order.

17. Explain why a diffraction grating is better than a double slit for measuring wavelengths accurately.

18. White light is incident on a diffraction grating. Describe qualitatively the pattern observed on a screen.

19. In a grating spectrum, which color is deviated more in a given order, red or blue? Explain your answer.

20. A grating produces a first-order maximum for light of wavelength 650 nm at an angle of 25.0°. Find the grating spacing and the number of lines per millimeter.

4.6 Solutions

Solution 1

Question. Define diffraction. Explain why diffraction is most noticeable when the size of an aperture is comparable with the wavelength.

Solution. Diffraction is the spreading or bending of waves when they pass through an aperture or around an obstacle.

Diffraction is most noticeable when the aperture size is comparable with the wavelength of the wave. If the aperture is much larger than the wavelength, most of the wave travels almost straight through, and the spreading is small. However, if the aperture width is of the same order as the wavelength, the wavelets from different parts of the opening spread significantly and overlap strongly. This produces an observable diffraction pattern.

Thus, diffraction becomes prominent when

$$a \sim \lambda,$$

where a is the aperture width and λ is the wavelength.

Solution 2

Question. Explain, using Huygens' principle, why light spreads after passing through a narrow slit.

Solution. According to Huygens' principle, every point on a wavefront acts as a source of secondary wavelets. When a plane wave reaches a narrow slit, only the part of the wavefront inside the slit passes through. Each point across the slit then acts as a source of secondary wavelets.

These secondary wavelets spread out into the region beyond the slit. Since the slit is narrow, the emerging wavefront is no longer simply plane; it spreads into regions that would not be reached if light travelled strictly in straight lines.

Therefore, light spreads after passing through a narrow slit because the wavelets emerging from different points of the slit propagate in different directions and overlap.

Solution 3

Question. In a single-slit diffraction experiment, explain why the central bright fringe is wider than the other bright fringes.

Solution. The central bright fringe lies between the first dark fringe above the central axis and the first dark fringe below the central axis. For a single slit of width a, the dark fringes occur when

$$a \sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots$$

The first minima occur at

$$a \sin\theta = \pm\lambda.$$

For small angles,

$$\sin\theta \approx \frac{y}{D},$$

so the first dark fringes are located at

$$y = \pm\frac{\lambda D}{a}.$$

Hence the width of the central bright fringe is

$$W_{\text{central}} = \frac{\lambda D}{a} - \left(-\frac{\lambda D}{a}\right) = \frac{2\lambda D}{a}.$$

The other bright fringes lie between successive dark fringes on one side of the central maximum. Their approximate width is about

$$\frac{\lambda D}{a}.$$

Thus, the central bright fringe is about twice as wide as the other bright fringes.

Solution 4

Question. Derive the condition

$$a\sin\theta = \lambda$$

for the first dark fringe in single-slit diffraction.

Solution. Consider a slit of width a. To locate the first dark fringe, divide the slit into two equal halves, each of width $a/2$.

Consider two rays travelling to a point P on the screen at an angle θ to the central axis. Let one ray come from the top of the slit and the other from the midpoint of the slit. The separation between these two points is $a/2$.

For a distant screen, the rays may be treated as parallel. The path difference between these two rays is

$$\Delta L = \frac{a}{2}\sin\theta.$$

For destructive interference, the two rays must arrive out of phase by π, corresponding to a path difference of $\lambda/2$. Thus,

$$\frac{a}{2}\sin\theta = \frac{\lambda}{2}.$$

Multiplying both sides by 2 gives

$$a\sin\theta = \lambda.$$

Therefore, the first dark fringe occurs when

$$\boxed{a\sin\theta = \lambda.}$$

Solution 5

Question. Derive the general condition

$$a\sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots$$

for the dark fringes in single-slit diffraction.

Solution. For the first dark fringe, the slit is divided into two equal parts, and corresponding rays from the two halves cancel.

For the second dark fringe, the slit is divided into four equal parts. Rays from adjacent pairs of zones cancel if the path difference between corresponding rays is $\lambda/2$.

For the mth dark fringe, the slit can be divided into $2m$ equal zones. The path difference between the top and bottom rays from the slit is

$$\Delta L = a \sin\theta.$$

For complete cancellation, this total path difference must be equal to an integer number of wavelengths:

$$\Delta L = m\lambda.$$

Therefore,

$$a \sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots$$

Hence, the single-slit dark fringes satisfy

$$\boxed{a \sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots}$$

Solution 6

Question. Light of wavelength 550 nm passes through a slit of width 0.100 mm. Find the angle of the first diffraction minimum.

Solution. For the first diffraction minimum,

$$a \sin\theta = \lambda.$$

Thus,

$$\sin\theta = \frac{\lambda}{a}.$$

Convert the given quantities to SI units:

$$\lambda = 550\,\text{nm} = 550 \times 10^{-9}\,\text{m},$$

$$a = 0.100\,\text{mm} = 0.100 \times 10^{-3}\,\text{m}.$$

Therefore,

$$\sin\theta = \frac{550 \times 10^{-9}}{0.100 \times 10^{-3}}.$$

$$\sin\theta = 5.50 \times 10^{-3}.$$

Hence,

$$\theta = \sin^{-1}(5.50 \times 10^{-3}).$$

For small angles,

$$\theta \approx 5.50 \times 10^{-3}\,\text{rad}.$$

In degrees,

$$\theta = 5.50 \times 10^{-3} \left(\frac{180^\circ}{\pi}\right) \approx 0.315^\circ.$$

Therefore,

$$\boxed{\theta \approx 5.50 \times 10^{-3}\,\text{rad} \approx 0.315^\circ.}$$

Solution 7

Question. A single slit of width 0.250 mm is illuminated by light of wavelength 600 nm. A screen is placed 1.50 m away. Find:

1. the distance from the center to the first dark fringe,
2. the width of the central bright fringe.

Solution. Given:

$$a = 0.250\,\text{mm} = 0.250 \times 10^{-3}\,\text{m},$$

$$\lambda = 600\,\text{nm} = 600 \times 10^{-9}\,\text{m},$$

$$D = 1.50\,\text{m}.$$

(a) Distance from the center to the first dark fringe

For small angles, the position of the first dark fringe is

$$y_1 = \frac{\lambda D}{a}.$$

Substituting,

$$y_1 = \frac{(600 \times 10^{-9})(1.50)}{0.250 \times 10^{-3}}.$$

$$y_1 = \frac{9.00 \times 10^{-7}}{2.50 \times 10^{-4}}.$$

$$y_1 = 3.60 \times 10^{-3}\,\text{m}.$$

Therefore,

$$\boxed{y_1 = 3.60\,\text{mm}.}$$

(b) Width of the central bright fringe

The central bright fringe extends from $-y_1$ to $+y_1$, so

$$W_{\text{central}} = 2y_1.$$

Thus,

$$W_{\text{central}} = 2(3.60\,\text{mm}) = 7.20\,\text{mm}.$$

Therefore,

$$\boxed{W_{\text{central}} = 7.20\,\text{mm}.}$$

Solution 8

Question. A single-slit diffraction pattern is formed on a screen 2.00 m away. The slit width is 0.150 mm and the wavelength is 500 nm. Find the positions of the first three dark fringes on one side of the central maximum.

Solution. For single-slit diffraction minima,

$$y_m = \frac{m\lambda D}{a}, \qquad m = 1, 2, 3, \ldots$$

Given:

$$D = 2.00\,\text{m},$$

$$a = 0.150\,\text{mm} = 0.150 \times 10^{-3}\,\text{m},$$

$$\lambda = 500\,\text{nm} = 500 \times 10^{-9}\,\text{m}.$$

First compute the spacing between successive minima:

$$\frac{\lambda D}{a} = \frac{(500 \times 10^{-9})(2.00)}{0.150 \times 10^{-3}}.$$

$$\frac{\lambda D}{a} = \frac{1.00 \times 10^{-6}}{1.50 \times 10^{-4}} = 6.67 \times 10^{-3}\,\text{m}.$$

Thus,

$$y_1 = 6.67\,\text{mm},$$

$$y_2 = 2(6.67\,\text{mm}) = 13.3\,\text{mm},$$

$$y_3 = 3(6.67\,\text{mm}) = 20.0\,\text{mm}.$$

Therefore, the first three dark fringes on one side of the central maximum are located at

$$\boxed{6.67\,\text{mm}, \quad 13.3\,\text{mm}, \quad 20.0\,\text{mm}.}$$

Solution 9

Question. Explain why narrowing a slit increases the spreading of the diffraction pattern.

Solution. For single-slit diffraction, the first dark fringe satisfies

$$a \sin\theta = \lambda.$$

Therefore,

$$\sin\theta = \frac{\lambda}{a}.$$

If the wavelength λ is fixed and the slit width a is decreased, the ratio λ/a increases. Hence $\sin\theta$ increases, and so the angle θ to the first minimum becomes larger.

Since the first dark fringes mark the edges of the central bright fringe, increasing θ makes the central bright fringe wider. Therefore, narrowing the slit increases the spreading of the diffraction pattern.

A narrower slit produces greater diffraction.

Solution 10

Question. In a double-slit experiment, the slit width is a and the slit separation is d. Explain the physical meaning of the factors

$$\cos^2\beta$$

and

$$\left(\frac{\sin\alpha}{\alpha}\right)^2$$

in the intensity expression.

Solution. For a double slit with finite slit width, the intensity is

$$I(\theta) = I_m \cos^2 \beta \left(\frac{\sin \alpha}{\alpha} \right)^2,$$

where

$$\beta = \frac{\pi d}{\lambda} \sin \theta$$

and

$$\alpha = \frac{\pi a}{\lambda} \sin \theta.$$

The factor

$$\cos^2 \beta$$

is the **interference factor**. It arises from the interference between light waves emerging from the two different slits. It determines the positions of the regularly spaced double-slit interference maxima and minima.

The factor

$$\left(\frac{\sin \alpha}{\alpha} \right)^2$$

is the **diffraction factor**. It arises because each slit has a finite width a, so each slit produces its own single-slit diffraction pattern. This factor forms the broad envelope that controls the brightness of the interference fringes.

Thus, the double-slit pattern with finite slit width consists of narrow interference fringes modulated by a broader single-slit diffraction envelope.

Solution 11

Question. A double-slit arrangement has $d = 0.400\,\text{mm}$ and $a = 0.100\,\text{mm}$. Determine the missing interference orders.

Solution. Missing orders occur when an interference maximum coincides with a diffraction minimum.

The interference maxima occur at

$$d\sin\theta = m\lambda.$$

The diffraction minima occur at

$$a\sin\theta = p\lambda, \qquad p = 1, 2, 3, \ldots$$

Dividing the first equation by the second gives

$$\frac{d}{a} = \frac{m}{p}.$$

Thus,

$$m = p\frac{d}{a}.$$

Given

$$d = 0.400\,\text{mm}, \qquad a = 0.100\,\text{mm}.$$

Therefore,

$$\frac{d}{a} = \frac{0.400}{0.100} = 4.$$

Hence,

$$m = 4p.$$

For

$$p = 1, 2, 3, \ldots,$$

the missing orders are

$$m = 4, 8, 12, \ldots$$

Therefore,

$$\boxed{\text{The fourth, eighth, twelfth, ...interference orders are missing.}}$$

Solution 12

Question. Define a diffraction grating. How does it differ from a double slit?

Solution. A diffraction grating is an optical device consisting of a large number of equally spaced parallel slits or grooves. These slits or grooves produce interference among many coherent waves.

A double slit has only two slits, while a diffraction grating has many slits, often hundreds or thousands per millimeter.

The main differences are:

- A double slit produces relatively broad interference fringes.
- A diffraction grating produces very narrow, sharp bright lines.
- A grating separates different wavelengths more clearly than a double slit.

- A grating is therefore more useful for accurate wavelength measurements and spectroscopy.

Solution 13

Question. A diffraction grating has 600 lines per millimeter. Calculate the grating spacing in meters.

Solution. The grating spacing is the distance between adjacent lines. If the grating has 600 lines per millimeter, then

$$d = \frac{1}{600}\,\text{mm}.$$

Thus,

$$d = 1.667 \times 10^{-3}\,\text{mm}.$$

Since

$$1\,\text{mm} = 10^{-3}\,\text{m},$$

we get

$$d = 1.667 \times 10^{-3} \times 10^{-3}\,\text{m}.$$

Therefore,

$$d = 1.667 \times 10^{-6}\,\text{m}.$$

Hence,

$$\boxed{d \approx 1.67 \times 10^{-6}\,\text{m}.}$$

Solution 14

Question. Light of wavelength 500 nm is incident normally on a grating with 400 lines per millimeter. Find the angle of the first-order maximum.

Solution. For a diffraction grating,

$$d \sin\theta = m\lambda.$$

For the first-order maximum,

$$m = 1.$$

The grating has 400 lines per millimeter, so

$$d = \frac{1}{400}\,\text{mm}.$$

Thus,

$$d = 2.50 \times 10^{-3}\,\text{mm} = 2.50 \times 10^{-6}\,\text{m}.$$

The wavelength is

$$\lambda = 500\,\text{nm} = 500 \times 10^{-9}\,\text{m}.$$

Using

$$\sin\theta = \frac{m\lambda}{d},$$

we have

$$\sin\theta = \frac{(1)(500\times10^{-9})}{2.50\times10^{-6}}.$$

$$\sin\theta = 0.200.$$

Therefore,

$$\theta = \sin^{-1}(0.200).$$

$$\theta \approx 11.5^\circ.$$

Hence,

$$\boxed{\theta \approx 11.5^\circ.}$$

Solution 15

Question. A grating has spacing $d = 1.67\times10^{-6}\,\text{m}$. The second-order maximum is observed at 40.0°. Determine the wavelength of the light.

Solution. The grating equation is

$$d\sin\theta = m\lambda.$$

Solving for λ gives

$$\lambda = \frac{d \sin \theta}{m}.$$

Given:

$$d = 1.67 \times 10^{-6}\,\text{m},$$

$$m = 2,$$

$$\theta = 40.0^\circ.$$

Thus,

$$\lambda = \frac{(1.67 \times 10^{-6}) \sin 40.0^\circ}{2}.$$

Since

$$\sin 40.0^\circ = 0.6428,$$

we get

$$\lambda = \frac{(1.67 \times 10^{-6})(0.6428)}{2}.$$

$$\lambda = \frac{1.073 \times 10^{-6}}{2} = 5.37 \times 10^{-7}\,\text{m}.$$

Therefore,

$$\boxed{\lambda \approx 5.37 \times 10^{-7}\,\text{m} = 537\,\text{nm}.}$$

Solution 16

Question. A diffraction grating has 500 lines per millimeter and is illuminated by light of wavelength 600 nm. Determine the maximum observable order.

Solution. The grating spacing is

$$d = \frac{1}{500}\,\text{mm}.$$

Thus,

$$d = 2.00 \times 10^{-3}\,\text{mm} = 2.00 \times 10^{-6}\,\text{m}.$$

The wavelength is

$$\lambda = 600\,\text{nm} = 600 \times 10^{-9}\,\text{m}.$$

For grating maxima,

$$d \sin\theta = m\lambda.$$

Since

$$\sin\theta \leq 1,$$

the maximum possible order satisfies

$$m\lambda \leq d.$$

Thus,

$$m \leq \frac{d}{\lambda}.$$

Substituting,

$$m \leq \frac{2.00 \times 10^{-6}}{600 \times 10^{-9}}.$$

$$m \leq 3.33.$$

The largest integer value is

$$m_{\max} = 3.$$

Therefore,

$$\boxed{m_{\max} = 3.}$$

Solution 17

Question. Explain why a diffraction grating is better than a double slit for measuring wavelengths accurately.

Solution. A diffraction grating has many slits, whereas a double slit has only two. Because many waves interfere in a grating, the principal maxima are very narrow and sharp. This makes their angular positions easier to measure accurately.

In a double-slit experiment, the bright fringes are broader and may overlap more strongly, especially if the light contains several wavelengths. In a grating, different wavelengths are separated into distinct bright lines. This makes it possible to determine wavelengths with greater precision.

Therefore, a diffraction grating is better for wavelength measurement because it produces

sharp, well-separated spectral lines.

Solution 18

Question. White light is incident on a diffraction grating. Describe qualitatively the pattern observed on a screen.

Solution. White light contains many wavelengths, ranging from violet to red. For a diffraction grating, the maxima satisfy

$$d \sin \theta = m\lambda.$$

For the zeroth order,

$$m = 0,$$

so

$$\theta = 0$$

for all wavelengths. Therefore, all colors overlap at the center, producing a central white maximum.

For higher orders, the angle depends on wavelength:

$$\sin \theta = \frac{m\lambda}{d}.$$

Thus, different wavelengths are diffracted at different angles. Violet light, which has a shorter wavelength, appears at a smaller angle, while red light, which has a longer wavelength, appears at a larger angle.

Therefore, the screen shows:

- a central white maximum,
- spectra on both sides of the central maximum,
- violet closer to the center,
- red farther from the center.

Solution 19

Question. In a grating spectrum, which color is deviated more in a given order, red or blue? Explain your answer.

Solution. For a diffraction grating,

$$d \sin \theta = m\lambda.$$

For a fixed grating spacing d and a fixed order m,

$$\sin \theta \propto \lambda.$$

Red light has a longer wavelength than blue light. Therefore, red light has a larger value of $\sin \theta$, and hence a larger diffraction angle θ.

Thus, red light is deviated more than blue light in a given order.

$$\boxed{\text{Red light is deviated more than blue light.}}$$

Solution 20

Question. A grating produces a first-order maximum for light of wavelength 650 nm at an angle of 25.0°. Find the grating spacing and the number of lines per millimeter.

Solution. The grating equation is

$$d \sin \theta = m\lambda.$$

For the first-order maximum,

$$m = 1.$$

Thus,

$$d = \frac{m\lambda}{\sin \theta}.$$

Given:

$$\lambda = 650\,\text{nm} = 650 \times 10^{-9}\,\text{m},$$

$$\theta = 25.0^\circ.$$

Therefore,

$$d = \frac{(1)(650 \times 10^{-9})}{\sin 25.0^\circ}.$$

Since

$$\sin 25.0^\circ = 0.4226,$$

we get

$$d = \frac{650 \times 10^{-9}}{0.4226}.$$

$$d = 1.54 \times 10^{-6}\,\text{m}.$$

Thus,

$$\boxed{d = 1.54 \times 10^{-6}\,\text{m}.}$$

Now find the number of lines per millimeter. First convert d to millimeters:

$$d = 1.54 \times 10^{-6}\,\text{m} = 1.54 \times 10^{-3}\,\text{mm}.$$

The number of lines per millimeter is

$$n = \frac{1}{d}.$$

Thus,

$$n = \frac{1}{1.54 \times 10^{-3}\,\text{mm}}.$$

$$n \approx 6.49 \times 10^{2}\,\text{lines/mm}.$$

Therefore,

$$\boxed{n \approx 649\,\text{lines/mm}.}$$

Chapter 5

Electrostatics

5.1 Introduction

Electrostatics is the branch of physics concerned with electric charges at rest. It deals with the forces, fields, potentials, and energies associated with stationary charges. The study of electrostatics is important because it provides the foundation for understanding electric circuits, capacitors, electric currents, electromagnetic waves, and many modern technologies.

Everyday electrostatic effects include the attraction of small pieces of paper to a rubbed plastic ruler, the crackling sound heard when removing a synthetic sweater, and lightning during a thunderstorm. In each case, electric charges are separated or redistributed, giving rise to electric forces.

In this chapter we shall study electric charge, Coulomb's law, the electric field, electric field lines, electric potential, electric potential energy, equipotential surfaces, and the relationship between electric field and electric potential.

5.2 Electric Charge

Electric charge is a fundamental property of matter. It is the physical property responsible for electric forces and electric interactions.

There are two types of electric charge:

- positive charge,
- negative charge.

Experiments show that charges interact according to the following rule:

Like charges repel, while unlike charges attract.

Thus, two positive charges repel each other, two negative charges also repel each other, but a positive charge and a negative charge attract each other.

5.2.1 Electric Charge in Matter

Atoms are made up of protons, neutrons, and electrons. Protons carry positive charge, electrons carry negative charge, and neutrons have no net electric charge. In a neutral atom, the number of protons is equal to the number of electrons, so the total charge is zero.

If an atom gains electrons, it becomes negatively charged. If it loses electrons, it becomes positively charged. Thus, most charging processes involve the transfer of electrons from one object to another.

5.2.2 Conductors and Insulators

Materials can be classified according to how easily electric charge moves through them.

Conductors

Conductors are materials through which electric charge can move rather freely. In metals, some electrons are weakly bound to atoms and can move through the material. These electrons are called conduction electrons.

Examples of conductors include:

- copper,
- aluminium,
- silver,
- the human body,
- tap water.

Metals are commonly used in electrical wiring because they allow charge to flow easily.

Insulators

Insulators, also called nonconductors, are materials through which electric charge cannot move freely. In an insulator, electrons are tightly bound to atoms and are not free to move through the material.

Examples of insulators include:

- rubber,
- plastic,
- glass,

- dry wood,
- chemically pure water.

Rubber and plastic are used to cover electric wires because they prevent electric charge from leaving the wire and passing into the surroundings.

Semiconductors

Semiconductors are materials whose electrical properties lie between those of conductors and insulators. Their ability to conduct charge can be controlled by temperature, impurities, or applied electric fields.

Examples of semiconductors include:

- silicon,
- germanium.

Semiconductors are essential in electronic devices such as diodes, transistors, solar cells, and computer chips.

Superconductors

Superconductors are materials that allow electric charge to move without resistance below a certain critical temperature. In a superconductor, electric current can persist for a very long time without energy loss.

Superconductors are used in powerful magnets, magnetic resonance imaging (MRI), particle accelerators, and research devices.

5.2.3 Quantization of Charge

Electric charge is quantized. This means that charge exists only in discrete amounts. Any observable charge q can be written as an integer multiple of the elementary charge e:

$$q = ne,$$

where

$$n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

and

$$e = 1.602 \times 10^{-19} \text{ C}.$$

The charge on a proton is

$$+e = +1.602 \times 10^{-19} \text{ C},$$

while the charge on an electron is

$$-e = -1.602 \times 10^{-19} \text{ C}.$$

Thus, a body cannot have a charge of $0.5e$ or $2.3e$. It must have a charge equal to an integer multiple of e.

5.2.4 Conservation of Charge

Electric charge is conserved. This means that the total charge of an isolated system remains constant. Charge can be transferred from one body to another, but it cannot be created or destroyed.

For example, when a glass rod is rubbed with silk, electrons may be transferred from the glass to the silk. The glass rod becomes positively charged and the silk becomes negatively charged. However, the total charge of the glass–silk system remains unchanged.

5.3 Coulomb's Law

When two charged particles are brought near each other, each particle exerts an electrostatic force on the other. This force may be attractive or repulsive depending on the signs of the charges.

If the charges have the same sign, they repel each other. If the charges have opposite signs, they attract each other.

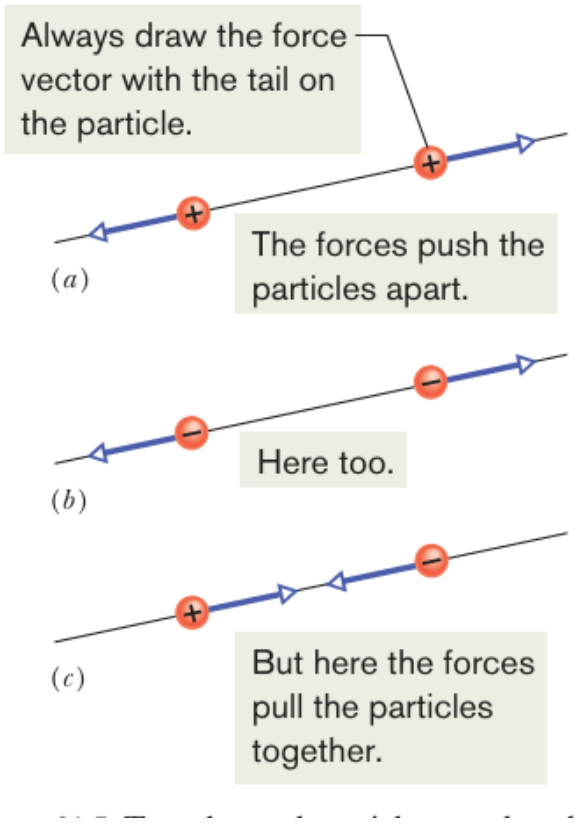


Figure 5.1: Forces between charged particles. Like charges repel, while unlike charges attract.

5.3.1 Statement of Coulomb's Law

Coulomb's law states that the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

If two point charges q_1 and q_2 are separated by a distance r, then the magnitude of the electrostatic force between them is

$$F = k\frac{|q_1 q_2|}{r^2},$$

where k is Coulomb's constant.

The value of Coulomb's constant is

$$k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2.$$

5.3.2 Vector Form of Coulomb's Law

Coulomb's law can also be written in vector form. Suppose charge q_2 exerts a force on charge q_1. Let $\hat{r}$ be a unit vector directed from q_2 toward q_1. Then

$$\vec{F}_{12} = k\frac{q_1 q_2}{r^2}\hat{r}.$$

Here, $\vec{F}_{12}$ is the force on q_1 due to q_2.

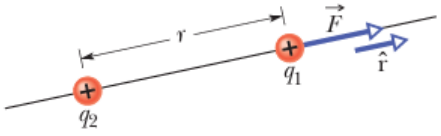


Figure 5.2: The electrostatic force on charge q_1 due to charge q_2.

The sign of the product $q_1 q_2$ determines whether the force is attractive or repulsive.

- If $q_1 q_2 > 0$, the charges have the same sign and the force is repulsive.

- If $q_1 q_2 < 0$, the charges have opposite signs and the force is attractive.

5.3.3 Permittivity of Free Space

For historical and practical reasons, Coulomb's constant is often written as

$$k = \frac{1}{4\pi\epsilon_0},$$

where ϵ_0 is called the permittivity of free space. Its value is

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N m}^2.$$

Thus, Coulomb's law may also be written as

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2}.$$

5.3.4 Principle of Superposition

The electrostatic force obeys the principle of superposition. If a charge is acted upon by several other charges, the net force on it is the vector sum of the individual forces.

If a charge q experiences forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \ldots$, then the net force is

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots .$$

This principle is essential when dealing with systems containing more than two charges.

Example 1: Force Between Two Point Charges

Two charges $q_1 = +3.0$ μC and $q_2 = +5.0$ μC are separated by a distance of 0.20 m. Find the magnitude of the electrostatic force between them.

Solution

Using Coulomb's law,

$$F = k\frac{|q_1 q_2|}{r^2}.$$

Substituting the values,

$$F = (8.99 \times 10^9)\frac{(3.0 \times 10^{-6})(5.0 \times 10^{-6})}{(0.20)^2}.$$

Therefore,

$$F = 3.37 \text{ N}.$$

Hence,

$$\boxed{F = 3.37 \text{ N}.}$$

Since both charges are positive, the force is repulsive.

5.4 The Electric Field

The concept of an electric field allows us to describe electric forces in terms of the space around charged objects. A charged object sets up an electric field in the surrounding region. Any other charge placed in that region experiences an electric force.

An electric field is a vector field because it has both magnitude and direction at every point in space.

5.4.1 Definition of Electric Field

To define the electric field at a point P, we imagine placing a small positive test charge q_0 at that point. If the test charge experiences an electric force $\vec{F}$, then the electric field at P is defined as

$$\vec{E} = \frac{\vec{F}}{q_0}.$$

Therefore,

$$\vec{F} = q_0\vec{E}.$$

The SI unit of electric field is

$$\text{N/C}.$$

Since

$$1 \text{ N/C} = 1 \text{ V/m},$$

the electric field may also be measured in volts per metre.

The test charge must be small so that it does not significantly disturb the charge distribution that produces the field.

5.4.2 Direction of the Electric Field

The direction of the electric field at a point is defined as the direction of the force that would act on a positive test charge placed at that point.

Thus:

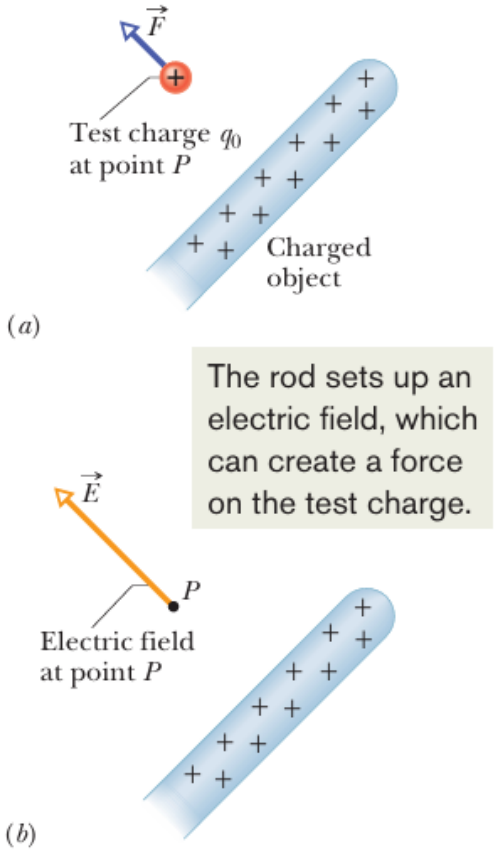


Figure 5.3: A charged object creates an electric field at point P. A positive test charge placed at P experiences a force.

- the electric field points away from positive charges,
- the electric field points toward negative charges.

If a negative charge is placed in an electric field, the force on it is opposite to the direction of the electric field because

$$\vec{F} = q\vec{E},$$

and q is negative.

5.4.3 Electric Field Lines

Electric field lines provide a useful way of representing electric fields visually. A field line is drawn so that the tangent to the line at any point gives the direction of the electric field at that point.

The rules for drawing electric field lines are:

1. The electric field vector at any point is tangent to the field line through that point.
2. The direction of a field line is the direction of the electric field.
3. Field lines originate on positive charges and terminate on negative charges.
4. The density of field lines represents the magnitude of the electric field.
5. Field lines never cross.

If field lines are close together, the electric field is strong. If they are far apart, the electric field is weak.

5.4.4 Electric Field Due to a Point Charge

Consider a point charge q. To find the electric field at a point a distance r from the charge, place a positive test charge q_0 at that point. From Coulomb's law, the force on the test charge is

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r^2}\hat{r}.$$

The electric field is

$$\vec{E} = \frac{\vec{F}}{q_0}.$$

Therefore,

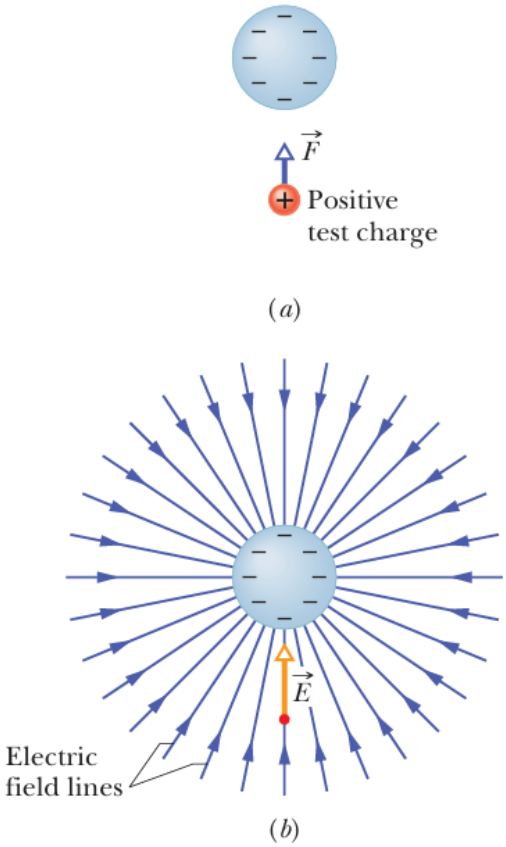


Figure 5.4: Electric field lines around a negatively charged sphere. The field lines point radially inward.

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}.$$

The magnitude of the electric field is

$$E = \frac{1}{4\pi\epsilon_0}\frac{|q|}{r^2}.$$

For a positive point charge, the electric field points radially outward. For a negative point charge, the electric field points radially inward.

5.4.5 Electric Field Due to Several Charges

Electric fields also obey the principle of superposition. If several charges produce electric fields at a point, the resultant electric field is the vector sum of the individual fields.

Thus,

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots .$$

This means that each charge contributes independently to the total electric field.

Example 2: Electric Field Due to a Point Charge

A charge $q = +2.0\ \mu$C is placed at the origin. Find the magnitude of the electric field at a point 0.50 m away.

Solution

The magnitude of the electric field due to a point charge is

$$E = k\frac{|q|}{r^2}.$$

Substituting,

$$E = (8.99 \times 10^9)\frac{2.0 \times 10^{-6}}{(0.50)^2}.$$

Thus,

$$E = 7.19 \times 10^4\ \text{N/C}.$$

Hence,

$$\boxed{E = 7.19 \times 10^4 \text{ N/C}.}$$

Since the charge is positive, the field points away from the charge.

5.5 Electric Potential and Electric Potential Energy

Electric force is a conservative force. This means that we can define an electric potential energy associated with the position of a charge in an electric field.

Electric potential energy is similar to gravitational potential energy. Just as a mass has gravitational potential energy due to its position in a gravitational field, a charge has electric potential energy due to its position in an electric field.

5.5.1 Electric Potential Energy

Consider a positive test charge q_0 placed at a point P in the electric field of a charged object. We choose the electric potential energy to be zero when the test charge is infinitely far away from the charged object.

If the electric force does work W in bringing the charge from infinity to point P, then the electric potential energy U at P is

$$U = -W.$$

The negative sign appears because electric potential energy is defined in relation to the work done by the electric force.

5.5.2 Electric Potential

The electric potential V at a point is defined as the electric potential energy per unit charge:

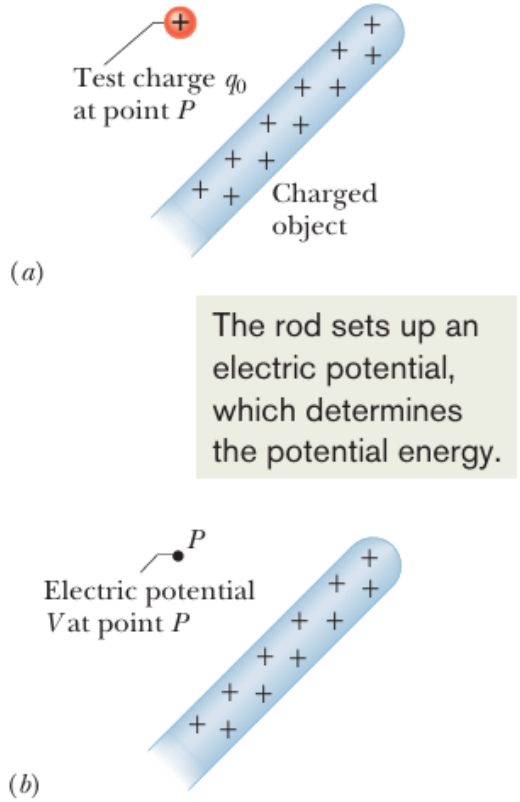


Figure 5.5: A charged object sets up an electric potential at point P.

$$V = \frac{U}{q_0}.$$

Equivalently, if $W_{\infty \to P}$ is the work done by the electric force in bringing the test charge from infinity to point P, then

$$V = -\frac{W_{\infty \to P}}{q_0}.$$

Electric potential is a scalar quantity. It has magnitude but no direction.

The SI unit of electric potential is the volt:

$$1 \text{ V} = 1 \text{ J/C}.$$

Thus, a potential of 1 V means that each coulomb of charge has 1 J of electric potential energy.

5.5.3 Potential Difference

If a charge moves from an initial point i to a final point f, the change in electric potential is

$$\Delta V = V_f - V_i.$$

If the charge has magnitude q, the corresponding change in electric potential energy is

$$\Delta U = q\Delta V.$$

Since the electric force is conservative, the work done by the electric field is

$$W = -\Delta U.$$

Therefore,

$$W = -q\Delta V.$$

If a positive charge moves to a lower potential, the electric field does positive work on it. If it moves to a higher potential, an external agent must do work against the electric field.

5.5.4 Electron-Volt

In atomic and nuclear physics, the joule is often too large to be convenient. A smaller unit of energy, called the electron-volt, is frequently used.

One electron-volt is the energy gained by a charge of magnitude e when it moves through a potential difference of 1 V:

$$1 \text{ eV} = e(1 \text{ V}).$$

Since

$$e = 1.602 \times 10^{-19} \text{ C},$$

we have

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}.$$

5.5.5 Equipotential Surfaces

An equipotential surface is a surface on which every point has the same electric potential.

If a charge moves between two points on the same equipotential surface, then

$$\Delta V = 0.$$

Therefore,

$$\Delta U = q\Delta V = 0.$$

The work done by the electric field is

$$W = -\Delta U = 0.$$

Thus, no work is done by the electric field when a charge moves along an equipotential surface.

Electric field lines are always perpendicular to equipotential surfaces. If the electric field had a component along an equipotential surface, work would be done in moving a charge along

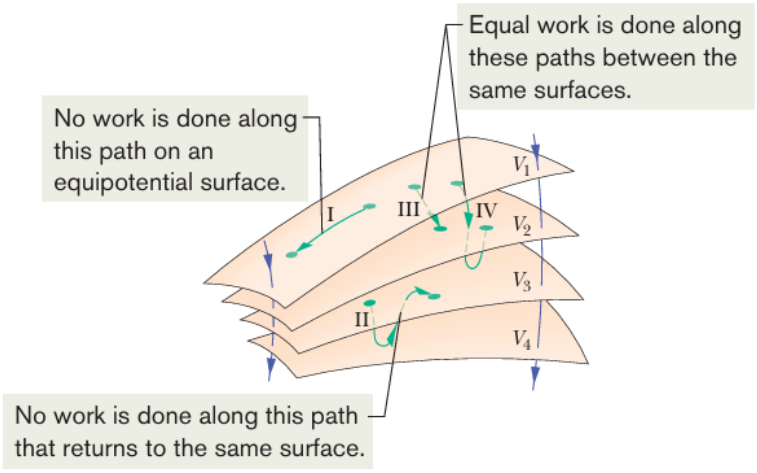


Figure 5.6: Equipotential surfaces. No work is done when a charge moves along a single equipotential surface.

that surface. Since no work is done, the field must be perpendicular to the surface.

5.6 Calculating Potential from the Electric Field

The electric potential difference between two points can be calculated from the electric field.

Consider a positive test charge q_0 moving from point i to point f through an electric field $\vec{E}$. At any point, the electric force on the charge is

$$\vec{F} = q_0 \vec{E}.$$

If the charge undergoes a small displacement $d\vec{s}$, the differential work done by the field is

$$dW = \vec{F} \cdot d\vec{s}.$$

Thus,

$$dW = q_0 \vec{E} \cdot d\vec{s}.$$

The total work done by the electric field as the charge moves from i to f is

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}.$$

Since

$$W = -\Delta U$$

and

$$\Delta U = q_0 \Delta V,$$

we obtain

$$q_0 \Delta V = -q_0 \int_i^f \vec{E} \cdot d\vec{s}.$$

Dividing by q_0,

$$\Delta V = V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}.$$

Therefore,

$$\boxed{V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}}$$

This important relation shows that the electric potential decreases in the direction of the electric field.

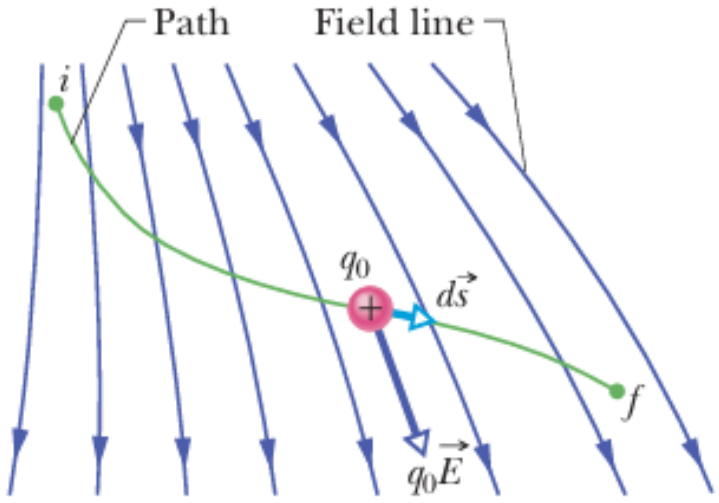


Figure 5.7: A test charge moving through an electric field from point i to point f.

5.6.1 Uniform Electric Field

For a uniform electric field, the magnitude and direction of $\vec{E}$ are constant. If the displacement is in the same direction as the field, then

$$\vec{E} \cdot d\vec{s} = E\, ds.$$

Thus,

$$\Delta V = -\int_i^f E\, ds.$$

Since E is constant,

$$\Delta V = -E \int_i^f ds.$$

Therefore,

$$\Delta V = -E\Delta x.$$

Hence,

$$\boxed{\Delta V = -E\Delta x.}$$

This means that when moving in the direction of the electric field, the electric potential decreases. When moving opposite to the field, the electric potential increases.

Equivalently, the electric field points from regions of higher potential toward regions of lower potential.

5.7 Potential Due to a Charged Particle

Consider a point charge q. We want to find the electric potential at a point P, a distance r from the charge. We choose the potential to be zero at infinity:

$$V(\infty) = 0.$$

The electric field due to the point charge is

$$E = \frac{1}{4\pi\epsilon_0}\frac{|q|}{r^2}.$$

Using

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s},$$

we move a positive test charge from r to infinity along a radial line. Thus,

$$V(\infty) - V(r) = -\int_r^\infty E\, dr.$$

Since $V(\infty) = 0$,

$$-V(r) = -\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\, dr.$$

Therefore,

$$V(r) = \int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\, dr.$$

Evaluating the integral,

$$V(r) = \frac{q}{4\pi\epsilon_0}\int_r^\infty \frac{1}{r^2}\, dr.$$

Since

$$\int \frac{1}{r^2}\, dr = -\frac{1}{r},$$

we obtain

$$V(r) = \frac{q}{4\pi\epsilon_0}\left[-\frac{1}{r}\right]_r^\infty.$$

Thus,

$$V(r) = \frac{q}{4\pi\epsilon_0}\left(0 + \frac{1}{r}\right).$$

Therefore,

$$\boxed{V(r) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}}$$

or

$$\boxed{V(r) = k\frac{q}{r}.}$$

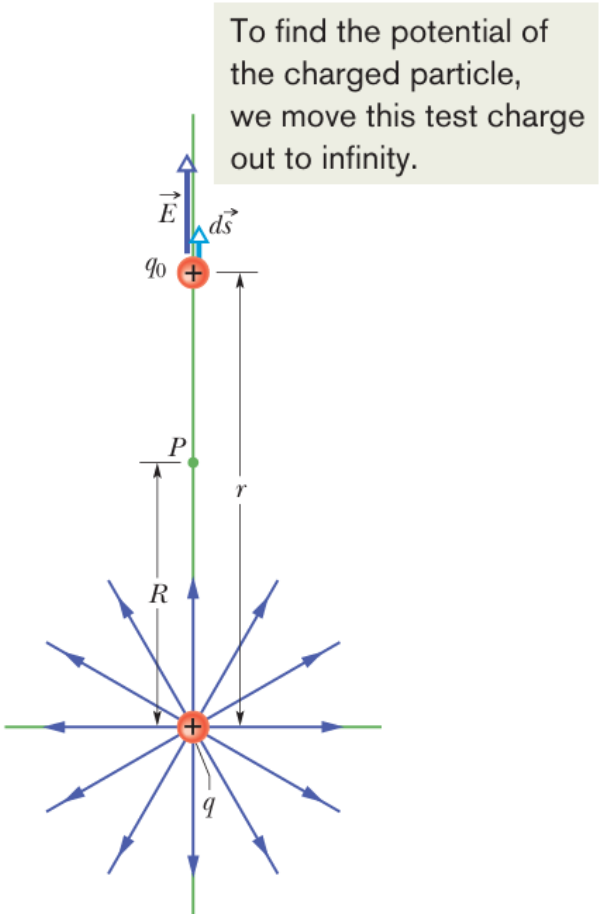


Figure 5.8: To find the potential due to a charged particle, a test charge is moved between point P and infinity.

A positive charge produces a positive electric potential, while a negative charge produces a negative electric potential.

5.7.1 Potential Due to a Group of Point Charges

Electric potential is a scalar quantity. Therefore, the total potential due to a group of point charges is the algebraic sum of the potentials due to each charge.

If charges $q_1, q_2, q_3, \ldots, q_n$ are located at distances $r_1, r_2, r_3, \ldots, r_n$ from a point P, then

$$V = V_1 + V_2 + V_3 + \cdots + V_n.$$

Thus,

$$\boxed{V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i}}$$

or

$$\boxed{V = k \sum_{i=1}^{n} \frac{q_i}{r_i}.}$$

Unlike electric field, electric potential does not require vector addition. Positive charges contribute positive potential, while negative charges contribute negative potential.

Example 3: Potential Due to a Point Charge

Find the electric potential at a point 0.30 m from a charge $q = +4.0\ \mu$C.

Solution

The potential due to a point charge is

$$V = k\frac{q}{r}.$$

Substituting,

$$V = (8.99 \times 10^9)\frac{4.0 \times 10^{-6}}{0.30}.$$

Thus,

$$V = 1.20 \times 10^5 \text{ V}.$$

Therefore,

$$\boxed{V = 1.20 \times 10^5 \text{ V}.}$$

Example 4: Potential Energy of a Charge in an Electric Potential

A charge $q = -2.0\ \mu\text{C}$ is placed at a point where the electric potential is $V = 300$ V. Find its electric potential energy.

Solution

Electric potential energy is

$$U = qV.$$

Substituting,

$$U = (-2.0 \times 10^{-6})(300).$$

Therefore,

$$U = -6.0 \times 10^{-4} \text{ J}.$$

Hence,

$$\boxed{U = -6.0 \times 10^{-4} \text{ J}.}$$

The negative sign shows that the charge has negative potential energy at that point.

5.8 Relationship Between Electric Field and Electric Potential

The electric field and electric potential are closely related. The electric field tells us the force per unit charge, while the electric potential tells us the potential energy per unit charge.

The relationship between them is

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}.$$

In one dimension, this can be written in differential form as

$$E_x = -\frac{dV}{dx}.$$

Thus, the electric field is the negative gradient of the electric potential. This means that the electric field points in the direction in which the potential decreases most rapidly.

5.9 Exercise Problems

1. A charge of $-3.0\ \mu$C is placed 0.50 m from a charge of $+2.0\ \mu$C. Determine the magnitude and direction of the force on each charge.

2. Of the charge Q initially on a tiny sphere, a portion q is to be transferred to a second, nearby sphere. Both can be treated as particles and are fixed with a certain separation. For what value of q/Q will the electrostatic force between the two spheres be maximized?

3. Two equally charged particles are held 3.2×10^{-3} m apart and then released from rest. The initial acceleration of the first particle is observed to be 7.0 m/s^2 and that of the second to be 9.0 m/s^2.If the mass of the first particle is 6.3×10^{-7} kg,what are (a) the mass of the second particle and (b) the magnitude of the charge of each particle?

4. Three particles are fixed on an x axis. Particle 1 of charge q_1 is at $x = -a$, and particle

2 of charge q2 is at $x = -a$. If their net electrostatic force on particle 3 of charge Q is to be zero, what must be the ratio $q1/q2$ when particle 3 is at (a) $x = +0.500a$ and (b) $x = +1.50a$?

5. The charges and coordinates of two charged particles held fixed in an xy plane are $q_1 = 3.0$ μC, $x_1 = 3.5$ cm, $y_1 = .50$ cm, and $q_2 = 4.0$ μC, $x_2 = 2.0$ cm, $y_2 = 1.5$ cm. Find the (a) magnitude and (b) direction of the electrostatic force on particle 2 due to particle 1.At what (c) x and (d) y coordinates should a third particle of charge $q_3 = 4.0$ μC be placed such that the net electrostatic force on particle 2 due to particles 1 and 3 is zero?

6. A positive test charge of 2.0×10^{-8} C experiences a force of 4.0×10^{-4} N at a certain point in an electric field. Find the electric field at that point.

7. An electron with a speed of 5.00×10^8 cm/s enters an electric field of magnitude 1.00×10^3 N/C, traveling along a field line in the direction that retards its motion. (a) How far will the electron travel in the field before stopping momentarily, and (b) how much time will have elapsed? (c) If the region containing the electric field is 8.00 mm long (too short for the electron to stop within it), what fraction of the electron's initial kinetic energy will be lost in that region?

8. Find the electric potential at a point 0.40 m from a point charge $q = +8.0$ μC.

9. An electron is released from rest in a uniform electric field of magnitude 2.00×10^4 N/C. Calculate the acceleration of the electron.(Ignore gravitation.)

10. At some instant the velocity components of an electron moving between two charged parallel plates are $v_x = 1.5 \times 10^5$ m/s and $v_y 3.0 \times 10^3$m/s. Suppose the electric field between the plates is uniform and given by $\vec{E} = (120\text{N/C})\hat{\text{j}}$. In unit-vector notation, what are (a) the electron's acceleration in that field and (b) the electron's velocity when its x coordinate has changed by 2.0 cm?

11. A charge of +3.0 μC is moved through a potential difference of 120 V. Find the change in electric potential energy.

12. Suppose that in a lightning flash the potential difference between a cloud and the ground is 1.0×10^9 V and the quantity of charge transferred is 30 C.(a) What is the change in energy of that transferred charge? (b) If all the energy released could be used to accelerate a 1000 kg car from rest,what would be its final speed?

13. An electron is accelerated through a potential difference of 500 V. Find the energy gained by the electron in electron-volts and in joules.

14. Explain why no work is done by the electric field when a charge moves along an equipotential surface.

15. A uniform electric field has magnitude 2.0×10^3 V/m. Calculate the potential difference between two points separated by 0.050 m in the direction of the field.

16. A spherical drop of water carrying a charge of 30 pC has a potential of 500 V at its surface (with V 0 at infinity). (a) What is the radius of the drop? (b) If two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop?

17. Three point charges are placed near a point P. The charges are $q_1 = +2.0\ \mu$C, $q_2 = -3.0\ \mu$C, and $q_3 = +4.0\ \mu$C. Their distances from P are 0.20 m, 0.30 m, and 0.40 m, respectively. Calculate the total electric potential at P.